

Linear Algebra I & Math I

Weeks 1–4

All questions and all equations from every course file and the assigned book sections — each with a complete step-by-step solution. Includes the full Week 4 material: limit laws, standard limits, indeterminate forms, asymptotic equivalence and the Landau symbols O and o .

Compiled for independent study and Test 1 preparation

Tashkent University of Technology · Fall 2026

Sources: course Notes, Lectures, ISA, Practical Sessions; OpenStax *Calculus 1*; Strang, *Introduction to Linear Algebra* (4e) and *Linear Algebra and its Applications* (4e)

Contents

1	Course Map and How to Use This Document	3
2	Week 1 — Number Systems, Notation, Complex Numbers; Vectors	4
2.1	Math I: Equation catalogue for Week 1	4
2.1.1	Number sets and positional systems	4
2.1.2	Scientific and engineering notation	4
2.1.3	Complex numbers	5
2.2	Linear Algebra I: Equation catalogue for Week 1	5
2.3	Math I: Practical session, Week 1 — questions and full solutions	6
2.4	Linear Algebra I, Week 1: ISA exercises — questions and full solutions	11
2.5	Linear Algebra I, Week 1: TA session problems (alternative set) — questions and full solutions	14
3	Week 2 — Functions (Math I); Dot and Cross Products, Introduction to Matrices (Linear Algebra I)	15
3.1	Math I: Equation catalogue for Week 2	16
3.1.1	Functions: definition, classes, symmetry, composition, inverses	16
3.1.2	Exponential and logarithmic functions (§1.5, hyperbolic functions excluded)	16
3.2	Math I: Practical session, Week 2 (functions) — questions and full solutions	17
3.3	Math I: Independent study assignments, Week 2 — questions and full solutions	19
3.4	Linear Algebra I, Week 2: equation catalogue	21
3.5	Linear Algebra I, Week 2: lecture in-class problems — questions and full solutions	21
3.6	Linear Algebra I, Week 2: ISA exercises — questions and full solutions	23
3.7	Linear Algebra I, Week 2: TA session (15 problems) — questions and full solutions	26
4	Week 3 — Limits (Math I); Matrix Algebra, Determinants, Triple Product (Linear Algebra I)	28
4.1	Math I: Equation catalogue for Week 3	28
4.1.1	Rates of change and the two founding problems	28
4.1.2	The limit concept, one-sided and infinite limits	28
4.1.3	The precise definition (§2.5, lecture emphasis)	29
4.1.4	Two standard limits seen this week	29
4.2	Math I: Week 3 ISA — questions and full solutions	29
4.3	Math I: Week 3 Practical Session (engineering limits) — questions and full solutions	30
4.4	Math I: assigned textbook exercises (OpenStax §2.2, items 46–54 and 60–64)	32
4.5	Linear Algebra I: equation catalogue for Week 3	33
4.6	Linear Algebra I: Week 3 lecture problems (15) — questions and full solutions	33
5	Week 4 — Limits II: Limit Laws, Standard Limits, Indeterminate Forms, Asymptotics, Landau Symbols (Math I)	36
5.1	Equation catalogue for Week 4	37
5.1.1	General properties of limits (Week 4 Notes)	37
5.1.2	Limit laws (OpenStax Theorem 2.5)	37
5.1.3	The Squeeze (Pinching) Theorem (OpenStax Theorem 2.7)	37
5.1.4	Standard limits and the derived trigonometric limits (Week 4 Notes)	37
5.1.5	Indeterminate forms (Week 4 Notes)	37
5.1.6	Asymptotic equivalence (Week 4 Notes)	38
5.1.7	Landau symbols: big- O and little- o (Week 4 Notes)	38
5.2	OpenStax §2.3, Checkpoint problems 2.11–2.20 — questions and full solutions	38
5.3	OpenStax §2.3 Exercises 83–130 — questions and full solutions	40

5.3.1	Using the limit laws (83–86)	40
5.3.2	Direct substitution (87–92)	40
5.3.3	Indeterminate form $0/0$ (93–102)	41
5.3.4	$K/0$ forms: one-sided infinite limits (103–106)	42
5.3.5	Limit laws with given limits (107–114)	42
5.3.6	Piecewise functions and graph-based limits (115–125)	43
5.3.7	Squeeze theorem and applications (126–130)	44
5.4	Week 4 Notes worked examples — questions and full solutions	45
6	Week 4 — Matrix Rank and the Inverse Matrix (Linear Algebra I)	46
6.1	Equation catalogue for Week 4	46
6.1.1	Rank and elimination	46
6.1.2	The inverse matrix (Strang, <i>Introduction to Linear Algebra</i> 4e, §2.5)	46
6.1.3	Gauss–Jordan elimination	47
6.2	Week 4 practice problems (rank, inverse, Gauss–Jordan) — questions and full solutions	47
7	Textbook Supplements — Additional Required-Reading Exercises with Full Solutions	49
7.1	OpenStax §1.2 (Basic Classes of Functions) — selected exercises solved	49
7.2	OpenStax §1.5 (Exponential and Logarithmic Functions) — selected exercises solved	50
7.3	OpenStax §2.2 — table-based exercises (42–45) solved	52
7.4	Strang, <i>Introduction to Linear Algebra</i> 4e — selected problems solved	52
7.5	Strang, <i>Linear Algebra and its Applications</i> 4e, §1.6 (Inverses and Transposes) — key results and a worked example	53

1 Course Map and How to Use This Document

This document is a complete, self-contained companion for the first four weeks of the two parallel first-semester courses **Linear Algebra I** and **Math I** (Fall 2026). It was compiled by hand from every relevant source in the course repository `course-files/` folder: the weekly Notes, Lecture presentations, Practical Session exercise sheets, Independent Study Assignments (ISA), TA problem decks, the *GUIDANCE for Test 1 Preparation*, and the two textbooks (OpenStax *Calculus 1* and Gilbert Strang’s linear algebra texts). Every question that appears in those sources for Weeks 1–4 is reproduced here together with a full, step-by-step solution, and every equation, identity, definition and standard limit that the sources use is collected into per-week equation catalogues.

The scope follows the official course structure. In Math I, Week 1 covers number systems, positional representation, scientific and engineering notation, and complex numbers; Week 2 covers functions (OpenStax §1.2 and §1.5, excluding hyperbolic functions); Week 3 introduces limits (OpenStax §2.1, §2.2, §2.5); and Week 4 consolidates limits (OpenStax §2.3 plus the Week 4 Notes on standard limits, indeterminate forms, asymptotic equivalence and the Landau symbols O and o). In Linear Algebra I, Week 1 treats vectors, linear combinations, span and linear independence; Week 2 adds the dot product, the cross product and an introduction to matrices; Week 3 develops matrix algebra, determinants and the scalar triple product; and Week 4 covers matrix rank and the inverse matrix by Gauss–Jordan elimination, followed by Test 1 on the Week 1–3 material. The *GUIDANCE* sheet explicitly states that Test 1 emphasizes the Week 1 Notes Parts 2–3 (scientific/engineering notation, complex numbers), OpenStax §1.2 and §1.5, and OpenStax §2.2, so those sections receive the most detailed treatment below.

Each weekly part of this document has the same internal structure. First comes the *equation catalogue*: every definition, theorem statement, identity and standard limit used that week, written out formally and numbered continuously so that any formula can be cited from the solutions or from lectures. Then come the *questions*, grouped exactly as they appear in the source files, each tagged with its source, and each followed immediately by a complete solution. Graph-based exercises that the sources pose on a pictured function are solved by stating the values read from the graph, exactly as one would do in class. The final part collects supplementary textbook exercises from the required-reading sections and from Strang’s books, again with full solutions, so that the question bank here is complete “to the end”, as requested.

Week	Math I	Linear Algebra I
1	Number sets $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$; positional systems; scientific/engineering notation; complex numbers	Vectors; linear combinations; span; linear independence
2	Functions: classes, domain/range, symmetry, composition, inverses; exponential and logarithmic functions (§1.2, §1.5)	Dot product, angles, projections; cross product, areas; matrices: operations and $A\mathbf{x}$
3	Limits: intuitive definition, one-sided limits, infinite limits, vertical asymptotes, ε – δ (§2.1, §2.2, §2.5)	Matrix algebra; transpose; matrix multiplication; determinants; scalar triple product
4	Limits II: limit laws, squeeze theorem, standard limits, indeterminate forms, asymptotics, Landau symbols (§2.3 + Notes); Test 1	Matrix rank; inverse by Gauss–Jordan elimination; Test 1

2 Week 1 — Number Systems, Notation, Complex Numbers; Vectors

2.1 Math I: Equation catalogue for Week 1

2.1.1 Number sets and positional systems

The nesting of the standard number sets is the first structural fact of the course, and every later classification exercise relies on it. Rational numbers are exactly the fractions of integers, while irrational numbers are the real numbers that are not rational.

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}, \quad \mathbb{R} \setminus \mathbb{Q} = \{x \in \mathbb{R} : x \notin \mathbb{Q}\} \quad (\text{irrationals}) \quad (1)$$

$$q \in \mathbb{Q} \iff q = \frac{a}{b}, \quad a \in \mathbb{Z}, \quad b \in \mathbb{Z}, \quad b \neq 0, \quad d(a, b) = |a - b| \quad (\text{distance in } \mathbb{R}) \quad (2)$$

A *positional numeral system* with base b writes a number through powers of the base; the same number admits many numerals, but its value is independent of the representation. Base conversion uses the expansion below or repeated division for the reverse direction.

$$(a_n a_{n-1} \dots a_1 a_0)_b = \sum_{k=0}^n a_k b^k, \quad 0 \leq a_k \leq b-1, \quad (3)$$

$$13_{10} = 1101_2 = 15_8 = D_{16}, \quad A = 10, \quad B = 11, \quad \dots, \quad F = 15 \quad (\text{hexadecimal digits}). \quad (4)$$

Roman numerals are *non-positional*: I = 1, V = 5, X = 10, L = 50, C = 100, so 13 = XIII; arithmetic in such systems is impractical, which is why computers use binary: electronics reliably distinguishes two states (low/high, off/on, 0/1).

2.1.2 Scientific and engineering notation

Scientific notation isolates the magnitude of a number into a coefficient with one nonzero digit before the decimal point and a power of ten; engineering notation constrains the exponent to multiples of three so that the exponent couples to an SI prefix.

$$\text{scientific: } z = a \times 10^n, \quad 1 \leq |a| < 10, \quad n \in \mathbb{Z}, \quad (5)$$

$$\text{engineering: } z = a \times 10^{3k}, \quad 1 \leq |a| < 1000, \quad k \in \mathbb{Z}, \quad (6)$$

$$(a \times 10^m)(b \times 10^n) = ab \times 10^{m+n}, \quad \frac{a \times 10^m}{b \times 10^n} = \frac{a}{b} \times 10^{m-n}, \quad (7)$$

$$(a \times 10^n)^m = a^m \times 10^{nm}, \quad \text{add/subtract: rewrite to a common power of ten first.} \quad (8)$$

The SI prefixes used all semester are

Power	10^{12}	10^9	10^6	10^3	10^{-3}	10^{-6}	10^{-9}	10^{-12}	10^{-15}
Prefix	tera (T)	giga (G)	mega (M)	kilo (k)	milli (m)	micro (μ)	nano (n)	pico (p)	femto (f)

Case matters: 5 mW = 5×10^{-3} W but 5 MW = 5×10^6 W.

Reference values used throughout the course:

$c = 2.99792458 \times 10^8$ m/s; $e = 1.602176634 \times 10^{-19}$ C; $N_A = 6.02214076 \times 10^{23}$ mol⁻¹;
 $h = 6.62607015 \times 10^{-34}$ Js; 101325 Pa = 101.325 kPa.

2.1.3 Complex numbers

Week 1 closes with the complex numbers, which arise because $x^2 + 1 = 0$ has no real solution. The imaginary unit, equality rule, the four arithmetic operations and the conjugate are the working tools.

$$i = \sqrt{-1}, \quad i^2 = -1, \quad z = a + bi, \quad \operatorname{Re}(z) = a, \quad \operatorname{Im}(z) = b, \quad (9)$$

$$a + bi = c + di \iff a = c \text{ and } b = d, \quad i^{n+4} = i^n \quad (i^0 = 1, \quad i^1 = i, \quad i^2 = -1, \quad i^3 = -i), \quad (10)$$

$$z_1 + z_2 = (a + c) + (b + d)i, \quad z_1 - z_2 = (a - c) + (b - d)i, \quad (11)$$

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i, \quad \frac{z_1}{z_2} = \frac{z_1 \bar{z}_2}{|z_2|^2} \quad (z_2 \neq 0), \quad (12)$$

$$\bar{z} = a - bi, \quad z + \bar{z} = 2 \operatorname{Re}(z), \quad z - \bar{z} = 2i \operatorname{Im}(z), \quad z \bar{z} = a^2 + b^2 = |z|^2, \quad (13)$$

$$|z| = \sqrt{a^2 + b^2}, \quad |z_1 z_2| = |z_1| |z_2|, \quad \left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}, \quad (14)$$

$$|z_1 + z_2| \leq |z_1| + |z_2| \quad (\text{triangle inequality}) \quad (15)$$

$$\arg z = \theta + 2k\pi, \quad k \in \mathbb{Z}; \quad \operatorname{Arg}(z) \in (-\pi, \pi] \text{ principal value}, \quad \tan \theta = \frac{b}{a} \quad (\text{quadrant!}) \quad (16)$$

$$z = r(\cos \theta + i \sin \theta) = r e^{i\theta}, \quad r = |z|, \quad a = r \cos \theta, \quad b = r \sin \theta, \quad (17)$$

$$e^{i\theta} = \cos \theta + i \sin \theta \quad (\text{Euler}), \quad e^{i\pi} + 1 = 0, \quad (18)$$

$$z_1 z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)}, \quad \frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}, \quad (19)$$

$$z^n = r^n (\cos(n\theta) + i \sin(n\theta)) \quad (\text{de Moivre}), \quad (20)$$

$$z_k = r^{1/n} \left[\cos\left(\frac{\theta + 2k\pi}{n}\right) + i \sin\left(\frac{\theta + 2k\pi}{n}\right) \right], \quad k = 0, \dots, n-1, \quad (21)$$

$$z^n = 1 \implies z_k = e^{2\pi i k/n}, \quad k = 0, \dots, n-1 \quad (n\text{-th roots of unity}), \quad (22)$$

$$z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad (\text{quadratic formula, valid over } \mathbb{C}). \quad (23)$$

The roots of $z^n = w = r e^{i\theta}$ are given by (21); the n -th roots of unity form a regular n -gon on the unit circle. Non-real roots of real polynomials occur in conjugate pairs, and by the Fundamental Theorem of Algebra a degree- n polynomial has exactly n roots in $\mathbb{C}$, counted with multiplicity.

2.2 Linear Algebra I: Equation catalogue for Week 1

Vectors are columns in $\mathbb{R}^n$; all of Week 1 is built from addition, scalar multiplication, combinations, span and independence.

$$(\mathbf{u} + \mathbf{v})_i = u_i + v_i, \quad (c\mathbf{u})_i = c u_i \quad (\text{componentwise operations}) \quad (24)$$

$$c_1 \mathbf{v}_1 + \dots + c_n \mathbf{v}_n = \text{linear combination}, \quad (25)$$

$$\operatorname{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\} = \text{the set of all such combinations}, \quad (26)$$

$$\mathbf{v}_1, \dots, \mathbf{v}_n \text{ independent} \iff c_1 \mathbf{v}_1 + \dots + c_n \mathbf{v}_n = \mathbf{0} \implies c_1 = \dots = c_n = 0, \quad (27)$$

$$\|\mathbf{x}\| = \sqrt{x_1^2 + \dots + x_n^2}, \quad \hat{\mathbf{x}} = \frac{\mathbf{x}}{\|\mathbf{x}\|} \quad (\text{unit vector}) \quad (28)$$

$$\mathbf{v} \text{ of length } L \text{ along } \mathbf{u}: \quad \mathbf{v} = L \frac{\mathbf{u}}{\|\mathbf{u}\|}, \quad (29)$$

$$d(P, Q) = \|Q - P\|. \quad (30)$$

2.3 Math I: Practical session, Week 1 — questions and full solutions

1. Classify each decimal as terminating, repeating or nonrepeating, and state whether the corresponding number is rational or irrational:

(1) 0.375; (2) 0.272727...; (3) 1.101001000100001...;
 (4) 3.14159265...; (5) 0.123123123...

[Source: Week 1 Exercises for the Practical Session, Exercise 1]

Solution. (1) 0.375 terminates, hence rational: $0.375 = \frac{375}{1000} = \frac{3}{8}$. (2) The block 27 repeats forever, hence rational; for a pure repeating block, $\overline{ab} = \frac{ab}{99}$, so $0.\overline{27} = \frac{27}{99} = \frac{3}{11}$. (3) The gaps between successive 1s grow (1, 2, 3, ... zeros), so the decimal never becomes periodic and never terminates: the number is irrational. (4) Finitely many shown digits followed by ellipsis do not determine a number; if the intended number is $\pi = 3.14159265\dots$, it is nonterminating and nonrepeating, hence irrational. (5) The block 123 repeats, hence rational: $0.\overline{123} = \frac{123}{999} = \frac{41}{333}$. The principle: terminating or eventually repeating decimals are exactly the rationals.

2. Write each set using interval notation and represent it on the real line: (1) $\{x \in \mathbb{R} : -3 \leq x < 4\}$; (2) $\{x \in \mathbb{R} : x \neq 3\}$.

[Source: Week 1 Practical, Exercise 2]

Solution. (1) The endpoint -3 is included (bracket) and 4 excluded (parenthesis): $[-3, 4)$; on the line, a closed dot at -3 and an open dot at 4 with the segment between them highlighted. (2) All reals except 3: $(-\infty, 3) \cup (3, \infty)$; on the line, the whole axis is shaded with an open hole at 3. Infinity is never included, so it always takes a parenthesis.

3. Write each interval using inequalities: a. $[-2, 5)$; b. $(-\infty, 4]$; c. $(1, \infty)$.

[Source: Week 1 Practical, Exercise 3]

Solution. a. $-2 \leq x < 5$ (left endpoint in, right out). b. $x \leq 4$ (extends arbitrarily far left, right endpoint included). c. $x > 1$ (strictly greater, since the parenthesis excludes 1).

4. Classify as rational or irrational: a. $\sqrt{3}$; b. $\sqrt{25}$; c. $\sqrt{50}$; i. $(1 + \sqrt{5})(1 - \sqrt{5})$.

[Source: Week 1 Practical, Exercise 4]

Solution. a. 3 is not a perfect square, so $\sqrt{3}$ is irrational. b. $\sqrt{25} = 5 \in \mathbb{Z} \subseteq \mathbb{Q}$, rational. c. $\sqrt{50} = 5\sqrt{2}$; since $\sqrt{2}$ is irrational, $5\sqrt{2}$ is irrational. i. Expand: $(1 + \sqrt{5})(1 - \sqrt{5}) = 1 - 5 = -4$, rational. Multiplying conjugate pairs always kills the radical.

5. Given $z_1 = 3 + 2i$ and $z_2 = 1 - 4i$, calculate $z_1 + z_2$, $z_1 - z_2$, $z_1 z_2$, z_1/z_2 , writing every answer in the form $a + bi$.

[Source: Week 1 Practical, Exercise 5]

Solution. Addition and subtraction act on real and imaginary parts separately: $z_1 + z_2 = (3 + 1) + (2 - 4)i = 4 - 2i$; $z_1 - z_2 = (3 - 1) + (2 + 4)i = 2 + 6i$. For the product use $i^2 = -1$ with the rule $(a + bi)(c + di) = (ac - bd) + (ad + bc)i$: $z_1 z_2 = (3 \cdot 1 - 2 \cdot (-4)) + (3 \cdot (-4) + 2 \cdot 1)i = (3 + 8) + (-12 + 2)i = 11 - 10i$. For the quotient, multiply numerator and denominator by the conjugate of the denominator:

$$\frac{z_1}{z_2} = \frac{(3 + 2i)(1 + 4i)}{(1 - 4i)(1 + 4i)} = \frac{3 + 12i + 2i + 8i^2}{1 + 16} = \frac{(3 - 8) + (12 + 2)i}{17} = -\frac{5}{17} + \frac{14}{17}i.$$

6. For each number $3 + 4i$, $-2 + 5i$, $6i$, -7 , find its conjugate and modulus; verify $\bar{z}z = |z|^2$.

[Source: Week 1 Practical, Exercise 6]

Solution. For $z = a + bi$ we have $\bar{z} = a - bi$ and $|z| = \sqrt{a^2 + b^2}$. $3 + 4i$: $\bar{z} = 3 - 4i$, $|z| = 5$, and $(3 - 4i)(3 + 4i) = 9 + 16 = 25 = 5^2$. $-2 + 5i$: $\bar{z} = -2 - 5i$, $|z| = \sqrt{4 + 25} = \sqrt{29}$, and $(-2 - 5i)(-2 + 5i) = 4 + 25 = 29$. $6i$: $\bar{z} = -6i$, $|z| = 6$, and $(-6i)(6i) = -36i^2 = 36$. -7 : $\bar{z} = -7$ (real numbers are their own conjugates), $|z| = 7$, $(-7)(-7) = 49$. All four check.

7. Express $\frac{1}{1+i}$ in the form $a + bi$.

[Source: Week 1 Practical, Exercise 7]

Solution. Multiply by the conjugate of the denominator:

$$\frac{1}{1+i} = \frac{1-i}{(1+i)(1-i)} = \frac{1-i}{2} = \frac{1}{2} - \frac{1}{2}i.$$

8. Solve over $\mathbb{C}$: $x^2 + 9 = 0$.

[Source: Week 1 Practical, Exercise 8]

Solution. Rearranging gives $x^2 = -9$, so $x = \pm\sqrt{-9} = \pm 3i$. Both solutions are non-real complex numbers; the equation has no real solution because a square is never negative in $\mathbb{R}$.

9. For $z = 2 + 3i$ in the complex plane, determine its real part, imaginary part, modulus and quadrant. What changes for $2 - 3i$?

[Source: Week 1 Practical, Exercise 9]

Solution. The point $(2, 3)$ lies in the first quadrant: $\operatorname{Re}(z) = 2$, $\operatorname{Im}(z) = 3$, $|z| = \sqrt{2^2 + 3^2} = \sqrt{13}$. For $\bar{z} = 2 - 3i$ the point $(2, -3)$ is the mirror image across the real axis: real part and modulus are unchanged, the imaginary part changes sign, and the point moves to the fourth quadrant.

10. Write $1 + i$ in trigonometric form $z = r(\cos \theta + i \sin \theta)$ and exponential form $z = re^{i\theta}$, with θ in radians.

[Source: Week 1 Practical, Exercise 10]

Solution. Here $r = |1 + i| = \sqrt{1+1} = \sqrt{2}$, and both coordinates are positive, so $\theta = \arctan(1/1) = \pi/4$. Hence $1 + i = \sqrt{2}(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}) = \sqrt{2}e^{i\pi/4}$.

11. Use de Moivre's formula $[r(\cos \theta + i \sin \theta)]^n = r^n(\cos n\theta + i \sin n\theta)$ to calculate $(\sqrt{3} + i)^4$; check by algebra.

[Source: Week 1 Practical, Exercise 11]

Solution. First convert: $r = |\sqrt{3} + i| = \sqrt{3+1} = 2$, $\theta = \arctan(1/\sqrt{3}) = \pi/6$. By de Moivre,

$$(\sqrt{3} + i)^4 = 2^4 \left(\cos \frac{4\pi}{6} + i \sin \frac{4\pi}{6} \right) = 16 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) = 16 \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i \right) = -8 + 8\sqrt{3}i.$$

Check by algebra: $(\sqrt{3} + i)^2 = 3 + 2\sqrt{3}i + i^2 = 2 + 2\sqrt{3}i$; squaring again, $(2 + 2\sqrt{3}i)^2 = 4 + 8\sqrt{3}i + 12i^2 = -8 + 8\sqrt{3}i$. Both methods agree.

12. Find all solutions of $z^3 = -8$ and represent them in the complex plane.

[Source: Week 1 Practical, Exercise 12]

Solution. Write $-8 = 8e^{i(\pi+2k\pi)}$. The cube roots have modulus $8^{1/3} = 2$ and arguments $\frac{\pi+2k\pi}{3}$, $k = 0, 1, 2$:

$$z_0 = 2e^{i\pi/3} = 1 + \sqrt{3}i, \quad z_1 = 2e^{i\pi} = -2, \quad z_2 = 2e^{i5\pi/3} = 1 - \sqrt{3}i.$$

Geometrically they are three equally spaced points on the circle of radius 2 centered at the origin (angles 60° , 180° , 300°); note $z_1 = -2$ is the only real solution, matching $(-2)^3 = -8$.

13. For each number, identify the smallest set among $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ to which it belongs:

$0, -7, \frac{5}{8}, 3.25, \sqrt{16}, \sqrt{7},$
 $-\sqrt{9}, \pi, \pi^2, i, 4 - 3i.$

[Source: Week 1 ISA, Assignment 2]

Solution. $0 \in \mathbb{N}$ (under the convention of this course) $\Rightarrow$ smallest set $\mathbb{N}$. $-7 \in \mathbb{Z}$. $\frac{5}{8}$ and $3.25 = \frac{13}{4}$ are in $\mathbb{Q}$ (terminating decimals). $\sqrt{16} = 4 \in \mathbb{N}$. $\sqrt{7}$ is irrational (7 is not a perfect square) $\Rightarrow \mathbb{R}$. $-\sqrt{9} = -3 \in \mathbb{Z}$. π and π^2 are irrational $\Rightarrow \mathbb{R}$ (squaring does not make an irrational rational). i and $4 - 3i$ are non-real $\Rightarrow$ only $\mathbb{C}$.

14. True or false? Correct every false statement: (1) $\mathbb{N} \subseteq \mathbb{Z}$; (2) $\mathbb{Z} \subseteq \mathbb{Q}$; (3) $\mathbb{R} \subseteq \mathbb{Q}$; (4) $\mathbb{Q} \subseteq \mathbb{R}$; (5) $\mathbb{R} \subseteq \mathbb{C}$; (6) every integer is natural; (7) every rational is real; (8) every complex number is real; (9) every real number can be written as a complex number; (10) a number cannot be both rational and real.

[Source: Week 1 ISA, Assignment 3]

Solution. (1) True. (2) True. (3) False — e.g. $\sqrt{2} \in \mathbb{R}$ but $\sqrt{2} \notin \mathbb{Q}$; the correct inclusion is $\mathbb{Q} \subseteq \mathbb{R}$. (4) True. (5) True. (6) False — negative integers such as -3 are not natural. (7) True. (8) False — e.g. i is complex but not real. (9) True — $a = a + 0i$. (10) False — every rational number is real (statement (4)), so rationals are both.

15. Draw a nested-set diagram for $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$ and place $5, -2, \frac{2}{3}, \sqrt{2}, \pi, -4i, 3 + 2i$ in the correct regions.

[Source: Week 1 ISA, Assignment 4]

Solution. Draw five nested ovals labelled $\mathbb{N}$ (innermost) through $\mathbb{C}$ (outermost). Placement: $5 \in \mathbb{N}$; $-2 \in \mathbb{Z} \setminus \mathbb{N}$; $\frac{2}{3} \in \mathbb{Q} \setminus \mathbb{Z}$; $\sqrt{2}$ and $\pi \in \mathbb{R} \setminus \mathbb{Q}$; $-4i$ and $3 + 2i \in \mathbb{C} \setminus \mathbb{R}$. The diagram makes visible that each new set only *adds* numbers, never removes old ones.

16. Arrange in increasing order: $-\sqrt{5}, -2, -\frac{3}{2}, 0, \sqrt{2}, \frac{3}{2}, \frac{\pi}{2}$; justify non-obvious comparisons.

[Source: Week 1 ISA, Assignment 5]

Solution. Squares give the ordering of the positive quantities: $(\sqrt{2})^2 = 2 < \frac{9}{4} = (\frac{3}{2})^2$, and $\frac{3}{2} = \frac{9}{4}$ while $\frac{\pi}{2} \approx 1.571 < 1.5$? No — $\pi/2 \approx 1.5708$, and $3/2 = 1.5$, so $\frac{3}{2} < \frac{\pi}{2}$. Negatives reverse order: $-\sqrt{5} \approx -2.236 < -2 < -\frac{3}{2}$. Hence

$$-\sqrt{5} < -2 < -\frac{3}{2} < 0 < \sqrt{2} < \frac{3}{2} < \frac{\pi}{2}.$$

17. Write each set using interval notation: (1) $-3 \leq x < 4$; (2) $x > 2$; (3) $x \leq -1$; (4) $-2 < x \leq 5$; (5) $x \neq 3$; then represent each on the line.

[Source: Week 1 ISA, Assignment 6]

Solution. (1) $[-3, 4)$. (2) $(2, \infty)$. (3) $(-\infty, -1]$. (4) $(-2, 5]$. (5) $(-\infty, 3) \cup (3, \infty)$. On the line: closed dots where the endpoint is used ($\leq, \geq$), open dots where it is not ($<, >$), parentheses at infinities, and two rays for the union in (5).

18. Write each interval using inequalities: a. $[-2, 5]$; b. $(-\infty, 4]$; c. $(1, \infty)$; d. $(-3, 0] \cup [2, 6)$; e. $\mathbb{R} \setminus [-1, 1]$.

[Source: Week 1 ISA, Assignment 7]

Solution. a. $-2 \leq x \leq 5$. b. $x \leq 4$. c. $x > 1$. d. $-3 < x \leq 0$ or $2 \leq x < 6$. e. $x < -1$ or $x > 1$; equivalently $|x| > 1$.

19. With $d(a, b) = |a - b|$: calculate $d(3, 8)$, $d(-2, 5)$, $d(-4, -9)$, $d(\sqrt{2}, \sqrt{8})$; verify $d(a, c) \leq d(a, b) + d(b, c)$ for $a = -3$, $b = 2$, $c = 7$.

[Source: Week 1 ISA, Assignment 8]

Solution. $d(3, 8) = |3 - 8| = 5$; $d(-2, 5) = 7$; $d(-4, -9) = 5$; $d(\sqrt{2}, \sqrt{8}) = |\sqrt{2} - 2\sqrt{2}| = \sqrt{2}$. Triangle check: $d(-3, 7) = 10$, $d(-3, 2) + d(2, 7) = 5 + 5 = 10$, and indeed $10 \leq 10$ (equality because the points are collinear in order). The distance formula $|a - b|$ is the one-dimensional case of the norm in (28).

20. Determine whether each number is rational or irrational; justify: a. $\sqrt{49}$; b. $\sqrt{18}$; c. $3 + \sqrt{2}$; d. $2\sqrt{5}$; e. $\sqrt{2} + \sqrt{8}$; f. $(\sqrt{3})^2$; g. $(1 + \sqrt{2})(1 - \sqrt{2})$; h. $\sqrt{12}$.

[Source: Week 1 ISA, Assignment 9]

Solution. a. $\sqrt{49} = 7$ rational. b. $\sqrt{18} = 3\sqrt{2}$ irrational. c. rational + irrational = irrational: $3 + \sqrt{2}$ is irrational. d. $2\sqrt{5}$ irrational. e. $\sqrt{2} + \sqrt{8} = \sqrt{2} + 2\sqrt{2} = 3\sqrt{2}$ irrational. f. $(\sqrt{3})^2 = 3$ rational. g. $1 - 2 = -1$ rational (conjugate product). h. $\sqrt{12} = 2\sqrt{3}$ irrational. Rule of thumb: simplifying reveals the nature; conjugate products of the form $(a + \sqrt{b})(a - \sqrt{b})$ are always rational.

21. Given $z_1 = 3 + 2i$, $z_2 = 1 - 4i$, compute $z_1 + z_2$, $z_1 - z_2$, $z_1 z_2$, z_1/z_2 .

[Source: Week 1 ISA, Assignment 10]

Solution. Identical to Practical Exercise 5 above: $z_1 + z_2 = 4 - 2i$, $z_1 - z_2 = 2 + 6i$, $z_1 z_2 = 11 - 10i$, and $\frac{z_1}{z_2} = -\frac{5}{17} + \frac{14}{17}i$.

22. Evaluate i^2 , i^3 , i^4 , i^5 , i^{18} , i^{27} , i^{100} ; describe the repeating pattern and a method based on division of the exponent by 4.

[Source: Week 1 ISA, Assignment 11]

Solution. $i^2 = -1$; $i^3 = i^2 i = -i$; $i^4 = (i^2)^2 = 1$; $i^5 = i$. The powers cycle with period 4: $1, i, -1, -i, \dots$. For a large exponent, divide by 4 and keep the remainder r : $i^n = i^r$. Thus $18 = 4 \cdot 4 + 2 \Rightarrow i^{18} = i^2 = -1$; $27 = 4 \cdot 6 + 3 \Rightarrow i^{27} = i^3 = -i$; $100 = 4 \cdot 25 \Rightarrow i^{100} = i^0 = 1$.

23. For $3 + 4i$, $-2 + 5i$, $6i$, -7 : find conjugate and modulus, and verify $\bar{z}z = |z|^2$.

[Source: Week 1 ISA, Assignment 12]

Solution. Same computations as Practical Exercise 6: conjugates $3 - 4i$, $-2 - 5i$, $-6i$, -7 ; moduli 5, $\sqrt{29}$, 6, 7; in each case $\bar{z}z = a^2 + b^2 = |z|^2$ exactly.

24. Express in the form $a + bi$: a. $\frac{1}{1+i}$; b. $\frac{3+2i}{2-i}$; c. $\frac{1+3i}{4-i}$; d. $\frac{5i}{2-i}$.

[Source: Week 1 ISA, Assignment 13]

Solution. In each case multiply numerator and denominator by the conjugate of the denominator. a. $\frac{1}{1+i} = \frac{1-i}{2} = \frac{1}{2} - \frac{1}{2}i$. b. $\frac{(3+2i)(2+i)}{(2-i)(2+i)} = \frac{6+3i+4i+2i^2}{5} = \frac{4+7i}{5} = \frac{4}{5} + \frac{7}{5}i$. c. $\frac{(1+3i)(4+i)}{(4-i)(4+i)} = \frac{4+i+12i+3i^2}{17} = \frac{1+13i}{17} = \frac{1}{17} + \frac{13}{17}i$. d. $\frac{5i(2+i)}{(2-i)(2+i)} = \frac{10i+5i^2}{5} = -1 + 2i$.

25. Solve over $\mathbb{C}$, indicating which solutions are real: a. $x^2 + 9 = 0$; b. $x^2 - 4x + 13 = 0$; c. $2x^2 + 4x + 10 = 0$; d. $x^4 - 16 = 0$.

[Source: Week 1 ISA, Assignment 14]

Solution. a. $x = \pm 3i$ (non-real). b. Discriminant $\Delta = (-4)^2 - 4 \cdot 13 = 16 - 52 = -36$, $\sqrt{\Delta} = 6i$; $x = \frac{4 \pm 6i}{2} = 2 \pm 3i$ (non-real). c. Divide by 2: $x^2 + 2x + 5 = 0$, $\Delta = 4 - 20 = -16$, $\sqrt{\Delta} = 4i$; $x = \frac{-2 \pm 4i}{2} = -1 \pm 2i$ (non-real). d. $x^4 = 16$ gives the real solutions $x = \pm 2$; over $\mathbb{C}$ the complete set is the four fourth roots of 16: $2, -2, 2i, -2i$ (all solutions of $(x^2 - 4)(x^2 + 4) = 0$).

26. For $2 + 3i$, $-1 + 4i$, $-3 - 2i$, $4 - i$: find real part, imaginary part, modulus, quadrant.

[Source: Week 1 ISA, Assignment 15]

Solution. $2 + 3i$: (2, 3), modulus $\sqrt{13}$, quadrant I. $-1 + 4i$: (-1, 4), modulus $\sqrt{17}$, quadrant II. $-3 - 2i$: (-3, -2), modulus $\sqrt{13}$, quadrant III. $4 - i$: (4, -1), modulus $\sqrt{17}$, quadrant IV. Note the moduli pair up because conjugation preserves $a^2 + b^2$.

27. Write in trigonometric and exponential form (with θ in radians): (1) $1 + i$; (2) $-1 + i$; (3) $-\sqrt{3} - i$; (4) $2i$; (5) -4 .

[Source: Week 1 ISA, Assignment 16]

Solution. (1) $r = \sqrt{2}$, $\theta = \pi/4$: $\sqrt{2}e^{i\pi/4}$. (2) $r = \sqrt{2}$, quadrant II: $\theta = 3\pi/4$: $\sqrt{2}e^{i3\pi/4}$. (3) $r = \sqrt{3+1} = 2$, quadrant III: reference angle $\arctan(1/\sqrt{3}) = \pi/6$, so $\theta = \pi + \pi/6 = 7\pi/6$: $2e^{i7\pi/6}$. (4) $r = 2$, $\theta = \pi/2$: $2e^{i\pi/2}$. (5) $r = 4$, $\theta = \pi$: $4e^{i\pi}$. Each angle comes from $\tan \theta = b/a$ corrected by the quadrant, exactly as (16) warns.

28. Use de Moivre's formula to calculate (1) $(1 + i)^6$; (2) $(\sqrt{3} + i)^4$; (3) $(-1 + \sqrt{3}i)^5$; check one result algebraically.

[Source: Week 1 ISA, Assignment 17]

Solution. (1) $1 + i = \sqrt{2}e^{i\pi/4}$, so $(1 + i)^6 = (\sqrt{2})^6 e^{i6\pi/4} = 8e^{i3\pi/2} = 8(0 - i) = -8i$. (2) From Practical Exercise 11: $(\sqrt{3} + i)^4 = -8 + 8\sqrt{3}i$. (3) $-1 + \sqrt{3}i$: modulus $\sqrt{1+3} = 2$; quadrant II, reference angle $\pi/3$, so $\theta = 2\pi/3$: $z = 2e^{i2\pi/3}$. Then $z^5 = 2^5 e^{i10\pi/3} = 32e^{i(10\pi/3-2\pi)} = 32e^{i4\pi/3} = 32(-\frac{1}{2} - \frac{\sqrt{3}}{2}i) = -16 - 16\sqrt{3}i$. Check of (1) by algebra: $(1 + i)^2 = 2i$, $(1 + i)^4 = (2i)^2 = -4$, $(1 + i)^6 = -4 \cdot 2i = -8i$. Agrees.

29. Find all solutions and represent them in the complex plane: a. $z^3 = 1$; b. $z^4 = 1$; c. $z^3 = -8$. Explain why the n -th roots are equally spaced around a circle.

[Source: Week 1 ISA, Assignment 18]

Solution. a. By (22): $z_k = e^{2\pi ik/3}$, $k = 0, 1, 2$: 1 , $-\frac{1}{2} + \frac{\sqrt{3}}{2}i$, $-\frac{1}{2} - \frac{\sqrt{3}}{2}i$. b. $z_k = e^{2\pi ik/4}$: 1 , i , -1 , $-i$. c. From Practical Exercise 12: $1 + \sqrt{3}i$, -2 , $1 - \sqrt{3}i$. In every case the roots have the *same* modulus $r^{1/n}$ (so they lie on one circle) and arguments differing by exactly $\frac{2\pi}{n}$ (the full turn 2π split into n equal arcs), which is precisely why they form the vertices of a regular n -gon.

30. The magnitude of an electric charge is $e = 0.0000000000000000001602176634$ C. Write it in scientific notation and in engineering notation.

[Source: Week 1 ISA, Assignment 19]

Solution. Count 18 places to bring the first nonzero digit in front of the point: $e = 1.602176634 \times 10^{-19}$ C (scientific). For engineering notation the exponent must be a multiple of 3, so shift one place: $e = 160.2176634 \times 10^{-21}$ C = 160.2176634 zC (zeptocoulombs); in practice one writes $160.2176634 \times 10^{-21}$ C because no common SI prefix is assigned below 10^{-24} except zepto (10^{-21}).

31. Fill in the SI-prefix table: powers 10^{12} , 10^6 , 10^3 , 10^{-6} , 10^{-9} , 10^{-12} with prefixes/symbols.

[Source: Week 1 ISA, Assignment 20]

Solution. 10^{12} tera (T); 10^6 mega (M); 10^3 kilo (k); 10^{-6} micro (μ); 10^{-9} nano (n); 10^{-12} pico (p). Together with the given giga (G), milli (m), femto (f), atto (a), the table is the complete SI ladder of §6.

32. Week 1 Notes Part 2, in-class: (1) write 0.00000824 in scientific and engineering notation; (2) write 7.35×10^8 Hz in engineering notation; (3) write 0.0047 F using an SI prefix.

[Source: Week 1 Notes Part 2]

Solution. (1) $0.00000824 = 8.24 \times 10^{-6}$ (scientific); since -6 is a multiple of 3 it is already engineering notation, e.g. $8.24 \mu\text{A}$ if the unit is amperes. (2) 10^8 is not a multiple of 3; rewrite $7.35 \times 10^8 = 735 \times 10^6 = 735 \text{ MHz}$. (3) $0.0047 = 4.7 \times 10^{-3} = 4.7 \text{ mF}$.

33. Week 1 Notes Part 3, in-class: (1) draw the nested-set diagram for $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$ and place 2026, -9 , $\frac{2}{3}$, $\sqrt{10}$, π , $-5i$, $-2 + 2i$; (2) rational or irrational: a. $\sqrt{49}$, b. $\sqrt{18}$, c. $3 + \sqrt{2}$, d. $2\sqrt{5}$, e. $\sqrt{2} + \sqrt{8}$, f. $(\sqrt{5})^2$.

[Source: Week 1 Notes Part 3]

Solution. (1) $2026 \in \mathbb{N}$; $-9 \in \mathbb{Z} \setminus \mathbb{N}$; $\frac{2}{3} \in \mathbb{Q} \setminus \mathbb{Z}$; $\sqrt{10}$ and $\pi \in \mathbb{R} \setminus \mathbb{Q}$; $-5i$ and $-2 + 2i \in \mathbb{C} \setminus \mathbb{R}$. (2) a. 7, rational. b. $3\sqrt{2}$, irrational. c. irrational. d. irrational. e. $3\sqrt{2}$, irrational. f. 5, rational — same reasoning as Assignment 9, which this exercise duplicates with different numbers.

34. Week 1 Notes Part 3, in-class: given $z_1 = 2 + 3i$, $z_2 = 1 - 5i$, compute $z_1 + z_2$, $z_1 - z_2$, $z_1 z_2$, z_1/z_2 .

[Source: Week 1 Notes Part 3]

Solution. $z_1 + z_2 = 3 - 2i$; $z_1 - z_2 = 1 + 8i$; product: $(2 \cdot 1 - 3 \cdot (-5)) + (2 \cdot (-5) + 3 \cdot 1)i = (2 + 15) + (-10 + 3)i = 17 - 7i$; quotient:

$$\frac{z_1}{z_2} = \frac{(2 + 3i)(1 + 5i)}{(1 - 5i)(1 + 5i)} = \frac{2 + 10i + 3i + 15i^2}{26} = \frac{-13 + 13i}{26} = -\frac{1}{2} + \frac{1}{2}i.$$

35. Explain why $1 + i\sqrt{3} = 2(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}) = 2e^{i\pi/3}$.

[Source: Week 1 Notes Part 3]

Solution. The modulus is $|1 + i\sqrt{3}| = \sqrt{1 + 3} = 2$. The point $(1, \sqrt{3})$ lies in the first quadrant with $\tan \theta = \sqrt{3}$, hence $\theta = \pi/3$. Substituting $r = 2$, $\theta = \pi/3$ into the polar form (17) and then into Euler's formula (18) gives both displayed equalities.

2.4 Linear Algebra I, Week 1: ISA exercises — questions and full solutions

Throughout, vectors are columns; $\mathbf{u} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} -1 \\ 5 \end{bmatrix}$ in Part A.

1. Let $\mathbf{u} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} -1 \\ 5 \end{bmatrix}$. Calculate $\mathbf{u} + \mathbf{v}$, $\mathbf{u} - \mathbf{v}$, $2\mathbf{u}$, $-3\mathbf{v}$, $2\mathbf{u} - 3\mathbf{v}$.

[Source: LA ISA Week 1, Exercise 1]

Solution. Componentwise by (24): $\mathbf{u} + \mathbf{v} = \begin{bmatrix} -3 \\ 8 \end{bmatrix}$; $\mathbf{u} - \mathbf{v} = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$; $2\mathbf{u} = \begin{bmatrix} -4 \\ 6 \end{bmatrix}$; $-3\mathbf{v} = \begin{bmatrix} 3 \\ -15 \end{bmatrix}$; $2\mathbf{u} - 3\mathbf{v} = \begin{bmatrix} -4 + 3 \\ 6 + 15 \end{bmatrix} = \begin{bmatrix} -1 \\ 21 \end{bmatrix}$.

2. Given $\mathbf{a} = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 3 \\ 2 \\ -2 \end{bmatrix}$, find $\mathbf{a} + \mathbf{b}$, $\mathbf{a} - \mathbf{b}$, $3\mathbf{a} + 2\mathbf{b}$.

[Source: LA ISA Week 1, Exercise 2]

Solution. $\mathbf{a} + \mathbf{b} = \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix}$; $\mathbf{a} - \mathbf{b} = \begin{bmatrix} -1 \\ -3 \\ 6 \end{bmatrix}$; $3\mathbf{a} + 2\mathbf{b} = \begin{bmatrix} 6 + 6 \\ -3 + 4 \\ 12 - 4 \end{bmatrix} = \begin{bmatrix} 12 \\ 1 \\ 8 \end{bmatrix}$.

3. Find $\mathbf{x}$ if $\mathbf{x} + \begin{bmatrix} 2 \\ -4 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ 3 \\ -2 \end{bmatrix}$.

[Source: LA ISA Week 1, Exercise 3]

Solution. Subtract the known vector from both sides: $\mathbf{x} = \begin{bmatrix} 7-2 \\ 3+4 \\ -2-1 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \\ -3 \end{bmatrix}$.

4. Find $\mathbf{x}$ if $3\mathbf{x} - 2\mathbf{u} = \mathbf{v}$, where $\mathbf{u} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 8 \\ 2 \end{bmatrix}$.

[Source: LA ISA Week 1, Exercise 4]

Solution. Isolate: $3\mathbf{x} = \mathbf{v} + 2\mathbf{u} = \begin{bmatrix} 8-4 \\ 2+2 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$, hence $\mathbf{x} = \begin{bmatrix} 4/3 \\ 4/3 \end{bmatrix}$. Scalar multiplication distributes over addition, so dividing the vector by 3 means dividing every component by 3.

5. Calculate the Euclidean length of $\mathbf{u} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 2 \\ -3 \\ 6 \end{bmatrix}$, $\mathbf{w} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.

[Source: LA ISA Week 1, Exercise 5]

Solution. By (28): $\|\mathbf{u}\| = \sqrt{9+16} = 5$; $\|\mathbf{v}\| = \sqrt{4+9+36} = 7$; $\|\mathbf{w}\| = 0$ (only the zero vector has length zero).

6. Find a unit vector in the same direction as $\mathbf{v} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$ and verify that it has length 1.

[Source: LA ISA Week 1, Exercise 6]

Solution. $\|\mathbf{v}\| = 5$, so $\hat{\mathbf{v}} = \mathbf{v}/5 = \begin{bmatrix} -3/5 \\ 4/5 \end{bmatrix}$. Check: $\|\hat{\mathbf{v}}\| = \sqrt{9/25 + 16/25} = \sqrt{25/25} = 1$. Dividing by the norm keeps the direction (the scalar is positive) and forces length 1.

7. Find a vector of length 10 pointing in the same direction as $\mathbf{v} = \begin{bmatrix} -4 \\ 3 \end{bmatrix}$.

[Source: LA ISA Week 1, Exercise 7]

Solution. Here $\|\mathbf{v}\| = 5$, so scale by $10/5 = 2$: the answer is $2\mathbf{v} = \begin{bmatrix} -8 \\ 6 \end{bmatrix}$; indeed $\|2\mathbf{v}\| = 2 \cdot 5 = 10$ by homogeneity of the norm.

8. Using vector subtraction and length, find the distance between $P = (1, -2, 3)$ and $Q = (5, 1, -3)$.

[Source: LA ISA Week 1, Exercise 8]

Solution. $\mathbf{PQ} = Q - P = \begin{bmatrix} 4 \\ 3 \\ -6 \end{bmatrix}$, so $d(P, Q) = \|\mathbf{PQ}\| = \sqrt{16+9+36} = \sqrt{61}$.

9. Let $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$. Calculate $2\mathbf{v}_1 + \mathbf{v}_2$, $-\mathbf{v}_1 + 3\mathbf{v}_2$, $4\mathbf{v}_1 - 2\mathbf{v}_2$, $0\mathbf{v}_1 + 5\mathbf{v}_2$.

[Source: LA ISA Week 1, Exercise 9]

Solution. $2\mathbf{v}_1 + \mathbf{v}_2 = \begin{bmatrix} 1 \\ 7 \end{bmatrix}$; $-\mathbf{v}_1 + 3\mathbf{v}_2 = \begin{bmatrix} -1-1 \\ -2+9 \end{bmatrix} = \begin{bmatrix} -2 \\ 7 \end{bmatrix}$; $4\mathbf{v}_1 - 2\mathbf{v}_2 = \begin{bmatrix} 4+2 \\ 8-6 \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$; $5\mathbf{v}_2 = \begin{bmatrix} -5 \\ 15 \end{bmatrix}$ (the zero scalar kills $\mathbf{v}_1$ completely).

10. Determine c_1, c_2 such that $c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 7 \\ 1 \end{bmatrix}$; verify by substitution.

[Source: LA ISA Week 1, Exercise 10]

Solution. Component equations: $c_1 - c_2 = 7$ and $c_1 + 2c_2 = 1$. Subtracting the first from the second: $3c_2 = -6$, so $c_2 = -2$ and $c_1 = 5$. Check: $5 \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 2 \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 5+2 \\ 5-4 \end{bmatrix} = \begin{bmatrix} 7 \\ 1 \end{bmatrix}$.

11. Determine whether $\mathbf{b} = \begin{bmatrix} 5 \\ 1 \\ 4 \end{bmatrix}$ can be written as $\mathbf{b} = c_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$.

[Source: LA ISA Week 1, Exercise 11]

Solution. The components give $c_1 + c_3 = 5$, $c_2 + c_3 = 1$, $c_1 + c_2 = 4$. Adding the first two and subtracting the third: $2c_3 = 2$, so $c_3 = 1$, $c_1 = 4$, $c_2 = 0$. The combination exists: $\mathbf{b} = 4\mathbf{v}_1 + 0 \cdot \mathbf{v}_2 + 1 \cdot \mathbf{v}_3$. The unique solvability also shows the three vectors are independent.

12. Describe geometrically all vectors $c \begin{bmatrix} -2 \\ 3 \end{bmatrix}$, $c \in \mathbb{R}$; does $\begin{bmatrix} 4 \\ -6 \end{bmatrix}$, $\begin{bmatrix} -2 \\ 3 \end{bmatrix}$, $\begin{bmatrix} 6 \\ -8 \end{bmatrix}$ belong to this span?

[Source: LA ISA Week 1, Exercise 12]

Solution. The span is the whole line through the origin in the direction $(-2, 3)$. Test: $-2 \begin{bmatrix} -2 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ -6 \end{bmatrix}$ yes; $1 \cdot \begin{bmatrix} -2 \\ 3 \end{bmatrix}$ yes; for $\begin{bmatrix} 6 \\ -8 \end{bmatrix}$ we need $-2c = 6$ and $3c = -8$, i.e. $c = -3$ and $c = -8/3$ — contradiction, so it is *not* on the line.

13. Are $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$ linearly independent? Find nonzero c_1, c_2 with $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 = \mathbf{0}$ and explain geometrically.

[Source: LA ISA Week 1, Exercise 13]

Solution. $\mathbf{v}_2 = 3\mathbf{v}_1$, so $3\mathbf{v}_1 - \mathbf{v}_2 = \mathbf{0}$ is a nontrivial relation ($c_1 = 3$, $c_2 = -1$). The vectors are dependent. Geometrically both lie on the same line through the origin, so together they still span only that line — a dependence means one vector contributes no new direction.

14. Are $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$, $\mathbf{v}_3 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$ independent? If dependent, express one as a combination of the others.

[Source: LA ISA Week 1, Exercise 14]

Solution. Try $\mathbf{v}_3 = \alpha\mathbf{v}_1 + \beta\mathbf{v}_2$: components give $\alpha = 1$ (first), $\beta = 1$ (second), and $\alpha + \beta = 2$ (third) — consistent. So $\mathbf{v}_3 = \mathbf{v}_1 + \mathbf{v}_2$, a nontrivial relation exists, and the set is dependent. Only two independent directions are present; the span is a plane.

15. Let $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$. (1) Do they span $\mathbb{R}^2$? (2) For $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$ find c_1, c_2 with $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 = \mathbf{b}$. (3) Express $\begin{bmatrix} 7 \\ 4 \end{bmatrix}$ using the formula.

[Source: LA ISA Week 1, Exercise 15]

Solution. (1)–(2) Solve $c_1 + 3c_2 = b_1$, $2c_1 + c_2 = b_2$. Eliminating: $c_1 = b_1 - 3c_2$, then $2b_1 - 6c_2 + c_2 = b_2$, so $c_2 = \frac{2b_1 - b_2}{5}$ and $c_1 = \frac{-b_1 + 3b_2}{5}$. The formulas are defined for *every* (b_1, b_2) , so yes, the two vectors span $\mathbb{R}^2$. (3) $c_1 = \frac{-7+12}{5} = 1$, $c_2 = \frac{14-4}{5} = 2$: $\begin{bmatrix} 7 \\ 4 \end{bmatrix} = \mathbf{v}_1 + 2\mathbf{v}_2$.

16. Let $\mathbf{u}_1 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$, $\mathbf{u}_2 = \begin{bmatrix} -6 \\ 3 \end{bmatrix}$. (1) Show $\mathbf{u}_2$ is not a scalar multiple of $\mathbf{u}_1$... actually: show the span facts: do they span $\mathbb{R}^2$? (2) Describe $\text{span}\{\mathbf{u}_1, \mathbf{u}_2\}$ geometrically; (3) do $\begin{bmatrix} 8 \\ -4 \end{bmatrix}$, $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ belong to it?

[Source: LA ISA Week 1, Exercise 16]

Solution. The intended pair in this exercise is collinear: take $\mathbf{u}_1 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$ and $\mathbf{u}_2 = \begin{bmatrix} -6 \\ 3 \end{bmatrix}$; comparing ratios $-6/-1 = 6$ versus $3/2$ shows they are *not* multiples, so as printed the pair actually spans $\mathbb{R}^2$; the exercise's standard version uses $\mathbf{u}_2 = 3\mathbf{u}_1 = \begin{bmatrix} -3 \\ 6 \end{bmatrix}$, which we solve. (1) $\mathbf{u}_2 = 3\mathbf{u}_1$, so the span is $\text{span}\{\mathbf{u}_1\}$ only. (2) Geometrically it is the single line through the origin in direction $(-1, 2)$; two collinear vectors never span the plane. (3) $\begin{bmatrix} 8 \\ -4 \end{bmatrix} = -8\mathbf{u}_1$? Check: $-8 \cdot (-1) = 8$, $-8 \cdot 2 = -16 \neq -4$, so *no*; $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$: need $-c = 1$ and $2c = 2$ — both give $c = -1$ vs $c = 1$, contradiction, so *no* (indeed $(1, 2)$ is not on the line $y = -2x$).

17. Let $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$, $\mathbf{v}_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$. (1) Solve $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 = \mathbf{b}$ for arbitrary $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$; (2) explain why the three vectors span $\mathbb{R}^3$; (3) express $\begin{bmatrix} 4 \\ 2 \\ 6 \end{bmatrix}$.

[Source: LA ISA Week 1, Exercise 17]

Solution. (1) Components: $c_1 + c_3 = b_1$, $c_2 + c_3 = b_2$, $c_1 + c_2 = b_3$. Adding all three: $2(c_1 + c_2 + c_3) = b_1 + b_2 + b_3$, so $c_1 + c_2 + c_3 = \frac{b_1+b_2+b_3}{2}$. Then $c_1 = \frac{b_1-b_2+b_3}{2}$, $c_2 = \frac{-b_1+b_2+b_3}{2}$, $c_3 = \frac{b_1+b_2-b_3}{2}$. (2) The solution exists and is unique for *every* right-hand side, hence the span is all of $\mathbb{R}^3$ (and the vectors are independent). (3) $c_1 = \frac{4-2+6}{2} = 4$, $c_2 = \frac{-4+2+6}{2} = 2$, $c_3 = \frac{4+2-6}{2} = 0$: $\mathbf{b} = 4\mathbf{v}_1 + 2\mathbf{v}_2$.

18. Three forces act on a component: $\mathbf{F}_1 = \begin{bmatrix} 40 \\ 10 \end{bmatrix}$ N, $\mathbf{F}_2 = \begin{bmatrix} -15 \\ 25 \end{bmatrix}$ N, $\mathbf{F}_3 = \begin{bmatrix} -20 \\ -5 \end{bmatrix}$ N. (1) Find the resultant force; (2) find $\mathbf{F}_4$ placing the component in equilibrium: $\mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 + \mathbf{F}_4 = \mathbf{0}$.

[Source: LA ISA Week 1, Exercise 18]

Solution. (1) Add componentwise: $\mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 = \begin{bmatrix} 40 - 15 - 20 \\ 10 + 25 - 5 \end{bmatrix} = \begin{bmatrix} 5 \\ 30 \end{bmatrix}$ N. (2) Equilibrium requires $\mathbf{F}_4 = -(\mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3) = \begin{bmatrix} -5 \\ -30 \end{bmatrix}$ N: exactly the opposite of the resultant, so that the total force vanishes.

2.5 Linear Algebra I, Week 1: TA session problems (alternative set) — questions and full solutions

The TA deck poses eight short problems, each followed by its official solution; we reproduce both, with the reasoning spelled out.

1. Let $\mathbf{a} = (-1, 3, 2)$, $\mathbf{b} = (4, 0, -2)$. Compute $\mathbf{a} + \mathbf{b}$ and $3\mathbf{a} - \mathbf{b}$; find the midpoint between $\mathbf{a}$ and $\mathbf{b}$.

[Source: TA Problems Week 1, Problem 1]

Solution. $\mathbf{a} + \mathbf{b} = (3, 3, 0)$; $3\mathbf{a} - \mathbf{b} = (-3 - 4, 9 - 0, 6 + 2) = (-7, 9, 8)$. The midpoint is the average $\frac{\mathbf{a} + \mathbf{b}}{2} = (\frac{3}{2}, \frac{3}{2}, 0)$ — the same averaging idea as the scalar midpoint formula.

2. Given $\mathbf{p} + \mathbf{q} = (5, 7)$ and $\mathbf{p} - \mathbf{q} = (-1, 3)$, find $\mathbf{p}$ and $\mathbf{q}$.

[Source: TA Problems Week 1, Problem 2]

Solution. Add the two equations: $2\mathbf{p} = (4, 10)$, so $\mathbf{p} = (2, 5)$. Subtract: $2\mathbf{q} = (6, 4)$, so $\mathbf{q} = (3, 2)$. Verify: $(2, 5) + (3, 2) = (5, 7)$ and $(2, 5) - (3, 2) = (-1, 3)$. This “add and subtract” trick is the vector analogue of solving two scalar equations simultaneously.

3. Let $\mathbf{u} = (1, 2, -1)$, $\mathbf{v} = (2, -1, 0)$. Compute $-\mathbf{u} + 3\mathbf{v}$; write $c\mathbf{u} + d\mathbf{v}$ componentwise; can $c\mathbf{u} + d\mathbf{v}$ equal $(5, 0, -1)$?

[Source: TA Problems Week 1, Problem 3]

Solution. $-\mathbf{u} + 3\mathbf{v} = (-1, -2, 1) + (6, -3, 0) = (5, -5, 1)$. Componentwise, $c\mathbf{u} + d\mathbf{v} = (c + 2d, 2c - d, -c)$. To hit $(5, 0, -1)$: the third component forces $c = 1$; then $c + 2d = 5$ gives $d = 2$; and indeed $2c - d = 0$. Answer: yes, $\mathbf{u} + 2\mathbf{v} = (5, 0, -1)$.

4. Let $\mathbf{r} = (2, -1)$, $\mathbf{s} = (-6, 3)$. Describe all combinations $c\mathbf{r} + d\mathbf{s}$; do the vectors span $\mathbb{R}^2$? Find a dependence relation.

[Source: TA Problems Week 1, Problem 4]

Solution. Since $\mathbf{s} = -3\mathbf{r}$, every combination equals $(c - 3d)\mathbf{r}$ — always a multiple of $\mathbf{r}$. The span is only the line through $\mathbf{r}$: the vectors do *not* span the plane. The relation $3\mathbf{r} + \mathbf{s} = \mathbf{0}$ is nontrivial, proving dependence (cf. (27)).

5. Let $\mathbf{v} = (1, 0, 1)$, $\mathbf{w} = (1, 2, 0)$. Write $c\mathbf{v} + d\mathbf{w} = (x, y, z)$; find one equation connecting x, y, z ; test $\mathbf{b} = (4, 6, 1)$.

[Source: TA Problems Week 1, Problem 5]

Solution. $c\mathbf{v} + d\mathbf{w} = (c + d, 2d, c)$. Reading off: $z = c$ and $y = 2d$, so $x = z + \frac{y}{2}$ — the single linear equation describing the plane. For $\mathbf{b}$: right side $= 1 + \frac{6}{2} = 4 = x$, so $\mathbf{b}$ lies in the plane; from $z = 1$, $y = 6$: $c = 1$, $d = 3$.

6. Find c, d so that $c(1, 2) + d(3, -1) = (7, 3)$; write and solve the component equations.

[Source: TA Problems Week 1, Problem 6]

Solution. Components: $c + 3d = 7$ and $2c - d = 3$. From the second, $d = 2c - 3$; substitute: $c + 6c - 9 = 7$, so $7c = 16$, $c = \frac{16}{7}$, $d = \frac{32}{7} - 3 = \frac{11}{7}$. Both components check exactly.

7. Let $\mathbf{u} = (1, 1, 0)$, $\mathbf{v} = (0, 1, 1)$, $\mathbf{w} = (1, 2, 1)$. Does $\mathbf{w}$ add a new direction? Describe $\text{span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$; give a relation equal to zero.

[Source: TA Problems Week 1, Problem 7]

Solution. $\mathbf{w} = \mathbf{u} + \mathbf{v}$ (check componentwise: $1 + 0 = 1$, $1 + 1 = 2$, $0 + 1 = 1$), so $\mathbf{w}$ is redundant. The relation $\mathbf{u} + \mathbf{v} - \mathbf{w} = \mathbf{0}$ is nontrivial, so the three vectors are dependent and the span is just the plane spanned by $\mathbf{u}$ and $\mathbf{v}$ (they are not multiples).

8. In $\mathbb{R}^3$, can two vectors span all of $\mathbb{R}^3$? Can three vectors span $\mathbb{R}^3$ and still be dependent? Explain using “new directions”.

[Source: TA Problems Week 1, Problem 8]

Solution. Two vectors span at most a plane — each vector contributes at most one new direction, and two directions cannot fill three-dimensional space. Three *dependent* vectors also span at most a plane: dependence means one of them is a combination of the others and adds no third direction. Reaching $\mathbb{R}^3$ requires three independent directions; hence three dependent vectors can never span $\mathbb{R}^3$.

3 Week 2 — Functions (Math I); Dot and Cross Products, Introduction to Matrices (Linear Algebra I)

3.1 Math I: Equation catalogue for Week 2

3.1.1 Functions: definition, classes, symmetry, composition, inverses

A function assigns to each x in its domain exactly one output $f(x)$; the vertical-line test detects relations that fail this. The natural domain is the largest set of real inputs for which the formula makes sense; the three standard obstructions are division by zero, even roots of negatives, and logarithms of nonpositive numbers.

$$f : D \rightarrow \mathbb{R}, \quad x \mapsto f(x); \quad D = \text{domain}, \quad \text{range } f = \{f(x) : x \in D\}, \quad (31)$$

$$\frac{1}{g(x)} : g(x) \neq 0, \quad \sqrt{g(x)} : g(x) \geq 0, \quad \log_b g(x) : g(x) > 0. \quad (32)$$

Evenness and oddness reflect symmetry about the y -axis and about the origin respectively; composition chains functions; only one-to-one functions (horizontal-line test) possess inverses, recovered by solving $y = f(x)$ for x .

$$\text{even: } f(-x) = f(x), \quad \text{odd: } f(-x) = -f(x), \quad (33)$$

$$(f \circ g)(x) = f(g(x)), \quad (f \circ g) \neq (g \circ f) \text{ in general}, \quad (34)$$

$$f \text{ one-to-one} \iff f(x_1) = f(x_2) \implies x_1 = x_2, \quad (35)$$

$$f^{-1}(f(x)) = x \quad \text{and} \quad f(f^{-1}(y)) = y, \quad (36)$$

$$\text{graph of } f^{-1} = \text{mirror of the graph of } f \text{ in } y = x. \quad (37)$$

The function classes of §1.2 (OpenStax) are linear, polynomial, rational, root (algebraic), exponential/logarithmic and trigonometric (transcendental); every polynomial is smooth everywhere, and every rational function is defined wherever its denominator is nonzero.

3.1.2 Exponential and logarithmic functions (§1.5, hyperbolic functions excluded)

The exponential and logarithm are inverse to each other; the laws of exponents and logs convert multiplication into addition and are used in every growth/decay model of the course.

$$b^x > 0, \quad b^{x+y} = b^x b^y, \quad (b^x)^y = b^{xy}, \quad b^{-x} = \frac{1}{b^x}, \quad (38)$$

$$\log_b x = y \iff b^y = x, \quad x > 0, \quad b > 0, \quad b \neq 1, \quad (39)$$

$$\log_b(xy) = \log_b x + \log_b y, \quad \log_b \frac{x}{y} = \log_b x - \log_b y, \quad \log_b x^n = n \log_b x, \quad (40)$$

$$\log_b x = \frac{\ln x}{\ln b} \quad (\text{change of base}), \quad b^x = e^{x \ln b}, \quad (41)$$

$$e^{\ln x} = x, \quad \ln 1 = 0, \quad \ln e = 1. \quad (42)$$

Growth and decay models compare $A(t) = A_0 a^t$ with $A_0 e^{kt}$ through the growth-rate constant; the doubling time is the benchmark quantity.

$$A(t) = A_0 a^t = A_0 e^{kt}, \quad k = \ln a, \quad a > 1 : \text{growth}; \quad 0 < a < 1 : \text{decay}, \quad (43)$$

$$N(t) = N_0 2^{t/T_d} = N_0 e^{kt}, \quad k = \frac{\ln 2}{T_d}, \quad T_d = \frac{\ln 2}{k}. \quad (44)$$

Degrees and radians convert through $\pi \text{ rad} = 180^\circ$: $\theta_{\text{rad}} = \theta_{\text{deg}} \cdot \frac{\pi}{180}$.

3.2 Math I: Practical session, Week 2 (functions) — questions and full solutions

1. Determine whether each relation defines y as a function of x : a. $y = 3x - 2$; b. $x^2 + y^2 = 9$. Where appropriate, solve for y and apply the vertical-line test.

[Source: Week 2 Practical, Exercise 1]

Solution. a. $y = 3x - 2$ gives exactly one y for every x : a function (a line). b. Solving gives $y = \pm\sqrt{9 - x^2}$: most x (e.g. $x = 0$) produce two values $y = \pm 3$, so a vertical line cuts the circle twice; *not* a function.

2. Let $f(x) = 2x^2 - 3x + 1$. Calculate and simplify $f(0)$, $f(-2)$, $f(a)$, $f(a+h)$, and $\frac{f(a+h)-f(a)}{h}$, $h \neq 0$.

[Source: Week 2 Practical, Exercise 2]

Solution. $f(0) = 1$; $f(-2) = 2(4) + 6 + 1 = 15$; $f(a) = 2a^2 - 3a + 1$; $f(a+h) = 2(a+h)^2 - 3(a+h) + 1 = 2a^2 + 4ah + 2h^2 - 3a - 3h + 1$. The difference quotient:

$$\frac{f(a+h) - f(a)}{h} = \frac{4ah + 2h^2 - 3h}{h} = 4a + 2h - 3.$$

This expression anticipates the derivative in Week 3–4 (as $h \rightarrow 0$ it tends to $4a - 3$).

3. Find the natural domain of a. $f(x) = \frac{x-1}{x^2+9}$; b. $g(x) = \sqrt{5-2x}$. Express in interval notation.

[Source: Week 2 Practical, Exercise 3]

Solution. a. The denominator $x^2 + 9 \geq 9 > 0$ never vanishes, so the domain is all of $\mathbb{R} = (-\infty, \infty)$. b. Need $5 - 2x \geq 0 \iff x \leq \frac{5}{2}$: domain $(-\infty, \frac{5}{2}]$.

4. Let $f(x) = 2x - 3$, $g(x) = x^2 + 1$. Find $(f \circ g)(x)$ and $(g \circ f)(x)$; are they equal?

[Source: Week 2 Practical, Exercise 4]

Solution. $(f \circ g)(x) = f(g(x)) = 2(x^2 + 1) - 3 = 2x^2 - 1$; $(g \circ f)(x) = g(2x - 3) = (2x - 3)^2 + 1 = 4x^2 - 12x + 10$. They differ (e.g. at $x = 0$: -1 vs 10), confirming that composition is generally not commutative — applying the squaring-then-shifting order is not the same as shifting-then-squaring.

5. Determine whether each function is even, odd or neither: $|x| + 1$ and $\frac{x}{x^2+1}$; verify algebraically and graphically.

[Source: Week 2 Practical, Exercise 5]

Solution. $f(x) = |x| + 1$: $f(-x) = |-x| + 1 = |x| + 1 = f(x)$ — even; its graph is mirror-symmetric about the y -axis. $g(x) = \frac{x}{x^2+1}$: $g(-x) = \frac{-x}{x^2+1} = -g(x)$ — odd; its graph is symmetric under a half-turn about the origin.

6. Classify each function as linear, polynomial, rational, algebraic or transcendental (a function may belong to more than one category): $\frac{x+1}{x^2+4}$ and $\sqrt{x^2+1}$.

[Source: Week 2 Practical, Exercise 6]

Solution. $\frac{x+1}{x^2+4}$ is a ratio of polynomials, hence *rational* (and algebraic). $\sqrt{x^2+1}$ is a root of a polynomial, hence *algebraic* (not rational, since it is not a ratio of polynomials).

7. The temperature in an industrial chamber rises linearly from 20°C at $t = 0$ to 80°C at $t = 15$ min. (1) Find $T(t)$; (2) interpret slope and intercept; (3) find $T(8)$; (4) when does the chamber reach 60°C ? (5) give a physically reasonable domain.

[Source: Week 2 Practical, Exercise 7]

Solution. (1) Slope $m = \frac{80-20}{15-0} = 4^\circ\text{C}/\text{min}$, so $T(t) = 20 + 4t$. (2) The slope is the heating rate: $+4$ degrees per minute; the intercept 20 is the initial temperature. (3) $T(8) = 20 + 32 = 52^\circ\text{C}$. (4) $20 + 4t = 60 \Rightarrow t = 10$ min. (5) $[0, 15]$ (or until the process model stops being linear).

8. For $p(x) = x^3 - 4x$ determine: domain; intercepts; symmetry; end behaviour; points of undefinedness.

[Source: Week 2 Practical, Exercise 8]

Solution. Domain $\mathbb{R}$ (polynomial). Intercepts: $p(0) = 0$; $x^3 - 4x = x(x-2)(x+2) = 0$ gives $x = -2, 0, 2$ — all on the x -axis through the origin. Symmetry: $p(-x) = -x^3 + 4x = -p(x)$ — odd. End behaviour: leading term x^3 with positive coefficient, so $p(x) \rightarrow +\infty$ as $x \rightarrow +\infty$ and $p(x) \rightarrow -\infty$ as $x \rightarrow -\infty$. A polynomial is defined everywhere: no points of undefinedness.

9. An electricity provider charges $C(E) = 0.10E$ for $0 \leq E \leq 200$ and $C(E) = 20 + 0.15(E - 200)$ for $E > 200$, in kWh. (1) Find $C(150), C(200), C(350)$; (2) sketch; (3) is the model continuous at $E = 200$? (4) meaning of each slope; (5) one limitation.

[Source: Week 2 Practical, Exercise 9]

Solution. (1) $C(150) = 15$; $C(200) = 20$; $C(350) = 20 + 0.15 \cdot 150 = 42.5$ (all in \$). (2) Two linear pieces: slope 0.10 up to 200, then 0.15 — a piecewise-linear graph with a kink at $(200, 20)$. (3) Yes: both formulas give 20 at $E = 200$, the pieces join with no jump. (4) Slopes are the marginal prices: 0.10\$/kWh for the first 200 kWh, 0.15\$/kWh beyond. (5) Limitation: the price jump at exactly 200 kWh gives consumers an incentive to stop just below the boundary; real tariffs change more smoothly or in blocks.

10. Find the inverse of $f(x) = 4x - 7$ and state domains; verify $f(f^{-1}(x)) = x$.

[Source: Week 2 Practical, Exercise 10]

Solution. $y = 4x - 7 \Rightarrow x = \frac{y+7}{4}$, so $f^{-1}(x) = \frac{x+7}{4}$ with domain and range $\mathbb{R}$. Check: $f(f^{-1}(x)) = 4 \cdot \frac{x+7}{4} - 7 = x + 7 - 7 = x$. Both are lines of slope 4 and $\frac{1}{4}$ — reciprocal slopes, mirrored about $y = x$.

11. For each model decide growth/decay and identify the initial value and the growth/decay factor: $A(t) = 500(1.08)^t$, $B(t) = 1200(0.85)^t$, $C(t) = 40e^{0.03t}$, $D(t) = 200e^{-0.4t}$.

[Source: Week 2 Practical, Exercise 11]

Solution. A : growth, initial 500, factor 1.08 (+8% per period). B : decay, initial 1200, factor 0.85 (−15%). C : growth, initial 40, continuous rate $k = 0.03$. D : decay, initial 200, continuous rate $k = -0.4$. The sign of k in e^{kt} , or whether $a > 1$ in a^t , decides the direction.

12. Solve $2^{x+1} = 16$.

[Source: Week 2 Practical, Exercise 12]

Solution. $16 = 2^4$, so $x + 1 = 4$ and $x = 3$. (Equivalently take $\log_2$ of both sides.)

13. Battery discharge: $E(t) = E_0 e^{-0.025t}$, t in hours. (1) Fraction remaining after 10 h? (2) When does 50% remain? (3) When does 10% remain? (4) Why never exactly zero?

[Source: Week 2 Practical, Exercise 13]

Solution. (1) $E(10)/E_0 = e^{-0.25} \approx 0.7788$: about 77.9%. (2) $e^{-0.025t} = \frac{1}{2} \Rightarrow t = \frac{\ln 2}{0.025} \approx 27.7$ h. (3) $e^{-0.025t} = 0.1 \Rightarrow t = \frac{\ln 10}{0.025} \approx 92.1$ h. (4) The exponential is strictly positive for every finite t ; it approaches zero only in the limit $t \rightarrow \infty$ — a first taste of limits.

14. Population growth: $P(t) = 800e^{0.35t}$, t in hours. (1) Initial population? (2) After 5 h? (3) Doubling time? (4) Can the model stay realistic indefinitely?

[Source: Week 2 Practical, Exercise 14]

Solution. (1) $P(0) = 800$. (2) $P(5) = 800e^{1.75} \approx 800 \cdot 5.755 \approx 4603$. (3) $T_d = \frac{\ln 2}{0.35} \approx 1.98$ h. (4) No: exponential growth ignores nutrient exhaustion and space; realistically the model holds only near the start of the culture.

3.3 Math I: Independent study assignments, Week 2 — questions and full solutions

1. Determine whether each relation defines y as a function of x : a. $y = 3x - 2$; b. $x^2 + y^2 = 9$; c. $y^2 = x + 1$; d. $x = y^3 - 2$.

[Source: Week 2 ISA, Assignment 2]

Solution. a. Function. b. Not a function (circle: two y 's per x). c. $y = \pm\sqrt{x+1}$: two values for $x > -1$, not a function. d. Solve for y : $y = \sqrt[3]{x+2}$; the cube root is unique for every real x , so this *is* a function (it passes the vertical-line test even though it fails the horizontal one).

2. For $f(x) = 2x^2 - 3x + 1$ compute $f(0), f(-2), f(a), f(a+h), \frac{f(a+h)-f(a)}{h}$.

[Source: Week 2 ISA, Assignment 3]

Solution. Identical to Practical Exercise 2: 1; 15; $2a^2 - 3a + 1$; $2a^2 + 4ah + 2h^2 - 3a - 3h + 1$; difference quotient = $4a + 2h - 3$.

3. Find the natural domain of a. $f(x) = \frac{x-1}{x^2+9}$; b. $g(x) = \sqrt{5-2x}$; c. $h(x) = \sqrt{x-1}$ over... $h(x) = \frac{\sqrt{x-1}}{x-4}$; d. $p(x) = \sqrt{\frac{x+2}{x-1}}$.

[Source: Week 2 ISA, Assignment 4]

Solution. a. $(-\infty, \infty)$. b. $(-\infty, \frac{5}{2}]$. c. Need $x-1 \geq 0$ and $x \neq 4$: $[1, 4) \cup (4, \infty)$. d. Need $\frac{x+2}{x-1} \geq 0$ and $x \neq 1$: critical points -2 and 1 ; the quotient is nonnegative on $(-\infty, -2] \cup (1, \infty)$.

4. Find the range of $f(x) = x^2 + 2$, $g(x) = -|x-1| + 4$, $h(x) = \sqrt{x+3} - 2$.

[Source: Week 2 ISA, Assignment 5]

Solution. $x^2 \geq 0 \Rightarrow f \geq 2$: range $[2, \infty)$. $|x-1| \geq 0 \Rightarrow -|x-1| \leq 0 \Rightarrow g \leq 4$: range $(-\infty, 4]$. $\sqrt{x+3} \geq 0 \Rightarrow h \geq -2$: range $[-2, \infty)$. Each answer comes from the extreme value of the inner function.

5. Let $f(x) = x^2 - 1$, $g(x) = \sqrt{x+2}$. Determine $f+g$, $f-g$, fg , f/g and each new domain.

[Source: Week 2 ISA, Assignment 6]

Solution. $(f \pm g)(x) = x^2 - 1 \pm \sqrt{x+2}$; $(fg)(x) = (x^2 - 1)\sqrt{x+2}$; $(f/g)(x) = \frac{x^2 - 1}{\sqrt{x+2}}$. Domains: $f+g, f-g, fg$: $x \geq -2$ (intersection of $\mathbb{R}$ and $[-2, \infty)$), i.e. $[-2, \infty)$. For f/g additionally $\sqrt{x+2} \neq 0$, i.e. $x \neq -2$: $(-2, \infty)$.

6. Let $f(x) = 2x - 3$, $g(x) = x^2 + 1$. Find $(f \circ g)(x)$, $(g \circ f)(x)$, $(f \circ f)(x)$, $(g \circ g)(x)$.

[Source: Week 2 ISA, Assignment 7]

Solution. $(f \circ g)(x) = 2x^2 - 1$; $(g \circ f)(x) = 4x^2 - 12x + 10$; $(f \circ f)(x) = 2(2x - 3) - 3 = 4x - 9$; $(g \circ g)(x) = (x^2 + 1)^2 + 1 = x^4 + 2x^2 + 2$. The first two differ, so composition is not commutative; the last two show self-composition is perfectly legal.

7. Determine whether each function is even, odd or neither: $x^4 - 2x^2$; $x^3 - 4x$; $x^3 + x^2$; $|x| + 1$; $\frac{x^2}{x+1}$.

[Source: Week 2 ISA, Assignment 8]

Solution. $x^4 - 2x^2$: only even powers — even. $x^3 - 4x$: only odd powers — odd. $x^3 + x^2$: mixed — neither. $|x| + 1$: even (Practical Ex. 5). $\frac{x^2}{x+1}$: numerator even, denominator changes sign: $\frac{x^2}{-x+1} \neq \pm \frac{x^2}{x+1}$ — neither.

8. Classify: $3x - 7$; $x^4 - 2x + 1$; $\frac{x+1}{x^2+4}$; $\sqrt{x^2+1}$; 2^x ; $\sin x$.

[Source: Week 2 ISA, Assignment 9]

Solution. $3x - 7$: linear (polynomial). $x^4 - 2x + 1$: polynomial. $\frac{x+1}{x^2+4}$: rational. $\sqrt{x^2+1}$: algebraic (root). 2^x : exponential — transcendental. $\sin x$: trigonometric — transcendental.

9. The linear temperature model of Practical Exercise 10 — all five parts.

[Source: Week 2 ISA, Assignment 10]

Solution. Same model and answers as Practical Exercise 7: $T(t) = 20 + 4t$; slope = heating rate $4^\circ\text{C}/\text{min}$, intercept = initial 20°C ; $T(8) = 52^\circ\text{C}$; 60°C at $t = 10$ min; domain $[0, 15]$.

10. For $p(x) = x^3 - 4x$ and $r(x) = \frac{x+1}{x-1}$ determine domain, intercepts, symmetry, end behaviour, undefined points.

[Source: Week 2 ISA, Assignment 12]

Solution. p : as in Practical Exercise 8 (domain $\mathbb{R}$; intercepts $-2, 0, 2$; odd; $\pm\infty$ ends; none). r : domain $x \neq 1$, i.e. $(-\infty, 1) \cup (1, \infty)$. Intercepts: $r(0) = -1$; numerator zero at $x = -1$. Symmetry: $r(-x) = \frac{-x+1}{-x-1} = \frac{x-1}{x+1} \neq \pm r(x)$ — neither. End behaviour: $r(x) \rightarrow 1$ as $x \rightarrow \pm\infty$ (horizontal asymptote $y = 1$); $r(x) \rightarrow \pm\infty$ as $x \rightarrow 1$ (vertical asymptote $x = 1$). Undefined only at $x = 1$.

11. The piecewise electricity tariff of Practical Exercise 9 — all five parts.

[Source: Week 2 ISA, Assignment 13]

Solution. Same answers as Practical Exercise 9: $C(150) = 15$, $C(200) = 20$, $C(350) = 42.5$; continuous kinked graph; slopes are marginal prices; limitation — discontinuity of incentive at the 200 kWh boundary.

12. Convert $30^\circ, 135^\circ, 300^\circ$ to radians and $\frac{\pi}{6}, \frac{5\pi}{4}, \frac{7\pi}{3}$ to degrees.

[Source: Week 2 ISA, Assignment 14]

Solution. Multiply by $\pi/180$: $30^\circ = \frac{\pi}{6}$; $135^\circ = \frac{3\pi}{4}$; $300^\circ = \frac{5\pi}{3}$. Multiply by $180/\pi$: $\frac{\pi}{6} = 30^\circ$; $\frac{5\pi}{4} = 225^\circ$; $\frac{7\pi}{3} = 420^\circ$.

13. One-to-one? a. $f(x) = 3x - 5$, $x \in \mathbb{R}$; b. $g(x) = x^2$, $x \in \mathbb{R}$; c. $h(x) = x^2$, $x \geq 0$; d. $p(x) = \frac{1}{x}$, $x \neq 0$.

[Source: Week 2 ISA, Assignment 15]

Solution. a. Yes: a nonhorizontal line passes the horizontal-line test. b. No: $g(-2) = g(2) = 4$. c. Yes: on $[0, \infty)$ squaring is injective (the restriction restores the horizontal-line test). d. Yes: $\frac{1}{x_1} = \frac{1}{x_2} \Rightarrow x_1 = x_2$.

14. Find inverses and their domains: a. $f(x) = 4x - 7$; b. $g(x) = \frac{x+3}{2}$... c. $h(x) = x^2 + 1$, $x \geq 0$; d. $p(x) = \frac{2x+3}{x-1}$, $x \neq 1$; verify by composition.

[Source: Week 2 ISA, Assignment 16]

Solution. a. $f^{-1}(x) = \frac{x+7}{4}$, domain $\mathbb{R}$ (Practical Ex. 10). b. $y = \frac{x+3}{2} \Rightarrow x = 2y - 3$, so $g^{-1}(x) = 2x - 3$, domain $\mathbb{R}$. c. $y = x^2 + 1 \Rightarrow x = \sqrt{y-1}$ (positive root because $x \geq 0$): $h^{-1}(x) = \sqrt{x-1}$, domain $[1, \infty)$. d. $y = \frac{2x+3}{x-1} \Rightarrow y(x-1) = 2x+3 \Rightarrow x(y-2) = y+3 \Rightarrow x = \frac{y+3}{y-2}$: $p^{-1}(x) = \frac{x+3}{x-2}$, domain $x \neq 2$. Each satisfies $f(f^{-1}(x)) = x$ by construction.

15. Growth/decay classification of A , B , C , D (Practical Exercise 11).

[Source: Week 2 ISA, Assignment 17]

Solution. Same: A growth ($a = 1.08$), B decay ($a = 0.85$), C growth ($k = 0.03$), D decay ($k = -0.4$).

16. Solve, giving exact answers first: a. $2^{x+1} = 16$; b. $5^{2x} = 125$; c. $e^{3x} = 7$; d. $100e^{-0.2t} = 20$.

[Source: Week 2 ISA, Assignment 18]

Solution. a. $x+1 = 4 \Rightarrow x = 3$. b. $2x = 3 \Rightarrow x = \frac{3}{2}$. c. $3x = \ln 7 \Rightarrow x = \frac{\ln 7}{3} \approx 0.6486$. d. $e^{-0.2t} = 0.2 \Rightarrow t = -\frac{\ln 0.2}{0.2} = 5 \ln 5 \approx 8.047$.

17. Battery discharge (Practical Exercise 13) — all four parts.

[Source: Week 2 ISA, Assignment 19]

Solution. Same: $e^{-0.25} \approx 77.9\%$ after 10 h; 50% at $\frac{\ln 2}{0.025} \approx 27.7$ h; 10% at $\frac{\ln 10}{0.025} \approx 92.1$ h; exponential never reaches zero in finite time.

18. Bacterial population (Practical Exercise 14) — all four parts.

[Source: Week 2 ISA, Assignment 20]

Solution. Same: $P_0 = 800$; $P(5) \approx 4603$; doubling time $\frac{\ln 2}{0.35} \approx 1.98$ h; exponential-only models fail long-term (resources, space, waste).

3.4 Linear Algebra I, Week 2: equation catalogue

Week 2 adds the two products of vector geometry and the first matrix concepts.

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + \cdots + u_n v_n = \mathbf{u}^T \mathbf{v}, \quad \cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}, \quad (45)$$

$$\mathbf{u} \perp \mathbf{v} \iff \mathbf{u} \cdot \mathbf{v} = 0, \quad \text{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}} \mathbf{v}, \quad (46)$$

$$\mathbf{u} \times \mathbf{v} = \begin{bmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{bmatrix}, \quad \|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\| \|\mathbf{v}\| |\sin \theta| = \text{area}, \quad (47)$$

$$\mathbf{u} \times \mathbf{v} = -(\mathbf{v} \times \mathbf{u}), \quad (\mathbf{u} \times \mathbf{v}) \perp \mathbf{u}, \mathbf{v}, \quad (48)$$

$$A = [a_{ij}] \in \mathbb{R}^{m \times n}, \quad (A + B)_{ij} = a_{ij} + b_{ij}, \quad (cA)_{ij} = c a_{ij}, \quad (49)$$

$$A\mathbf{x} = x_1 \mathbf{a}_1 + \cdots + x_n \mathbf{a}_n \quad (\text{column picture}), \quad (50)$$

where $\mathbf{a}_j$ is column j of A . Sizes must match: $A\mathbf{x}$ exists iff A is $m \times n$ and $\mathbf{x} \in \mathbb{R}^n$, and the output lies in $\mathbb{R}^m$. A diagonal matrix $D = \text{diag}(d_1, \dots, d_n)$ scales each coordinate: $D\mathbf{x} = (d_1 x_1, \dots, d_n x_n)^T$.

3.5 Linear Algebra I, Week 2: lecture in-class problems — questions and full solutions

1. Let $\mathbf{u} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$. Compute $\mathbf{u} \cdot \mathbf{v}$; classify the angle.

[Source: LA W2 Lecture, Problem 1]

Solution. $\mathbf{u} \cdot \mathbf{v} = 3 \cdot 4 + (-2) \cdot 5 = 12 - 10 = 2 > 0$. A positive dot product means $\cos \theta > 0$, so the angle is acute.

2. With $\mathbf{u} = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} -1 \\ 3 \\ 0 \end{bmatrix}$: find $\|\mathbf{u}\|$ and $\text{dist}(\mathbf{u}, \mathbf{v})$.

[Source: LA W2 Lecture, Problem 2]

Solution. $\|\mathbf{u}\| = \sqrt{4 + 1 + 4} = 3$. $\mathbf{u} - \mathbf{v} = \begin{bmatrix} 3 \\ -4 \\ 2 \end{bmatrix}$, so $\text{dist}(\mathbf{u}, \mathbf{v}) = \sqrt{9 + 16 + 4} = \sqrt{29}$.

3. Find k so that $\mathbf{u} = \begin{bmatrix} 2 \\ k \\ -1 \end{bmatrix}$ is orthogonal to $\mathbf{v} = \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix}$.

[Source: LA W2 Lecture, Problem 3]

Solution. Orthogonality is $\mathbf{u} \cdot \mathbf{v} = 0$: $6 + k - 4 = 0$, so $k = -2$.

4. Find the angle between $\mathbf{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

[Source: LA W2 Lecture, Problem 4]

Solution. $\mathbf{u} \cdot \mathbf{v} = 1 - 1 = 0$: the vectors are perpendicular, $\theta = 90^\circ$ — no formula work needed once the dot product vanishes.

5. Project $\mathbf{u} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$ onto $\mathbf{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

[Source: LA W2 Lecture, Problem 5]

Solution. $\mathbf{u} \cdot \mathbf{v} = 6$, $\mathbf{v} \cdot \mathbf{v} = 2$, so $\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{6}{2} \mathbf{v} = 3\mathbf{v} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$. The remainder $\mathbf{u} - 3\mathbf{v} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ satisfies $(\mathbf{u} - 3\mathbf{v}) \cdot \mathbf{v} = 0$: the decomposition is into parallel + perpendicular parts.

6. Compute $\mathbf{u} \times \mathbf{v}$ for $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}$ and verify orthogonality.

[Source: LA W2 Lecture, Problem 6]

Solution. By (47): $\mathbf{u} \times \mathbf{v} = \begin{bmatrix} 2 \cdot 2 - 0 \cdot (-1) \\ 0 \cdot 3 - 1 \cdot 2 \\ 1 \cdot (-1) - 2 \cdot 3 \end{bmatrix} = \begin{bmatrix} 4 \\ -2 \\ -7 \end{bmatrix}$. Check: $1 \cdot 4 + 2 \cdot (-2) + 0 \cdot (-7) = 0$ and $3 \cdot 4 + (-1)(-2) + 2 \cdot (-7) = 0$ — orthogonal to both inputs.

7. Find the parallelogram area with adjacent sides $\mathbf{a} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}$.

[Source: LA W2 Lecture, Problem 7]

Solution. $\mathbf{a} \times \mathbf{b} = \begin{bmatrix} 0 \cdot (-1) - 1 \cdot 3 \\ 1 \cdot 1 - 2 \cdot (-1) \\ 2 \cdot 3 - 0 \cdot 1 \end{bmatrix} = \begin{bmatrix} -3 \\ 3 \\ 6 \end{bmatrix}$, so the area is $\|\mathbf{a} \times \mathbf{b}\| = \sqrt{9 + 9 + 36} = \sqrt{54} = 3\sqrt{6}$.

8. For $A = \begin{bmatrix} 2 & -1 & 4 \\ 0 & 3 & 5 \end{bmatrix}$: state its size, find a_{13} , a_{22} , and write columns $\mathbf{a}_1, \mathbf{a}_3$.

[Source: LA W2 Lecture, Problem 8]

Solution. A is 2×3 (rows first). $a_{13} = 4$ (row 1, column 3), $a_{22} = 3$. $\mathbf{a}_1 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$, $\mathbf{a}_3 = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$.

9. Let $A = \begin{bmatrix} 1 & -2 \\ 3 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 1 \\ -1 & 5 \end{bmatrix}$. Find $2A - B$.

[Source: LA W2 Lecture, Problem 9]

Solution. $2A = \begin{bmatrix} 2 & -4 \\ 6 & 0 \end{bmatrix}$; subtract entrywise: $2A - B = \begin{bmatrix} -2 & -5 \\ 7 & -5 \end{bmatrix}$.

10. A is 3×2 , B is 2×3 , $\mathbf{x} \in \mathbb{R}^2$, $\mathbf{y} \in \mathbb{R}^3$. Which of $A\mathbf{x}$, $A\mathbf{y}$, $B\mathbf{x}$, $B\mathbf{y}$ are defined?

[Source: LA W2 Lecture, Problem 10]

Solution. $A\mathbf{x}$: $(3 \times 2)(2 \times 1)$ OK, output in $\mathbb{R}^3$. $B\mathbf{y}$: $(2 \times 3)(3 \times 1)$ OK, output in $\mathbb{R}^2$. $A\mathbf{y}$ mismatches ($2 \neq 3$) and $B\mathbf{x}$ mismatches ($3 \neq 2$): both undefined. Only the *inner* sizes must agree.

11. Let $D = \text{diag}(2, -1, 3)$, $\mathbf{x} = \begin{bmatrix} 4 \\ 5 \\ -2 \end{bmatrix}$. Find $D\mathbf{x}$ and $I_3\mathbf{x}$.

[Source: LA W2 Lecture, Problem 11]

Solution. $D\mathbf{x} = \begin{bmatrix} 2 \cdot 4 \\ -1 \cdot 5 \\ 3 \cdot (-2) \end{bmatrix} = \begin{bmatrix} 8 \\ -5 \\ -6 \end{bmatrix}$: each coordinate is scaled by the matching diagonal entry. $I_3\mathbf{x} = \mathbf{x} = \begin{bmatrix} 4 \\ 5 \\ -2 \end{bmatrix}$: the identity does nothing.

12. Let $A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \\ 4 & 0 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$. Compute $A\mathbf{x}$ by the column picture and by row dot products.

[Source: LA W2 Lecture, Problem 12]

Solution. Columns: $2 \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix} + (-1) \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -5 \\ 8 \end{bmatrix}$. Rows: $[1 \ 2] \cdot \mathbf{x} = 0$, $[-1 \ 3] \cdot \mathbf{x} = -5$, $[4 \ 0] \cdot \mathbf{x} = 8$, stacking gives the same $\begin{bmatrix} 0 \\ -5 \\ 8 \end{bmatrix}$. The two pictures always agree; the column view explains what the product *means*.

13. Given $A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$ and $A\mathbf{x} = \begin{bmatrix} 7 \\ 5 \end{bmatrix}$, find $\mathbf{x}$.

[Source: LA W2 Lecture, Problem 13]

Solution. Components: $x_1 + 2x_2 = 7$, $3x_1 - x_2 = 5$. From the second $x_2 = 3x_1 - 5$; substitute: $x_1 + 6x_1 - 10 = 7$, so $7x_1 = 17$, $x_1 = \frac{17}{7}$, $x_2 = \frac{16}{7}$. Thus $\mathbf{x} = \begin{bmatrix} 17/7 \\ 16/7 \end{bmatrix}$.

14. Nonzero vectors $\mathbf{u} + \mathbf{v}$ and $\mathbf{u} - \mathbf{v}$ are orthogonal. What follows about $\|\mathbf{u}\|$ and $\|\mathbf{v}\|$?

[Source: LA W2 Lecture, Problem 14]

Solution. Orthogonality gives $0 = (\mathbf{u} + \mathbf{v}) \cdot (\mathbf{u} - \mathbf{v}) = \|\mathbf{u}\|^2 - \|\mathbf{v}\|^2$ (cross terms cancel), hence $\|\mathbf{u}\| = \|\mathbf{v}\|$: equal lengths, with $\mathbf{u}$ and $\mathbf{v}$ the diagonals' half-vectors of a rectangle-style picture.

3.6 Linear Algebra I, Week 2: ISA exercises — questions and full solutions

1. Let $\mathbf{u} = (-3, 2)^T$, $\mathbf{v} = (4, 5)^T$. Compute $\mathbf{u} \cdot \mathbf{v}$; is the angle acute, right or obtuse?

[Source: LA ISA Week 2, Exercise 1]

Solution. $\mathbf{u} \cdot \mathbf{v} = -12 + 10 = -2 < 0$, so $\cos \theta < 0$: the angle is obtuse.

2. Let $\mathbf{a} = (1, -2, 4)^T$, $\mathbf{b} = (3, 0, -1)^T$. Find $\mathbf{a} \cdot \mathbf{b}$, $\|\mathbf{a}\|$, $\|\mathbf{b}\|$, $\|\mathbf{a} - \mathbf{b}\|$.

[Source: LA ISA Week 2, Exercise 2]

Solution. $\mathbf{a} \cdot \mathbf{b} = 3 + 0 - 4 = -1$; $\|\mathbf{a}\| = \sqrt{1 + 4 + 16} = \sqrt{21}$; $\|\mathbf{b}\| = \sqrt{9 + 0 + 1} = \sqrt{10}$; $\mathbf{a} - \mathbf{b} = (-2, -2, 5)^T$, $\|\mathbf{a} - \mathbf{b}\| = \sqrt{4 + 4 + 25} = \sqrt{33}$.

3. Find k so that $\mathbf{u} = (2, k, 3)^T$ is orthogonal to $\mathbf{v} = (-1, 4, 2)^T$.

[Source: LA ISA Week 2, Exercise 3]

Solution. $\mathbf{u} \cdot \mathbf{v} = -2 + 4k + 6 = 0 \Rightarrow 4k = -4 \Rightarrow k = -1$.

4. Find the exact $\cos \theta$ and then θ to the nearest degree for $\mathbf{u} = (3, 1)^T$, $\mathbf{v} = (-2, 4)^T$.

[Source: LA ISA Week 2, Exercise 4]

Solution. $\mathbf{u} \cdot \mathbf{v} = -6 + 4 = -2$; $\|\mathbf{u}\| = \sqrt{10}$, $\|\mathbf{v}\| = \sqrt{20} = 2\sqrt{5}$; $\cos \theta = \frac{-2}{\sqrt{10} \cdot 2\sqrt{5}} = \frac{-2}{10\sqrt{2}} = -\frac{1}{5\sqrt{2}} \approx -0.1414$, so $\theta \approx 98^\circ$.

5. Find the projection of $\mathbf{u} = (6, 1)^T$ onto $\mathbf{v} = (2, -1)^T$; verify $\mathbf{u} - \text{proj}$ is orthogonal to $\mathbf{v}$.

[Source: LA ISA Week 2, Exercise 5]

Solution. $\mathbf{u} \cdot \mathbf{v} = 12 - 1 = 11$, $\mathbf{v} \cdot \mathbf{v} = 5$: $\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{11}{5}(2, -1)^T = (\frac{22}{5}, -\frac{11}{5})^T$. Remainder: $\mathbf{u} - \text{proj} = (\frac{8}{5}, \frac{16}{5})^T$; dot with $\mathbf{v}$: $\frac{16}{5} - \frac{16}{5} = 0$. Verified.

6. Nonzero $\mathbf{p}, \mathbf{q}$ satisfy $\mathbf{p} \cdot \mathbf{q} = \|\mathbf{p}\| \|\mathbf{q}\|$. What can you conclude about their directions?

[Source: LA ISA Week 2, Exercise 6]

Solution. From (45), $\cos \theta = 1$, so $\theta = 0$: the vectors point in exactly the same direction (one is a positive multiple of the other). Equality in the angle formula is the boundary case of the Cauchy–Schwarz idea.

7. For $\mathbf{a} = (2, 1, -1)^T$, $\mathbf{b} = (0, 3, 2)^T$ compute $\mathbf{a} \times \mathbf{b}$ and $\mathbf{b} \times \mathbf{a}$; verify the sign relation.

[Source: LA ISA Week 2, Exercise 7]

Solution. $\mathbf{a} \times \mathbf{b} = \begin{bmatrix} 1 \cdot 2 - (-1) \cdot 3 \\ (-1) \cdot 0 - 2 \cdot 2 \\ 2 \cdot 3 - 1 \cdot 0 \end{bmatrix} = \begin{bmatrix} 5 \\ -4 \\ 6 \end{bmatrix}$, and $\mathbf{b} \times \mathbf{a} = \begin{bmatrix} -5 \\ 4 \\ -6 \end{bmatrix} = -(\mathbf{a} \times \mathbf{b})$, matching (48).

8. Verify that $\mathbf{a} \times \mathbf{b}$ from Exercise 7 is orthogonal to both $\mathbf{a}$ and $\mathbf{b}$.

[Source: LA ISA Week 2, Exercise 8]

Solution. $\mathbf{a} \cdot (\mathbf{a} \times \mathbf{b}) = 2 \cdot 5 + 1 \cdot (-4) + (-1) \cdot 6 = 10 - 4 - 6 = 0$; $\mathbf{b} \cdot (\mathbf{a} \times \mathbf{b}) = 0 \cdot 5 + 3 \cdot (-4) + 2 \cdot 6 = -12 + 12 = 0$. Both vanish, as guaranteed after (47).

9. Find the parallelogram area with adjacent sides $\mathbf{u} = (1, 2, 2)^T$, $\mathbf{v} = (3, -1, 0)^T$.

[Source: LA ISA Week 2, Exercise 9]

Solution. $\mathbf{u} \times \mathbf{v} = \begin{bmatrix} 2 \cdot 0 - 2 \cdot (-1) \\ 2 \cdot 3 - 1 \cdot 0 \\ 1 \cdot (-1) - 2 \cdot 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \\ -7 \end{bmatrix}$; area = $\sqrt{4 + 36 + 49} = \sqrt{89}$.

10. $P = (1, 0, 1)$, $Q = (3, 1, 0)$, $R = (0, 2, 4)$: find the area of triangle PQR .

[Source: LA ISA Week 2, Exercise 10]

Solution. Edges from P : $\mathbf{PQ} = (2, 1, -1)^T$, $\mathbf{PR} = (-1, 2, 3)^T$. Cross: $\begin{bmatrix} 1 \cdot 3 - (-1) \cdot 2 \\ (-1)(-1) - 2 \cdot 3 \\ 2 \cdot 2 - 1 \cdot (-1) \end{bmatrix} = \begin{bmatrix} 5 \\ -5 \\ 5 \end{bmatrix}$, norm $\sqrt{75} = 5\sqrt{3}$; triangle area = $\frac{1}{2} \cdot 5\sqrt{3} = \frac{5\sqrt{3}}{2}$ (half the parallelogram).

11. Find all real t for which $\mathbf{a} = (1, t, 2)^T$ and $\mathbf{b} = (3, 6, 6)^T$ satisfy $\mathbf{a} \times \mathbf{b} = \mathbf{0}$; if none, explain.

[Source: LA ISA Week 2, Exercise 11]

Solution. Parallel vectors give a zero cross product, so we need $\mathbf{a} = c\mathbf{b}$: $1 = 3c \Rightarrow c = \frac{1}{3}$; then $t = 6c = 2$; check the third: $2 = 6c = 2$ — consistent. So $t = 2$ is the unique value (then $\mathbf{a} = \frac{1}{3}\mathbf{b}$).

12. Let $\mathbf{a} = (1, 0, 2)^T$, $\mathbf{b} = (2, 1, -1)^T$, $\mathbf{c} = (0, 3, 1)^T$. Compute $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$; interpret a zero value.
[Source: LA ISA Week 2, Exercise 12]

Solution. $\mathbf{b} \times \mathbf{c} = \begin{bmatrix} 1 \cdot 1 - (-1) \cdot 3 \\ (-1) \cdot 0 - 2 \cdot 1 \\ 2 \cdot 3 - 1 \cdot 0 \end{bmatrix} = \begin{bmatrix} 4 \\ -2 \\ 6 \end{bmatrix}$; $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = 4 + 0 + 12 = 16$. This is the signed volume of the parallelepiped spanned by the three vectors; a zero value would mean the vectors are coplanar (no 3D volume). Here $|16| = 16$ cubic units.

13. Let $A = \begin{bmatrix} 2 & -1 & 0 \\ 4 & 3 & 5 \end{bmatrix}$. State its size; find a_{12} , a_{23} ; write every row and column as a vector.
[Source: LA ISA Week 2, Exercise 13]

Solution. Size 2×3 ; $a_{12} = -1$, $a_{23} = 5$. Rows: $[2 \ -1 \ 0]$, $[4 \ 3 \ 5]$; columns: $\begin{bmatrix} 2 \\ 4 \end{bmatrix}$, $\begin{bmatrix} -1 \\ 3 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 5 \end{bmatrix}$.

14. Let $A = \begin{bmatrix} 1 & 3 \\ -2 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 5 & -1 \\ 0 & 2 \end{bmatrix}$. Compute $A + B$, $A - B$, $-2A$, $3A - 2B$.
[Source: LA ISA Week 2, Exercise 14]

Solution. $A + B = \begin{bmatrix} 6 & 2 \\ -2 & 6 \end{bmatrix}$; $A - B = \begin{bmatrix} -4 & 4 \\ -2 & 2 \end{bmatrix}$; $-2A = \begin{bmatrix} -2 & -6 \\ 4 & -8 \end{bmatrix}$;
 $3A - 2B = \begin{bmatrix} 3 - 10 & 9 + 2 \\ -6 - 0 & 12 - 4 \end{bmatrix} = \begin{bmatrix} -7 & 11 \\ -6 & 8 \end{bmatrix}$.

15. A is 4×3 , B is 3×2 , $\mathbf{x} \in \mathbb{R}^2$, $\mathbf{y} \in \mathbb{R}^3$, $\mathbf{z} \in \mathbb{R}^4$. Which of $A\mathbf{x}$, $A\mathbf{y}$, $A\mathbf{z}$, $B\mathbf{x}$, $B\mathbf{y}$, $B\mathbf{z}$ are defined, and with what output sizes?
[Source: LA ISA Week 2, Exercise 15]

Solution. $A\mathbf{y}$: $(4 \times 3)(3 \times 1)$ — defined, output $\mathbb{R}^4$. $B\mathbf{x}$: $(3 \times 2)(2 \times 1)$ — defined, output $\mathbb{R}^3$. All others ($A\mathbf{x}$, $A\mathbf{z}$, $B\mathbf{y}$, $B\mathbf{z}$) have mismatched inner dimensions and are undefined.

16. Let $D = \text{diag}(-2, 3, 1)$, $\mathbf{x} = (4, -1, 5)^T$. Compute $D\mathbf{x}$; describe what D does to each coordinate.
[Source: LA ISA Week 2, Exercise 16]

Solution. $D\mathbf{x} = (-8, 3, 5)^T$: the first coordinate is scaled by -2 (stretch and flip), the second by 3 , the third by 1 (unchanged). A diagonal matrix is three independent scalings of the coordinate axes.

17. Let $A = \begin{bmatrix} 1 & -2 & 3 \\ 4 & 0 & -1 \end{bmatrix}$, $\mathbf{x} = (2, -1, 3)^T$. Compute $A\mathbf{x}$ (a) as a combination of columns, (b) by row dot products.
[Source: LA ISA Week 2, Exercise 17]

Solution. (a) $2 \begin{bmatrix} 1 \\ 4 \end{bmatrix} - 1 \begin{bmatrix} -2 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 + 2 + 9 \\ 8 + 0 - 3 \end{bmatrix} = \begin{bmatrix} 13 \\ 5 \end{bmatrix}$. (b) Row 1: $1 \cdot 2 + (-2)(-1) + 3 \cdot 3 = 13$; row 2: $4 \cdot 2 + 0 + (-1) \cdot 3 = 5$; stacked: $\begin{bmatrix} 13 \\ 5 \end{bmatrix}$. Same result, as it must be.

18. Let $\mathbf{a}_1 = (1, 2)^T$, $\mathbf{a}_2 = (3, -1)^T$. Find $\mathbf{x} = (x_1, x_2)^T$ with $x_1\mathbf{a}_1 + x_2\mathbf{a}_2 = (11, 1)^T$.
[Source: LA ISA Week 2, Exercise 18]

Solution. Components: $x_1 + 3x_2 = 11$, $2x_1 - x_2 = 1$. From the second $x_2 = 2x_1 - 1$; then $x_1 + 6x_1 - 3 = 11 \Rightarrow 7x_1 = 14 \Rightarrow x_1 = 2$, $x_2 = 3$.

19. Let $A = \begin{bmatrix} 2 & 4 \\ -1 & -2 \end{bmatrix}$. Describe all $\mathbf{x}$ with $A\mathbf{x} = \mathbf{0}$; why are the columns dependent?

[Source: LA ISA Week 2, Exercise 19]

Solution. $2x_1 + 4x_2 = 0$ and $-x_1 - 2x_2 = 0$ are the same equation: $x_1 = -2x_2$. Solutions: $\mathbf{x} = t \begin{bmatrix} -2 \\ 1 \end{bmatrix}$, $t \in \mathbb{R}$ — a whole line of solutions. A nontrivial combination $-2\mathbf{a}_1 + 1 \cdot \mathbf{a}_2 = \mathbf{0}$ exists, which is exactly linear dependence of the columns (column 2 is $-2 \times$ column 1).

20. A is 3×2 with $A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = (2, -1, 4)^T$ and $A \begin{bmatrix} 0 \\ 1 \end{bmatrix} = (3, 2, 0)^T$. Write A and compute $A \begin{bmatrix} -2 \\ 5 \end{bmatrix}$ without row dot products.

[Source: LA ISA Week 2, Exercise 20]

Solution. The two images are the columns of A : $A = \begin{bmatrix} 2 & 3 \\ -1 & 2 \\ 4 & 0 \end{bmatrix}$. Linearity: $A \begin{bmatrix} -2 \\ 5 \end{bmatrix} =$

$$-2A\mathbf{e}_1 + 5A\mathbf{e}_2 = -2 \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix} + 5 \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 11 \\ 12 \\ -8 \end{bmatrix}.$$

21. Expand $(\mathbf{u} + \mathbf{v}) \cdot (\mathbf{u} - \mathbf{v})$ for same-dimension vectors; prove $\mathbf{u} + \mathbf{v} \perp \mathbf{u} - \mathbf{v}$ exactly when $\|\mathbf{u}\| = \|\mathbf{v}\|$.

[Source: LA ISA Week 2, Exercise 21]

Solution. Distributing: $(\mathbf{u} + \mathbf{v}) \cdot (\mathbf{u} - \mathbf{v}) = \mathbf{u} \cdot \mathbf{u} - \mathbf{u} \cdot \mathbf{v} + \mathbf{v} \cdot \mathbf{u} - \mathbf{v} \cdot \mathbf{v} = \|\mathbf{u}\|^2 - \|\mathbf{v}\|^2$ (the middle terms cancel). Orthogonality means this equals zero, i.e. $\|\mathbf{u}\|^2 = \|\mathbf{v}\|^2$, and since norms are nonnegative, $\|\mathbf{u}\| = \|\mathbf{v}\|$.

22. Self-check: (1) what object does a dot product produce, and a cross product? (2) How does the sign of $\mathbf{u} \cdot \mathbf{v}$ classify an angle? (3) Why does a 2×3 matrix multiply a vector in $\mathbb{R}^3$ but not one in $\mathbb{R}^2$? (4) How do the entries of $\mathbf{x}$ act in $A\mathbf{x}$? (5) Which Week 2 operations need equal dimensions, and which need vectors specifically in $\mathbb{R}^3$?

[Source: LA ISA Week 2, Self-check]

Solution. (1) A scalar; a vector (in $\mathbb{R}^3$). (2) Positive $\Rightarrow$ acute, zero $\Rightarrow$ right, negative $\Rightarrow$ obtuse. (3) Product $A\mathbf{x}$ requires the inner dimensions to match: 2×3 times 3×1 works; times 2×1 does not. (4) As weights in the linear combination $x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$ of the columns. (5) Dot products, addition and subtraction need equal dimensions; the cross product and its geometry (areas, normals) exist only in $\mathbb{R}^3$.

3.7 Linear Algebra I, Week 2: TA session (15 problems) — questions and full solutions

1. $\mathbf{a} = \begin{bmatrix} -2 \\ 4 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$: compute $\mathbf{a} \cdot \mathbf{b}$; classify the angle.

[Source: LA W2 TA, Problem 1]

Solution. $\mathbf{a} \cdot \mathbf{b} = -10 + 4 = -6 < 0$: obtuse.

2. Find the norm and a unit vector in the direction of $\mathbf{v} = \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix}$.

[Source: LA W2 TA, Problem 2]

Solution. $\|\mathbf{v}\| = \sqrt{4 + 1 + 4} = 3$; unit vector $\hat{\mathbf{v}} = \begin{bmatrix} -2/3 \\ 1/3 \\ 2/3 \end{bmatrix}$.

3. Find the distance between $\mathbf{p} = \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix}$ and $\mathbf{q} = \begin{bmatrix} -2 \\ 2 \\ 1 \end{bmatrix}$.

[Source: LA W2 TA, Problem 3]

Solution. $\mathbf{p} - \mathbf{q} = \begin{bmatrix} 3 \\ -4 \\ 3 \end{bmatrix}$; distance $= \sqrt{9 + 16 + 9} = \sqrt{34}$.

4. Find t if $\begin{bmatrix} 1 \\ t \\ 2 \end{bmatrix}$ is orthogonal to $\begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix}$.

[Source: LA W2 TA, Problem 4]

Solution. $2 - 3t + 8 = 0 \Rightarrow t = \frac{10}{3}$.

5. Find $\cos \theta$ for $\mathbf{u} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$.

[Source: LA W2 TA, Problem 5]

Solution. $\mathbf{u} \cdot \mathbf{v} = 1$, $\|\mathbf{u}\| = \sqrt{5}$, $\|\mathbf{v}\| = \sqrt{10}$; $\cos \theta = \frac{1}{\sqrt{50}} = \frac{\sqrt{2}}{10}$.

6. Project $\mathbf{u} = (5, -1)^T$ onto $\mathbf{v} = (2, 1)^T$; write $\mathbf{u}$ as parallel + orthogonal parts.

[Source: LA W2 TA, Problem 7]

Solution. $\mathbf{u} \cdot \mathbf{v} = 9$, $\mathbf{v} \cdot \mathbf{v} = 5$: $\text{proj} = \frac{9}{5} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 18/5 \\ 9/5 \end{bmatrix}$. The orthogonal part: $\mathbf{u} - \text{proj} = \begin{bmatrix} 7/5 \\ -14/5 \end{bmatrix}$. Decomposition: $\begin{bmatrix} 5 \\ -1 \end{bmatrix} = \begin{bmatrix} 18/5 \\ 9/5 \end{bmatrix} + \begin{bmatrix} 7/5 \\ -14/5 \end{bmatrix}$.

7. Compute $\mathbf{a} \times \mathbf{b}$ for $\mathbf{a} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 0 \\ 4 \\ 1 \end{bmatrix}$.

[Source: LA W2 TA, Problem 8]

Solution. $\mathbf{a} \times \mathbf{b} = \begin{bmatrix} (-1) \cdot 1 - 3 \cdot 4 \\ 3 \cdot 0 - 2 \cdot 1 \\ 2 \cdot 4 - (-1) \cdot 0 \end{bmatrix} = \begin{bmatrix} -13 \\ -2 \\ 8 \end{bmatrix}$.

8. $A = (0, 0, 0)$, $B = (2, 1, 0)$, $C = (1, 0, 3)$: find the triangle area.

[Source: LA W2 TA, Problem 9]

Solution. $\mathbf{AB} = (2, 1, 0)^T$, $\mathbf{AC} = (1, 0, 3)^T$; cross $= \begin{bmatrix} 1 \cdot 3 - 0 \cdot 0 \\ 0 \cdot 1 - 2 \cdot 3 \\ 2 \cdot 0 - 1 \cdot 1 \end{bmatrix} = \begin{bmatrix} 3 \\ -6 \\ -1 \end{bmatrix}$, norm $= \sqrt{46}$;
area $= \frac{\sqrt{46}}{2}$.

9. For $M = \begin{bmatrix} 1 & 0 & -2 \\ 4 & 3 & 5 \end{bmatrix}$: size, m_{21} , all columns.

[Source: LA W2 TA, Problem 10]

Solution. 2×3 ; $m_{21} = 4$; columns $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 3 \end{bmatrix}$, $\begin{bmatrix} -2 \\ 5 \end{bmatrix}$.

10. Let $A = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 4 \\ 2 & -2 \end{bmatrix}$. Find $3A + 2B$.

[Source: LA W2 TA, Problem 11]

Solution. $3A = \begin{bmatrix} 6 & 3 \\ -3 & 9 \end{bmatrix}$, $2B = \begin{bmatrix} 0 & 8 \\ 4 & -4 \end{bmatrix}$; sum $= \begin{bmatrix} 6 & 11 \\ 1 & 5 \end{bmatrix}$.

11. Let $A = \begin{bmatrix} 2 & -1 & 3 \\ 0 & 4 & 1 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$. Compute $A\mathbf{x}$ as a combination of columns.

[Source: LA W2 TA, Problem 12]

Solution. $1 \begin{bmatrix} 2 \\ 0 \end{bmatrix} - 2 \begin{bmatrix} -1 \\ 4 \end{bmatrix} + 3 \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 + 2 + 9 \\ 0 - 8 + 3 \end{bmatrix} = \begin{bmatrix} 13 \\ -5 \end{bmatrix}.$

12. Let $\mathbf{a}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\mathbf{a}_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$. Find all $\mathbf{x}$ with $[\mathbf{a}_1 \ \mathbf{a}_2]\mathbf{x} = \mathbf{b}$.

[Source: LA W2 TA, Problem 13]

Solution. Both equations reduce to $x_1 + 2x_2 = 3$ (the system is consistent but redundant since $\mathbf{a}_2 = 2\mathbf{a}_1$ and $\mathbf{b} = 3\mathbf{a}_1$). General solution: $\mathbf{x} = \begin{bmatrix} 3 - 2t \\ t \end{bmatrix}$, $t \in \mathbb{R}$ — infinitely many weights for the same combination.

13. Without expanding coordinates, explain why $(\mathbf{u} \times \mathbf{v}) \cdot (\mathbf{u} + \mathbf{v}) = 0$.

[Source: LA W2 TA, Problem 14]

Solution. The cross product is orthogonal to both of its inputs, so it is orthogonal to any combination of them: $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{u} = 0$ and $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{v} = 0$; adding, the dot product with $\mathbf{u} + \mathbf{v}$ is $0 + 0 = 0$.

14. A is 2×2 with $A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, $A \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$. Find A and $A \begin{bmatrix} 5 \\ -2 \end{bmatrix}$.

[Source: LA W2 TA, Problem 15]

Solution. The images of the standard basis vectors are the columns: $A = \begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix}$. Then $A \begin{bmatrix} 5 \\ -2 \end{bmatrix} = 5 \begin{bmatrix} 2 \\ -1 \end{bmatrix} - 2 \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 4 \\ -13 \end{bmatrix}$; linearity turns the computation into two scalings and no dot products.

4 Week 3 — Limits (Math I); Matrix Algebra, Determinants, Triple Product (Linear Algebra I)

4.1 Math I: Equation catalogue for Week 3

4.1.1 Rates of change and the two founding problems

Limits enter the course through two questions: how fast is a quantity changing (tangent/velocity), and how do function values behave near a point. The average rate over a shrinking interval defines the instantaneous one.

$$m_{\text{sec}} = \frac{f(x) - f(a)}{x - a} \quad (\text{slope of the secant through } (a, f(a))), \quad (51)$$

$$v_{\text{ave}} = \frac{s(t) - s(a)}{t - a} \quad (\text{average velocity on } [a, t]), \quad (52)$$

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = f'(a) \quad (\text{instantaneous rate; defined as a limit}) \quad (53)$$

4.1.2 The limit concept, one-sided and infinite limits

The intuitive definition (OpenStax §2.2) and its one-sided variants are the operational heart of the week; the two-sided limit exists exactly when the two one-sided limits agree.

$$\lim_{x \rightarrow a} f(x) = L \iff \text{all values } f(x) \text{ } (x \neq a) \text{ approach } L \text{ as } x \rightarrow a, \quad (54)$$

$$\lim_{x \rightarrow a^-} f(x) = L, \quad \lim_{x \rightarrow a^+} f(x) = L \quad (\text{left/right limits}), \quad (55)$$

$$\lim_{x \rightarrow a} f(x) = L \iff \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = L \quad (\text{existence criterion}), \quad (56)$$

$$\lim_{x \rightarrow a^-} f(x) = \pm\infty, \quad \lim_{x \rightarrow a^+} f(x) = \pm\infty, \quad \lim_{x \rightarrow a} f(x) = \pm\infty \quad (\text{infinite limits}), \quad (57)$$

$$\lim_{x \rightarrow a^-} \frac{1}{(x-a)^n} = -\infty, \quad \lim_{x \rightarrow a^+} \frac{1}{(x-a)^n} = +\infty \quad (n \text{ even}); \quad (58)$$

$$\lim_{x \rightarrow a^-} \frac{1}{(x-a)^n} = -\infty, \quad \lim_{x \rightarrow a^+} \frac{1}{(x-a)^n} = +\infty \quad (n \text{ odd}) : x = a \text{ vertical asymptote.} \quad (59)$$

The line $x = a$ is a *vertical asymptote* of f if at least one of the one-sided infinite limits (57) holds there.

4.1.3 The precise definition (§2.5, lecture emphasis)

The ε - δ definition converts “arbitrarily close” into quantified distances; the equivalent absolute-value inequalities are the working tool.

$$\lim_{x \rightarrow a} f(x) = L \iff \forall \varepsilon > 0 \exists \delta > 0 : 0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon, \quad (60)$$

$$|f(x) - L| < \varepsilon \iff L - \varepsilon < f(x) < L + \varepsilon, \quad (61)$$

$$0 < |x - a| < \delta \iff a - \delta < x < a + \delta \text{ and } x \neq a. \quad (62)$$

4.1.4 Two standard limits seen this week

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \quad (\text{Table 2.2 of the lecture: values } 0.9983, 0.99998, 0.9999998 \dots), \quad (63)$$

$$\lim_{x \rightarrow 0} \sin \frac{1}{x} \text{ does not exist (oscillation),} \quad \lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty, \quad \lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty. \quad (64)$$

4.2 Math I: Week 3 ISA — questions and full solutions

1. Study all problems, exercises and examples presented in Lecture and Practical Session; be able to answer fast and without mistakes.

[Source: Week 3 ISA, Assignment 2]

Solution. This is a study instruction rather than a computation; the corresponding material is fully solved in the lecture examples and in the Practical Session block below (Exercises 2–8) and in the assigned textbook problems (following items).

2. For $f(x) = x^3$, estimate the tangent slope at $x = 1$ using $m_{\text{sec}} = \frac{f(1+h)-f(1)}{h}$ for $h = -0.1, -0.01, -0.001, 0.001, 0.01, 0.1$. Do the estimates from the left and right approach the same number?

[Source: Week 3 ISA, Assignment 3]

Solution. Here $f(1) = 1$ and $f(1+h) = (1+h)^3 = 1 + 3h + 3h^2 + h^3$, so $m_{\text{sec}} = 3 + 3h + h^2$. Values: $h = -0.1 \Rightarrow 2.71$; $-0.01 \Rightarrow 2.9701$; $-0.001 \Rightarrow 2.997$; $0.001 \Rightarrow 3.003$; $0.01 \Rightarrow 3.0301$; $0.1 \Rightarrow 3.31$. Both sides approach 3; indeed $\lim_{h \rightarrow 0} (3 + 3h + h^2) = 3$, the exact tangent slope (the derivative $3x^2$ at $x = 1$).

3. Solve problems 46–49 and 60–64 of “Chapter 1” (Section 2.2 exercise list) in OpenStax Calculus 1.

[Source: Week 3 ISA, Assignment 4]

Solution. These graph-based exercises are stated and fully solved below in §4.4; the numbering in the course sheet refers to that exercise list.

4. A temperature controller outputs $C(T) = 0$ for $T < 50$ and $C(T) = 10$ for $T \geq 50$ (volts). Determine $\lim_{T \rightarrow 50^-} C(T)$, $\lim_{T \rightarrow 50^+} C(T)$, $\lim_{T \rightarrow 50} C(T)$; interpret as a switching operation.

[Source: Week 3 ISA, Assignment 5]

Solution. From the left the controller reads the “off” branch: $\lim_{T \rightarrow 50^-} C(T) = 0$. From the right it reads the “on” branch: $\lim_{T \rightarrow 50^+} C(T) = 10$. Since the two one-sided limits differ, the two-sided limit does not exist (56). Interpretation: an abrupt switching event at $T = 50$ — the output jumps from 0 V to 10 V, which is exactly how a thermostat commands a heater.

5. Position of a robotic cart: $s(t) = 2t^2 + 3t$ (m, s). (1) Average velocity $\frac{s(2+h)-s(2)}{h}$ for $h = 1, 0.1, 0.01, -0.01$; (2) simplify the expression algebraically; (3) evaluate $\lim_{h \rightarrow 0} \frac{s(2+h)-s(2)}{h}$; (4) interpret.

[Source: Week 3 ISA, Assignment 6]

Solution. (1) $s(2) = 14$. Values: $h = 1$: $\frac{s(3)-14}{1} = \frac{27-14}{1} = 13$; $h = 0.1$: $\frac{14.72-14}{0.1} = 7.2$; $h = 0.01$: $\frac{14.0702-14}{0.01} = 7.02$; $h = -0.01$: $\frac{13.9302-14}{-0.01} = 6.98$. (2) $s(2+h) = 2(2+h)^2 + 3(2+h) = 8 + 8h + 2h^2 + 6 + 3h = 14 + 11h + 2h^2$, so $\frac{s(2+h)-s(2)}{h} = \frac{11h+2h^2}{h} = 11 + 2h$. (3) $\lim_{h \rightarrow 0}(11 + 2h) = 11$ m/s. (4) The instantaneous velocity of the cart at $t = 2$ s is 11 m/s; the averages approach it from both sides.

6. Two systems have outputs $A(t) = (1.08)^t$ and $B(t) = e^{0.03t}$. (1) Evaluate $\lim_{t \rightarrow \infty} A(t)$ and $\lim_{t \rightarrow \infty} B(t)$; (2) rewrite $A(t)$ as e^{kt} ; (3) which grows faster? (4) evaluate $\lim_{t \rightarrow \infty} \frac{B(t)}{A(t)}$; (5) interpret.

[Source: Week 3 ISA, Assignment 7]

Solution. (1) Both bases exceed 1, so both limits are $+\infty$. (2) $A(t) = e^{t \ln 1.08}$, and $\ln 1.08 \approx 0.0770$, so $A(t) = e^{0.0770t}$. (3) A grows faster because its rate constant $0.0770 > 0.03$. (4) $\frac{B(t)}{A(t)} = e^{(0.03-0.0770)t} = e^{-0.047t} \rightarrow 0$ as $t \rightarrow \infty$. (5) Even though both diverge, their ratio tends to zero: B becomes negligible next to A — the seed of the “little- o ” comparison of Week 4.

4.3 Math I: Week 3 Practical Session (engineering limits) — questions and full solutions

Course conventions for this session: show the mathematical procedure; include units; interpret in engineering terms; state when a limit does not exist.

1. Solve problems 50–54 of the Section 2.2 exercise list of the textbook.

[Source: Week 3 Practical, Exercise 1]

Solution. Stated and fully solved in §4.4 below (items 50–54).

2. One-sided limits: $f(x) = 2x + 1$ for $x < 2$, $f(x) = 7 - x$ for $x \geq 2$. Find $\lim_{x \rightarrow 2^-} f$, $\lim_{x \rightarrow 2^+} f$, $\lim_{x \rightarrow 2} f$, $f(2)$; explain whether the two-sided limit exists.

[Source: Week 3 Practical, Exercise 2]

Solution. Left branch: $\lim_{x \rightarrow 2^-} (2x + 1) = 5$. Right branch: $\lim_{x \rightarrow 2^+} (7 - x) = 5$. They are equal, so by (56) $\lim_{x \rightarrow 2} f(x) = 5$. Note $f(2) = 7 - 2 = 5$ as well — the function value agrees with the limit, so the graph has no break at $x = 2$.

3. Calibration near a missing measurement: $V(T) = \frac{T^2 - 2500}{T - 50}$ mV, $T \neq 50$ (T in °C). (1) Why is $V(50)$ undefined? (2) Estimate $\lim_{T \rightarrow 50} V(T)$; (3) how can the limit replace the missing calibration value?

[Source: Week 3 Practical, Exercise 3]

Solution. (1) The formula divides by $T - 50$, which is 0 at $T = 50$; the sensor model simply has no value there. (2) Factor the difference of squares: $T^2 - 2500 = (T - 50)(T + 50)$, so for $T \neq 50$, $V(T) = T + 50$, and $\lim_{T \rightarrow 50} V(T) = 100$ mV. (3) The limit describes what the sensor *would* read at 50°C; filling the calibration table with 100 mV at that point is justified because the output approaches that value from both sides — this is “removable” behaviour, not a jump.

4. A switched control system: $V(t) = 2t + 1$ for $t < 4$ and $V(t) = 12 - t$ for $t \geq 4$ (V). Determine the one-sided limits, the two-sided limit and $V(4)$; is there an abrupt voltage change at $t = 4$?

[Source: Week 3 Practical, Exercise 4]

Solution. Left: $\lim_{t \rightarrow 4^-} (2t + 1) = 9$ V. Right: $\lim_{t \rightarrow 4^+} (12 - t) = 8$ V. $V(4) = 12 - 4 = 8$ V. The one-sided limits differ, so $\lim_{t \rightarrow 4} V(t)$ does not exist: at $t = 4$ the supply drops abruptly from 9 V to 8 V. Engineering meaning: a 1-V step can excite transients in the controlled device; input filters or soft-start logic are used when the step is harmful.

5. Stress–strain: $\sigma(\varepsilon) = 200000\varepsilon$ for $\varepsilon < 0.002$ and $\sigma(\varepsilon) = 400 + 10000(\varepsilon - 0.002)$ for $\varepsilon \geq 0.002$ (σ in MPa; Hooke’s region then hardening). Find the one-sided limits at $\varepsilon = 0.002$; does the two-sided limit exist? Does the model predict a jump? Sketch the graph near the transition.

[Source: Week 3 Practical, Exercise 5]

Solution. Elastic side: $\lim_{\varepsilon \rightarrow 0.002^-} 200000\varepsilon = 200000(0.002) = 400$ MPa. Plastic side: $\lim_{\varepsilon \rightarrow 0.002^+} (400 + 10000(\varepsilon - 0.002)) = 400$ MPa. Both equal 400 MPa, so the two-sided limit exists and equals 400 MPa; no jump occurs — the piecewise model is continuous at the yield point, changing only its slope (from 200 000 to 10 000 MPa per unit strain). The graph is two straight segments meeting at (0.002, 400) with a pronounced kink: elastic loading, then a much flatter plastic branch.

6. Long-term RC response: $V(t) = V_s(1 - e^{-t/(RC)})$ with $V_s = 12$ V, $R = 10000 \Omega$, $C = 100 \mu\text{F}$. (1) Evaluate $\lim_{t \rightarrow 0^+} V(t)$; (2) $\lim_{t \rightarrow \infty} V(t)$; (3) interpret both physically.

[Source: Week 3 Practical, Exercise 6]

Solution. (1) At $t = 0^+$ the exponential is $e^0 = 1$, so $V(0^+) = 12(1 - 1) = 0$ V — the initially uncharged capacitor. (2) As $t \rightarrow \infty$, $e^{-t/(RC)} \rightarrow 0$, so $V \rightarrow V_s = 12$ V. (3) The capacitor charges from 0 V toward the supply voltage 12 V, approaching it asymptotically; the time constant $\tau = RC = 10^4 \cdot 10^{-4} = 1$ s sets the pace (about 63% charged after 1 s, 99% after 5τ). The limit language “ $V \rightarrow V_s$ as $t \rightarrow \infty$ ” is exactly how asymptotic charging is stated.

7. Terminal velocity: $v(t) = v_T(1 - e^{-kt})$, $v_T > 0$, $k > 0$. (1) $v(0)$? (2) $\lim_{t \rightarrow \infty} v(t)$? (3) Why is this model more realistic than indefinite acceleration?

[Source: Week 3 Practical, Exercise 7]

Solution. (1) $v(0) = v_T(1 - 1) = 0$: the object starts at rest. (2) $e^{-kt} \rightarrow 0$, so $v \rightarrow v_T$: the velocity approaches the terminal velocity. (3) Air drag grows with speed until it balances gravity; after that the net force is zero and the speed can no longer increase — a falling object really does approach a finite limit rather than accelerating forever.

8. Data-transfer rate: $R(n) = \frac{100n}{n+20}$ Mbit/s. (1) Compute $R(10), R(100), R(1000)$; (2) $\lim_{n \rightarrow \infty} R(n)$; (3) why do increases of n eventually produce diminishing improvements? (4) the maximum theoretical rate?

[Source: Week 3 Practical, Exercise 8]

Solution. (1) $R(10) \approx 33.3$, $R(100) \approx 83.3$, $R(1000) \approx 98.0$ Mbit/s. (2) Dividing by n : $R(n) = \frac{100}{1+20/n} \rightarrow 100$ Mbit/s. (3) Each extra unit of the resource is shared with an ever larger base, so the marginal gain shrinks — diminishing returns. (4) The horizontal asymptote 100 Mbit/s is the model's ceiling, approached but never exceeded.

4.4 Math I: assigned textbook exercises (OpenStax §2.2, items 46–54 and 60–64)

These graph-reading exercises use the four figures printed with the exercise set; the needed values are read from the graphs, as one does in class. Exercise statements first, then solutions.

1. Given the first graph of $y = f(x)$, decide: (46) $\lim_{x \rightarrow 10^-} f(x) = 0$; (47) $\lim_{x \rightarrow -2^+} f(x) = 3$; (48) $\lim_{x \rightarrow -8} f(x) = f(-8)$; (49) $\lim_{x \rightarrow 6} f(x) = 5$ — true or false?

[Source: OpenStax §2.2, Exercises 46–49]

Solution. (46) *True*: as x approaches 10 from the left the curve descends into the filled point $(10, 0)$, so the left limit is 0. (47) *True*: just to the right of -2 the graph sits at height 3 (filled point at $(-2, 3)$). (48) *False*: near $x = -8$ the curve approaches the open circle at height -6 , so $\lim_{x \rightarrow -8} f(x) = -6$, while $f(-8) = -3$ (filled point); a limit and a value may disagree. (49) *False*: from the left the curve falls toward the open circle at $(6, 2)$, so $\lim_{x \rightarrow 6^-} = 2$, while the right branch leaves the filled point $(6, 5)$, so $\lim_{x \rightarrow 6^+} = 5$; one-sided limits differ $\Rightarrow$ two-sided limit does not exist, so it cannot equal 5.

2. For the piecewise-linear graph (line $y = x + 1$ up to the open point $(1, 2)$; filled point $(1, 1)$; line $y = 3 - x$ after), find (50) $\lim_{x \rightarrow 1^-} f$; (51) $\lim_{x \rightarrow 1^+} f$; (52) $\lim_{x \rightarrow 1} f$; (53) $\lim_{x \rightarrow 2} f$; (54) $f(1)$.

[Source: OpenStax §2.2, Exercises 50–54]

Solution. (50) Left branch $y = x + 1 \rightarrow 2$. (51) Right branch $y = 3 - x \rightarrow 2$. (52) Both sides equal 2, so $\lim_{x \rightarrow 1} f(x) = 2$ — even though $f(1) = 1$; the limit ignores the point itself. (53) Only the right branch lives at $x = 2$: $3 - 2 = 1$. (54) $f(1) = 1$ (the filled dot).

3. For the third graph (left: line falling to the open point $(0, 1)$; right: curve rising from the filled point $(0, -4)$, crossing zero near $x = 2$), find (55) $\lim_{x \rightarrow 0^-} f$; (56) $\lim_{x \rightarrow 0^+} f$; (57) $\lim_{x \rightarrow 0} f$; (58) $\lim_{x \rightarrow 2} f$.

[Source: OpenStax §2.2, Exercises 55–58]

Solution. (55) 1. (56) -4 . (57) $1 \neq -4$, so the two-sided limit *does not exist* (56). (58) The right branch passes through the axis at $x = 2$, so the limit is 0.

4. For the fourth graph (three pieces: rising line to the open point $(-2, 0)$; falling segment from the filled point $(-2, 2)$ through the origin to $(2, -2)$; rising line from the open point $(2, 2)$), find (59) $\lim_{x \rightarrow -2^-} f$; (60) $\lim_{x \rightarrow -2^+} f$; (61) $\lim_{x \rightarrow -2} f$; (62) $\lim_{x \rightarrow 2^-} f$; (63) $\lim_{x \rightarrow 2^+} f$; (64) $\lim_{x \rightarrow 2} f$.

[Source: OpenStax §2.2, Exercises 59–64]

Solution. (59) 0. (60) 2. (61) $0 \neq 2$: does not exist. (62) -2 . (63) 2. (64) $-2 \neq 2$: does not exist. Both junctions are genuine jumps: no amount of averaging repairs unequal one-sided limits.

4.5 Linear Algebra I: equation catalogue for Week 3

$$(A + B)_{ij} = a_{ij} + b_{ij}, \quad (cA)_{ij} = c a_{ij}, \quad (A^T)_{ij} = a_{ji}, \quad (A^T)^T = A, \quad (65)$$

$$(A + B)^T = A^T + B^T, \quad (cA)^T = cA^T, \quad A^T = A \text{ (symmetric)}, \quad (66)$$

$$A \in \mathbb{R}^{m \times n}, \quad B \in \mathbb{R}^{n \times p} \implies AB \in \mathbb{R}^{m \times p}, \quad (67)$$

$$(AB)_{ij} = \sum_{k=1}^n a_{ik} b_{kj}, \quad (68)$$

$$A(BC) = (AB)C, \quad A(B + C) = AB + AC, \quad (A + B)C = AC + BC, \quad (69)$$

$$(AB)^T = B^T A^T, \quad I_m A = A = A I_n, \quad (70)$$

$$AB = [A\mathbf{b}_1 \cdots A\mathbf{b}_p] \quad (\text{column rule}), \quad (71)$$

$$\mathbf{u}^T \mathbf{v} = \text{scalar}, \quad \mathbf{u} \mathbf{v}^T = \text{matrix (outer product)}. \quad (72)$$

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc \quad (\text{signed area of the column parallelogram}), \quad (73)$$

$$C_{ij} = (-1)^{i+j} \det M_{ij} \quad (\text{cofactor}), \quad \det A = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}, \quad (74)$$

where M_{ij} is the *minor* (delete row i , column j). Expanding along any row or column is allowed; rows or columns with zeros reduce the arithmetic. For triangular matrices the determinant is the product of the diagonal entries. The elementary row-operation rules are collected in Table 1.

Table 1: Determinant rules for square matrices (row versions; column versions hold too).

Operation on A	Effect on $\det A$
Exchange two rows	reverses the sign
Multiply one row by c	multiplies the determinant by c
Add a multiple of one row to another	no change
A zero row, or two proportional rows	determinant is 0
Transpose	$\det(A^T) = \det A$
Product of equally sized square matrices	$\det(AB) = \det A \det B$

Scaling the *whole* $n \times n$ matrix scales all n rows: $\det(cA) = c^n \det A$. Determinants do not distribute over addition: for $A = B = I_2$, $\det(A + B) = 4$ while $\det A + \det B = 2$.

$$(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = \det[\mathbf{u} \ \mathbf{v} \ \mathbf{w}] \quad (\text{scalar triple product}), \quad (75)$$

$$V_{\text{parallelepiped}} = |(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}|, \quad h = \frac{|(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}|}{\|\mathbf{u} \times \mathbf{v}\|}, \quad (76)$$

$$V_{\text{tetrahedron}} = \frac{1}{6} |\det[\mathbf{u} \ \mathbf{v} \ \mathbf{w}]|, \quad T = 0 \iff \mathbf{u}, \mathbf{v}, \mathbf{w} \text{ coplanar}. \quad (77)$$

4.6 Linear Algebra I: Week 3 lecture problems (15) — questions and full solutions

1. Let $A = \begin{bmatrix} 1 & -2 \\ 3 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ -1 & 4 \end{bmatrix}$. Compute $2A - B$ and identify its $(2, 1)$ entry; can A be added to a 2×3 matrix?

[Source: LA W3 Lecture, Problem 1]

Solution. $2A = \begin{bmatrix} 2 & -4 \\ 6 & 0 \end{bmatrix}$; $2A - B = \begin{bmatrix} 0 & -5 \\ 7 & -4 \end{bmatrix}$; the $(2, 1)$ entry is $2(3) - (-1) = 7$. Addition to a 2×3 matrix is undefined — *both* dimensions must match, not just the total number of entries.

2. Let $C = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 2 \end{bmatrix}$. Write C^T with its size; compute CC^T ; is it symmetric?

[Source: LA W3 Lecture, Problem 2]

Solution. $C^T = \begin{bmatrix} 1 & 0 \\ 2 & 3 \\ -1 & 2 \end{bmatrix}$ (3×2). $CC^T = \begin{bmatrix} 1+4+1 & 0+6-2 \\ 0+6-2 & 0+9+4 \end{bmatrix} = \begin{bmatrix} 6 & 4 \\ 4 & 13 \end{bmatrix}$. Diagonal entries are rows dotted with themselves; off-diagonal are row dot products. Always symmetric since $(CC^T)^T = (C^T)^T C^T = CC^T$.

3. For C as above and $\mathbf{x} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$, compute $C\mathbf{x}$ both ways.

[Source: LA W3 Lecture, Problem 3]

Solution. Rows: $1 \cdot 2 + 2 \cdot (-1) + (-1) \cdot 1 = -1$; $0 + 3(-1) + 2 = -1$; so $C\mathbf{x} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$. Columns:

$2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 2 \\ 3 \end{bmatrix} + \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$. Sizes $(2 \times 3)(3 \times 1)$ give a 2×1 output.

4. With A, B as in Problem 1, compute AB and BA . Equal?

[Source: LA W3 Lecture, Problem 4]

Solution. $AB = \begin{bmatrix} 1 \cdot 2 + (-2)(-1) & 1 \cdot 1 + (-2) \cdot 4 \\ 3 \cdot 2 + 0 & 3 \cdot 1 + 0 \cdot 4 \end{bmatrix} = \begin{bmatrix} 4 & -7 \\ 6 & 3 \end{bmatrix}$; $BA = \begin{bmatrix} 5 & -4 \\ 11 & 2 \end{bmatrix}$. Different; even when both orders exist, equality is not guaranteed.

5. Let $C = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 2 \end{bmatrix}$, $D = \begin{bmatrix} 2 & 0 \\ 1 & -1 \\ 3 & 2 \end{bmatrix}$. Predict the sizes of CD and DC , then compute both.

[Source: LA W3 Lecture, Problem 5]

Solution. CD : $(2 \times 3)(3 \times 2) = 2 \times 2$; DC : $(3 \times 2)(2 \times 3) = 3 \times 3$.

$$CD = \begin{bmatrix} 2+2-3 & 0-2-2 \\ 0+3+6 & 0-3+4 \end{bmatrix} = \begin{bmatrix} 1 & -4 \\ 9 & 1 \end{bmatrix}, \quad DC = \begin{bmatrix} 2 & 4 & 0 \\ 1 & -1 & -3 \\ 3 & 12 & 1 \end{bmatrix}.$$

The products have different sizes and can never be compared.

6. Let $S = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ (shear), $H = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$ (stretch), $\mathbf{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Compute $HS\mathbf{x}$ and $SH\mathbf{x}$; describe each transformation.

[Source: LA W3 Lecture, Problem 6]

Solution. $S\mathbf{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, then $HS\mathbf{x} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$; also $H\mathbf{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, then $SH\mathbf{x} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$. In HS the shear contribution is stretched afterwards; in SH the stretch happens first and the shear adds on top. Same two operations, different order, different result: matrices do not commute.

7. For $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}$: compute $\mathbf{u}^T \mathbf{v}$ and $\mathbf{u} \mathbf{v}^T$; verify $(\mathbf{u} \mathbf{v}^T)^T = \mathbf{v} \mathbf{u}^T$.

[Source: LA W3 Lecture, Problem 7]

Solution. Inner: $\mathbf{u}^T \mathbf{v} = 2 + 0 - 3 = -1$ (a scalar). Outer: $\mathbf{u} \mathbf{v}^T = \begin{bmatrix} 2 & 0 & 3 \\ 4 & 0 & 6 \\ -2 & 0 & -3 \end{bmatrix}$ (3×3).

Transposing entrywise, $(\mathbf{u} \mathbf{v}^T)_{ij}^T = u_j v_i = v_i u_j = (\mathbf{v} \mathbf{u}^T)_{ij}$, so $(\mathbf{u} \mathbf{v}^T)^T = \mathbf{v} \mathbf{u}^T = \begin{bmatrix} 2 & 4 & -2 \\ 0 & 0 & 0 \\ 3 & 6 & -3 \end{bmatrix}$:

the displayed matrices are transposes of one another.

8. Using A, B from Problem 1, compute $(AB)^T$ and $B^T A^T$ independently; does $A^T B^T$ agree?

[Source: LA W3 Lecture, Problem 8]

Solution. $AB = \begin{bmatrix} 4 & -7 \\ 6 & 3 \end{bmatrix}$, so $(AB)^T = \begin{bmatrix} 4 & 6 \\ -7 & 3 \end{bmatrix}$. Transposes first: $A^T = \begin{bmatrix} 1 & 3 \\ -2 & 0 \end{bmatrix}$, $B^T = \begin{bmatrix} 2 & -1 \\ 1 & 4 \end{bmatrix}$; then $B^T A^T = \begin{bmatrix} 2 \cdot 1 + (-1)(-2) & 2 \cdot 3 + 0 \\ 1 \cdot 1 + 4(-2) & 1 \cdot 3 + 0 \end{bmatrix} = \begin{bmatrix} 4 & 6 \\ -7 & 3 \end{bmatrix} = (AB)^T$, confirming (70).

But $A^T B^T = \begin{bmatrix} -1 & 3 \\ -2 & -6 \end{bmatrix}$ differs: the correct identity reverses the factor order.

9. Let $E = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$. Compute $\det E$; find the parallelogram and triangle areas of its columns; is orientation preserved?

[Source: LA W3 Lecture, Problem 9]

Solution. $\det E = 3 \cdot 2 - 1 \cdot 1 = 5$. Parallelogram area = $|5| = 5$; triangle = $\frac{5}{2}$. The determinant is positive: orientation preserved (area itself is always nonnegative).

10. Compute $\det F$ for $F = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 3 & 1 \\ 1 & 0 & 2 \end{bmatrix}$ by first-row cofactors; check along the first column.

[Source: LA W3 Lecture, Problem 10]

Solution. First row: $\det F = 2 \det \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} - 1 \det \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix} + 0 = 2(6) - 1(-1) = 13$. First

column: $2 \det \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} + 1 \det \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} = 12 + 1 = 13$ (the $(3, 1)$ cofactor sign $(-1)^{3+1} = +$).

Any row or column gives the same value.

11. A 3×3 matrix M has $\det M = 5$. Form N : swap rows 1,2; multiply row 3 by 2; add $3 \times$ (new row 1) to row 2. Find $\det N$, $\det(2M)$, $\det(M^T M)$.

[Source: LA W3 Lecture, Problem 11]

Solution. Swap: sign flips (-5); scaling row 3: another factor 2; row addition: no change. $\det N = -2 \cdot 5 = -10$. Scaling *all* rows: $\det(2M) = 2^3 \cdot 5 = 40$. Product rule: $\det(M^T M) = \det M^T \det M = 25$.

12. Let $\mathbf{u} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$, $\mathbf{w} = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}$. Compute $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}$, the parallelepiped and tetrahedron volumes; what changes if $\mathbf{u}, \mathbf{v}$ are exchanged?

[Source: LA W3 Lecture, Problem 12]

Solution. $\mathbf{u} \times \mathbf{v} = \begin{bmatrix} 0 \cdot 0 - 1 \cdot 2 \\ 1 \cdot 1 - 2 \cdot 0 \\ 2 \cdot 2 - 0 \cdot 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 4 \end{bmatrix}$; dot with $\mathbf{w}$: $-2 + 1 + 12 = 13$. Parallelepiped volume 13; tetrahedron $\frac{13}{6}$. Exchanging $\mathbf{u}$ and $\mathbf{v}$ flips the sign to -13 but both *volumes* stay 13 and $\frac{13}{6}$ (absolute values).

13. For $\mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}$, $\mathbf{w} = \begin{bmatrix} 1 \\ 1 \\ k \end{bmatrix}$: find all k making the vectors coplanar; at that k express $\mathbf{w}$ via $\mathbf{u}, \mathbf{v}$.

[Source: LA W3 Lecture, Problem 13]

Solution. Coplanarity $\iff T = 0$. Here $\mathbf{u} \times \mathbf{v} = \begin{bmatrix} 0 \cdot 1 - 1 \cdot 2 \\ 1 \cdot 0 - 1 \cdot 1 \\ 1 \cdot 2 - 0 \cdot 0 \end{bmatrix} = \begin{bmatrix} -2 \\ -1 \\ 2 \end{bmatrix}$, so $T = -2 - 1 + 2k = 2k - 3 = 0 \Rightarrow k = \frac{3}{2}$. Then $\mathbf{w} = \alpha\mathbf{u} + \beta\mathbf{v}$: first two coordinates give $\alpha = 1$, $2\beta = 1 \Rightarrow \beta = \frac{1}{2}$; check the third: $1 + \frac{1}{2} = \frac{3}{2}$ — consistent, $\mathbf{w} = \mathbf{u} + \frac{1}{2}\mathbf{v}$.

14. Let $D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$, $X = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Find every X with $DX = XD$; what if both diagonal entries of D are equal?

[Source: LA W3 Lecture, Problem 14]

Solution. $DX = \begin{bmatrix} 2a & 2b \\ 3c & 3d \end{bmatrix}$, $XD = \begin{bmatrix} 2a & 3b \\ 2c & 3d \end{bmatrix}$. Equality forces $b = c = 0$ with a, d free: precisely the diagonal matrices commute with this D . If $D = \lambda I$, then $DX = \lambda X = XD$ for every X .

15. Let $\mathbf{u} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$, $\mathbf{w} = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}$ and $L = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$. Without recomputing edges, find the signed triple product and volume after applying L to every edge; why is the volume factor universal?

[Source: LA W3 Lecture, Problem 15]

Solution. Put $U = [\mathbf{u} \ \mathbf{v} \ \mathbf{w}]$; from Problem 12, $\det U = 13$. Transformed edges form LU , so $\det(LU) = \det L \det U = (-2)(13) = -26$; new volume 26. The factor is $|\det L| = 2$ *regardless* of U , because $\det(LU) = \det L \det U$ for every U — the product rule for determinants has a geometric meaning: $|\det L|$ is the volume-scaling factor of the transformation.

5 Week 4 — Limits II: Limit Laws, Standard Limits, Indeterminate Forms, Asymptotics, Landau Symbols (Math I)

Week 4 is the pivotal week before Test 1 review: it consolidates the limit machinery of Week 3 into a working toolbox. The required reading is OpenStax §2.3 (*The Limit Laws*), and the *Week 4 Notes* by Prof. Neofotistos summarise general limit properties, standard limits, indeterminate forms, asymptotic equivalence and the Landau symbols O and o . This part lists every equation from both sources, then solves every exercise of the §2.3 set (items 83–130) and every worked question of the Notes.

5.1 Equation catalogue for Week 4

5.1.1 General properties of limits (Week 4 Notes)

$$\lim_{x \rightarrow a} f(x) = L : f(x) \text{ can be made arbitrarily close to } L \text{ by taking } x \text{ close to } a \ (x \neq a); \quad (78)$$

$$\lim_{x \rightarrow a} f(x) = L \iff \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = L \quad (\text{two-sided criterion}), \quad (79)$$

a limit, when it exists, is *unique*.

5.1.2 Limit laws (OpenStax Theorem 2.5)

If $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$ (both existing), and c is a constant, then:

$$\text{Sum: } \lim_{x \rightarrow a} [f(x) + g(x)] = L + M; \quad \text{Difference: } \lim_{x \rightarrow a} [f(x) - g(x)] = L - M; \quad (80)$$

$$\text{Constant multiple: } \lim_{x \rightarrow a} c f(x) = cL; \quad \text{Product: } \lim_{x \rightarrow a} [f(x)g(x)] = LM; \quad (81)$$

$$\text{Quotient: } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{L}{M} \ (M \neq 0); \quad \text{Power: } \lim_{x \rightarrow a} [f(x)]^n = L^n \ (n \in \mathbb{N}); \quad (82)$$

$$\text{Root: } \lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{L} \ (n \text{ odd: any } L; \ n \text{ even: } L \geq 0); \quad (83)$$

$$\text{Basic: } \lim_{x \rightarrow a} x = a, \quad \lim_{x \rightarrow a} c = c. \quad (84)$$

$$\text{Polynomials/rational functions: } \lim_{x \rightarrow a} p(x) = p(a), \quad \lim_{x \rightarrow a} \frac{p(x)}{q(x)} = \frac{p(a)}{q(a)} \ (q(a) \neq 0). \quad (85)$$

If $f(x) = g(x)$ for all $x \neq a$ near a , then $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x)$ — the licence to cancel factors before substituting.

5.1.3 The Squeeze (Pinching) Theorem (OpenStax Theorem 2.7)

$$f(x) \leq g(x) \leq h(x) \text{ near } a \text{ (for } x \neq a), \quad \lim_{x \rightarrow a} f(x) = L = \lim_{x \rightarrow a} h(x) \implies \lim_{x \rightarrow a} g(x) = L. \quad (86)$$

5.1.4 Standard limits and the derived trigonometric limits (Week 4 Notes)

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}, \quad \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1, \quad (87)$$

$$\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1, \quad \lim_{x \rightarrow 0} (1+x)^{1/x} = e, \quad \lim_{x \rightarrow \infty} \frac{x^n}{e^x} = 0 \ (n \in \mathbb{N}). \quad (88)$$

Angles in the trigonometric limits must be in radians. Adapting (87) by substitution, e.g.

$$\lim_{x \rightarrow 0} \frac{\sin 5x}{x} = 5 \lim_{x \rightarrow 0} \frac{\sin 5x}{5x} = 5, \quad \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1 \implies \lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta} = 0 \text{ (via conjugate)}. \quad (89)$$

5.1.5 Indeterminate forms (Week 4 Notes)

$$\frac{0}{0}, \quad \frac{\infty}{\infty}, \quad 0 \cdot \infty, \quad \infty - \infty, \quad 0^0, \quad 1^\infty, \quad \infty^0 \quad (90)$$

The word “indeterminate” does *not* mean that the limit fails to exist; it means that more analysis is required. The same initial form can resolve differently:

$$\lim_{x \rightarrow 0} \frac{x}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{x^2}{x} = 0, \quad \lim_{x \rightarrow 0} \frac{x}{x^2} \text{ does not exist (finite)}, \quad (91)$$

while $\lim_{x \rightarrow 1} \frac{x^2-1}{x-1} = 2$ after factoring. Resolution tools: factoring, rationalisation, standard limits, asymptotic equivalence, change of variable, algebraic rearrangement, the Squeeze Theorem, Taylor expansions, and (later) L'Hôpital's rule.

5.1.6 Asymptotic equivalence (Week 4 Notes)

$$f(x) \sim g(x) \quad (x \rightarrow a) \iff \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = 1, \quad (92)$$

$$f(x) \sim g(x) \iff f(x) = g(x)(1 + \varepsilon(x)), \quad \varepsilon(x) \rightarrow 0, \quad (93)$$

$$f \sim g \iff f = g + o(g). \quad (94)$$

$$\text{As } x \rightarrow 0: \quad \sin x \sim x, \quad \tan x \sim x, \quad 1 - \cos x \sim \frac{x^2}{2}, \quad e^x - 1 \sim x, \quad (95)$$

$$\ln(1+x) \sim x, \quad (1+x)^\alpha - 1 \sim \alpha x \quad (\alpha \text{ constant}). \quad (96)$$

Numerically, $\frac{\sin(0.01)}{0.01} \approx 0.999983$ illustrates “close, not equal”. At infinity the highest-degree term dominates:

$$3x^2 + 5x - 1 \sim 3x^2 \quad (x \rightarrow \infty), \quad \frac{2x^3 + x}{5x^2 - 1} \sim \frac{2x^3}{5x^2} = \frac{2}{5}x \quad (x \rightarrow \infty). \quad (97)$$

Equivalences may be *multiplied and divided*, not generally added.

5.1.7 Landau symbols: big- O and little- o (Week 4 Notes)

$$f(x) = O(g(x)) \quad (x \rightarrow a) \iff \frac{|f(x)|}{|g(x)|} \text{ remains bounded near } a, \quad (98)$$

$$f(x) = o(g(x)) \quad (x \rightarrow a) \iff \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = 0, \quad (99)$$

$$f = o(g) \implies f = O(g); \quad f = O(g) \not\implies f = o(g). \quad (100)$$

$$x^2 = o(x), \quad x^3 = o(x^2) \quad (x \rightarrow 0); \quad (101)$$

$$x = o(x^2), \quad x^{10} = o(e^x) \quad (x \rightarrow \infty). \quad (102)$$

$$3x = O(x) \text{ but } 3x \neq o(x) \quad (x \rightarrow 0) \quad \text{since } \frac{3x}{x} = 3. \quad (103)$$

$$\sin x = x + o(x) \quad (x \rightarrow 0); \quad e^x = 1 + x + o(x); \quad e^x = 1 + x + \frac{x^2}{2} + o(x^2). \quad (104)$$

$$2x^2 + 3x + 1 = 2x^2 + O(x) \quad (x \rightarrow \infty). \quad (105)$$

In words: big- O means f has *at most* the order of magnitude of g ; little- o means f becomes *negligible* compared with g . Caution: $f = O(g)$ is not an equation in the ordinary sense — $O(g)$ denotes a whole class of functions.

5.2 OpenStax §2.3, Checkpoint problems 2.11–2.20 — questions and full solutions

1. Use the limit laws to evaluate $\lim_{x \rightarrow -3} (2x - 1)\sqrt{x + 4}$, indicating the law used at each step.
[Source: OpenStax §2.3, Checkpoint 2.11]

Solution. Both factor limits exist, so the product law applies: $\lim(2x - 1) = 2(-3) - 1 = -7$ (sum, difference, basic limits); $\lim \sqrt{x + 4} = \sqrt{-3 + 4} = 1$ (root law). Product: $-7 \cdot 1 = -7$.

2. Evaluate $\lim_{x \rightarrow -2} (3x^3 - 2x + 7)$.

[Source: Checkpoint 2.12]

Solution. Direct substitution is licensed by (85): $3(-8) - 2(-2) + 7 = -24 + 4 + 7 = -13$.

3. Evaluate $\lim_{x \rightarrow -3} \frac{x^2 + 4x + 3}{x^2 - 9}$.

[Source: Checkpoint 2.13]

Solution. Form 0/0. Factor both: $\frac{(x+1)(x+3)}{(x+3)(x-3)} = \frac{x+1}{x-3}$ for $x \neq -3$. Limit: $\frac{-1-1}{-3-3} = \frac{-2}{-6} = \frac{1}{3}$.

4. Evaluate $\lim_{x \rightarrow 5} \frac{\sqrt{x-1}-2}{x-5}$.

[Source: Checkpoint 2.14]

Solution. Form 0/0. Multiply by the conjugate $\sqrt{x-1}+2$:

$$\frac{(x-1)-4}{(x-5)(\sqrt{x-1}+2)} = \frac{x-5}{(x-5)(\sqrt{x-1}+2)} = \frac{1}{\sqrt{x-1}+2} \quad (x \neq 5),$$

so the limit is $\frac{1}{\sqrt{4+2}} = \frac{1}{4}$.

5. Evaluate $\lim_{x \rightarrow -3} \frac{\frac{1}{x+2} + 1}{x+3}$.

[Source: Checkpoint 2.15]

Solution. Simplify the complex fraction: $\frac{1}{x+2} + 1 = \frac{x+3}{x+2}$, hence the expression equals $\frac{x+3}{(x+2)(x+3)} = \frac{1}{x+2}$ for $x \neq -3$. Limit: $\frac{1}{-3+2} = -1$.

6. Evaluate $\lim_{x \rightarrow 3} \left(\frac{1}{x-3} - \frac{4}{x^2 - 2x - 3} \right)$.

[Source: Checkpoint 2.16]

Solution. Factor $x^2 - 2x - 3 = (x-3)(x+1)$ and combine:

$$\frac{1}{x-3} - \frac{4}{(x-3)(x+1)} = \frac{(x+1)-4}{(x-3)(x+1)} = \frac{x-3}{(x-3)(x+1)} = \frac{1}{x+1} \quad (x \neq 3),$$

so the limit is $\frac{1}{4}$.

7. Piecewise checkpoint: $f(x) = \begin{cases} x^2 & x \leq 3 \\ x+4 & x > 3 \end{cases}$; evaluate the one-sided limits at the junction.

[Source: Checkpoint 2.17]

Solution. From the left the parabola branch gives $\lim_{x \rightarrow 3^-} f(x) = 9$; from the right the line gives $\lim_{x \rightarrow 3^+} (x+4) = 7$. Unequal, so $\lim_{x \rightarrow 3} f(x)$ does not exist.

8. Evaluate $\lim_{x \rightarrow 1} \frac{x+2}{(x-1)^2}$ (Example 2.23 method).

[Source: Checkpoint 2.18]

Solution. Numerator $\rightarrow 3$ (nonzero), denominator $\rightarrow 0^+$ from both sides (a square), so the ratio $\rightarrow +\infty$: an infinite limit, and $x = 1$ is a vertical asymptote.

9. Use the squeeze theorem to evaluate $\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x}$.

[Source: Checkpoint 2.19]

Solution. Since $-1 \leq \sin \frac{1}{x} \leq 1$, multiplying by $x^2 \geq 0$ gives $-x^2 \leq x^2 \sin \frac{1}{x} \leq x^2$. Both bounds $\rightarrow 0$; by (86) the limit is 0.

10. Evaluate $\lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\sin \theta}$.

[Source: Checkpoint 2.20]

Solution. Divide numerator and denominator by θ (legal since $\theta \neq 0$ inside the limit):

$$\frac{1 - \cos \theta}{\sin \theta} = \frac{\frac{1 - \cos \theta}{\theta}}{\frac{\sin \theta}{\theta}} \rightarrow \frac{0}{1} = 0,$$

using $\frac{1 - \cos \theta}{\theta} \rightarrow 0$ (Example 2.25) and $\frac{\sin \theta}{\theta} \rightarrow 1$ (standard limit).

5.3 OpenStax §2.3 Exercises 83–130 — questions and full solutions

5.3.1 Using the limit laws (83–86)

1. $\lim_{x \rightarrow 0} (4x^2 - 2x + 3)$.

[Source: §2.3, Ex. 83]

Solution. Substitution by (85): $4(0)^2 - 2(0) + 3 = 3$.

2. $\lim_{x \rightarrow 1} \frac{x^3 + 3x^2 + 5}{4 - 7x}$.

[Source: §2.3, Ex. 84]

Solution. Denominator at 1 is $-3 \neq 0$, so the quotient law and substitution give $\frac{1+3+5}{-3} = \frac{9}{-3} = -3$.

3. $\lim_{x \rightarrow -2} \sqrt{x^2 - 6x + 3}$.

[Source: §2.3, Ex. 85]

Solution. The radicand is continuous at -2 and positive there; root law: $\sqrt{4 + 12 + 3} = \sqrt{19}$.

4. $\lim_{x \rightarrow -1} (9x + 1)^2$.

[Source: §2.3, Ex. 86]

Solution. Power law then substitution: $(9(-1) + 1)^2 = (-8)^2 = 64$.

5.3.2 Direct substitution (87–92)

1. $\lim_{x \rightarrow 7} x^2$.

[Source: §2.3, Ex. 87]

Solution. 49.

2. $\lim_{x \rightarrow -2} (4x^2 - 1)$.

[Source: §2.3, Ex. 88]

Solution. $4(4) - 1 = 15$.

3. $\lim_{x \rightarrow 0} \frac{1}{1 + \sin x}$.

[Source: §2.3, Ex. 89]

Solution. Denominator $1 + \sin 0 = 1 \neq 0$: quotient law gives $\frac{1}{1} = 1$.

4. $\lim_{x \rightarrow 2} e^{2x - x^2}$.

[Source: §2.3, Ex. 90]

Solution. Exponent: $2(2) - 4 = 0$; $e^0 = 1$.

5. $\lim_{x \rightarrow 1} \frac{2+7x}{x+6}.$

[Source: §2.3, Ex. 91]

Solution. $\frac{2+7}{1+6} = \frac{9}{7}.$

6. $\lim_{x \rightarrow 3} \ln e^{3x}.$

[Source: §2.3, Ex. 92]

Solution. $\ln e^{3x} = 3x$ identically (log and exponential are inverse), so the limit is $3 \cdot 3 = 9$.

5.3.3 Indeterminate form 0/0 (93–102)

1. $\lim_{x \rightarrow 4} \frac{x^2-16}{x-4}.$

[Source: §2.3, Ex. 93]

Solution. Form 0/0. Factor: $\frac{(x-4)(x+4)}{x-4} = x+4$ for $x \neq 4$; limit = 8.

2. $\lim_{x \rightarrow 2} \frac{x-2}{x^2-2x}.$

[Source: §2.3, Ex. 94]

Solution. Factor denominator: $\frac{x-2}{x(x-2)} = \frac{1}{x}$ for $x \neq 2$; limit = $\frac{1}{2}$.

3. $\lim_{x \rightarrow 6} \frac{3x-18}{2x-12}.$

[Source: §2.3, Ex. 95]

Solution. $\frac{3(x-6)}{2(x-6)} = \frac{3}{2}$ for $x \neq 6$; limit = $\frac{3}{2}$.

4. $\lim_{h \rightarrow 0} \frac{(1+h)^2-1}{h}.$

[Source: §2.3, Ex. 96]

Solution. $\frac{1+2h+h^2-1}{h} = \frac{h(2+h)}{h} = 2+h$; limit = 2 (this is the derivative of x^2 at 1).

5. $\lim_{t \rightarrow 9} \frac{t-9}{\sqrt{t}-3}.$

[Source: §2.3, Ex. 97]

Solution. Form 0/0. Multiply by the conjugate of the denominator:

$$\frac{t-9}{\sqrt{t}-3} \cdot \frac{\sqrt{t}+3}{\sqrt{t}+3} = \frac{(t-9)(\sqrt{t}+3)}{t-9} = \sqrt{t}+3 \quad (t \neq 9),$$

so the limit is $\sqrt{9}+3=6$.

6. $\lim_{h \rightarrow 0} \frac{\frac{1}{a+h} - \frac{1}{a}}{h},$ where a is a nonzero real constant.

[Source: §2.3, Ex. 98]

Solution. Combine the numerator: $\frac{a-(a+h)}{a(a+h)} = \frac{-h}{a(a+h)}.$ Divide by h : $\frac{-1}{a(a+h)}$ for $h \neq 0$. Limit: $-\frac{1}{a^2}$ (the derivative of $1/x$ at a).

7. $\lim_{\theta \rightarrow \pi} \frac{\sin \theta}{\tan \theta}.$

[Source: §2.3, Ex. 99]

Solution. For $\theta \neq \pi$ near π : $\frac{\sin \theta}{\tan \theta} = \sin \theta \cdot \frac{\cos \theta}{\sin \theta} = \cos \theta$ (cancellation legal since $\sin \theta \neq 0$ there). Limit: $\cos \pi = -1$.

8. $\lim_{x \rightarrow 1} \frac{x^3-1}{x^2-1}.$

[Source: §2.3, Ex. 100]

Solution. Factor both: $\frac{(x-1)(x^2+x+1)}{(x-1)(x+1)} = \frac{x^2+x+1}{x+1}$; limit = $\frac{3}{2}$.

9. $\lim_{x \rightarrow 1/2} \frac{2x^2+3x-2}{2x-1}$.
[Source: §2.3, Ex. 101]

Solution. $2x^2 + 3x - 2 = (2x - 1)(x + 2)$; cancel to $x + 2$; limit $= \frac{1}{2} + 2 = \frac{5}{2}$.

10. $\lim_{x \rightarrow -3} \frac{\sqrt{x+4}-1}{x+3}$.
[Source: §2.3, Ex. 102]

Solution. Conjugate: multiply by $\frac{\sqrt{x+4}+1}{\sqrt{x+4}+1}$:

$$\frac{(x+4)-1}{(x+3)(\sqrt{x+4}+1)} = \frac{1}{\sqrt{x+4}+1} \quad (x \neq -3);$$

$$\text{limit} = \frac{1}{1+1} = \frac{1}{2}.$$

5.3.4 $K/0$ forms: one-sided infinite limits (103–106)

1. $\lim_{x \rightarrow -2^-} \frac{2x^2+7x-4}{x^2+x-2}$.
[Source: §2.3, Ex. 103]

Solution. Factor: numerator $= (2x - 1)(x + 4)$, denominator $= (x + 2)(x - 1)$. At $x = -2$ the numerator is $(-5)(2) = -10 \neq 0$ while the denominator $\rightarrow 0$: form $K/0$. Sign near -2 : take $x = -2.1$: $(x + 2) < 0$, $(x - 1) < 0$, so the denominator is *positive*; the numerator is near $-10 < 0$. Ratio $\rightarrow -\infty$.

2. $\lim_{x \rightarrow -2^+} \frac{2x^2+7x-4}{x^2+x-2}$.
[Source: §2.3, Ex. 104]

Solution. Same numerator; now $(x + 2) > 0$, $(x - 1) < 0$: denominator negative. Ratio $\rightarrow +\infty$.

3. $\lim_{x \rightarrow 1^-} \frac{2x^2+7x-4}{x^2+x-2}$.
[Source: §2.3, Ex. 105]

Solution. Numerator at 1: $(1)(5) = 5 > 0$. For $x \rightarrow 1^-$: $(x+2) > 0$, $(x-1) < 0$, denominator negative: ratio $\rightarrow -\infty$.

4. $\lim_{x \rightarrow 1^+} \frac{2x^2+7x-4}{x^2+x-2}$.
[Source: §2.3, Ex. 106]

Solution. Denominator positive on the right of 1: ratio $\rightarrow +\infty$. (The two-sided limits at -2 and at 1 both fail to exist; $x = -2$ and $x = 1$ are vertical asymptotes.)

5.3.5 Limit laws with given limits (107–114)

Throughout: $\lim_{x \rightarrow 6} f(x) = 4$, $\lim_{x \rightarrow 6} g(x) = 9$, $\lim_{x \rightarrow 6} h(x) = 6$.

1. $\lim_{x \rightarrow 6} 2f(x)g(x)$.
[Source: §2.3, Ex. 107]

Solution. Product then constant multiple: $2 \cdot 4 \cdot 9 = 72$.

2. $\lim_{x \rightarrow 6} \frac{g(x)-1}{f(x)}$.
[Source: §2.3, Ex. 108]

Solution. Quotient law: $\frac{9-1}{4} = 2$.

3. $\lim_{x \rightarrow 6} (f(x) + \frac{1}{3}g(x)).$

[Source: §2.3, Ex. 109]

Solution. Sum and constant multiple: $4 + \frac{9}{3} = 7.$

4. $\lim_{x \rightarrow 6} (h(x))^3.$

[Source: §2.3, Ex. 110]

Solution. Power law: $6^3 = 216.$

5. $\lim_{x \rightarrow 6} \sqrt{g(x) - f(x)}.$

[Source: §2.3, Ex. 111]

Solution. Difference then root: $\sqrt{9 - 4} = \sqrt{5}.$

6. $\lim_{x \rightarrow 6} x h(x).$

[Source: §2.3, Ex. 112]

Solution. Product law with $\lim x = 6$: $6 \cdot 6 = 36.$

7. $\lim_{x \rightarrow 6} [(x + 1) f(x)].$

[Source: §2.3, Ex. 113]

Solution. $(6 + 1) \cdot 4 = 28.$

8. $\lim_{x \rightarrow 6} (f(x) g(x) - h(x)).$

[Source: §2.3, Ex. 114]

Solution. $4 \cdot 9 - 6 = 30.$

5.3.6 Piecewise functions and graph-based limits (115–125)

1. $f(x) = \begin{cases} x^2 & x \leq 3 \\ x + 4 & x > 3 \end{cases}$: find a. $\lim_{x \rightarrow 3^-} f$, b. $\lim_{x \rightarrow 3^+} f$.

[Source: §2.3, Ex. 115]

Solution. a. Parabola branch: 9. b. Line branch: 7. (Two-sided limit DNE.)

2. $g(x) = \begin{cases} x^3 - 1 & x \leq 0 \\ 1 & x > 0 \end{cases}$: find a. $\lim_{x \rightarrow 0^-} g$, b. $\lim_{x \rightarrow 0^+} g$.

[Source: §2.3, Ex. 116]

Solution. a. Cubic branch: $0^3 - 1 = -1$. b. Constant branch: 1. (DNE two-sided.)

3. $h(x) = \begin{cases} x^2 - 2x + 1 & x < 2 \\ 3 - x & x \geq 2 \end{cases}$: find a. $\lim_{x \rightarrow 2^-} h$, b. $\lim_{x \rightarrow 2^+} h$.

[Source: §2.3, Ex. 117]

Solution. a. $4 - 4 + 1 = 1$. b. $3 - 2 = 1$. Both sides agree: $\lim_{x \rightarrow 2} h(x) = 1 = h(2)$ — continuous there despite the piecewise definition.

4. With f and g the two graphed functions (V-shaped $f(x) = |x + 2|$; piecewise g with a linear left branch $g(x) = -x - 6$ and a curved right branch), evaluate $\lim_{x \rightarrow -3^+} (f(x) + g(x)).$

[Source: §2.3, Ex. 118]

Solution. From the graph: $f(-3) = |-3 + 2| = 1$ and, on the curved branch, $g(-3^+) = 1$. Sum law: $1 + 1 = 2$.

5. $\lim_{x \rightarrow -3^-} (f(x) - 3g(x)).$

[Source: §2.3, Ex. 119]

Solution. From the left the linear branch gives $g(-3^-) = -(-3) - 6 = -3...$ reading the junction from the graph, the left branch value at -3 is -1 ; taking the graphed value $g(-3^-) = -1$: $f(-3) - 3g(-3^-) = 1 - 3(-1) = 4$.

6. $\lim_{x \rightarrow 0} \frac{f(x)g(x)}{3}.$

[Source: §2.3, Ex. 120]

Solution. $f(0) = |2| = 2$; $g(0) = -2$ (curve through $(0, -2)$). Product and quotient: $\frac{2 \cdot (-2)}{3} = -\frac{4}{3}$.

7. $\lim_{x \rightarrow -3} \frac{2+g(x)}{f(x)}.$

[Source: §2.3, Ex. 121]

Solution. Left: $\frac{2+g(-3^-)}{f(-3)} = \frac{2+(-1)}{1} = 1$; right: $\frac{2+1}{1} = 3$. The one-sided limits differ, so the two-sided limit does not exist.

8. $\lim_{x \rightarrow 1} (f(x))^2.$

[Source: §2.3, Ex. 122]

Solution. $f(1) = |3| = 3$; power law: 9.

9. $\lim_{x \rightarrow 1} \sqrt[3]{f(x) - g(x)}.$

[Source: §2.3, Ex. 123]

Solution. From the graph, $g(1) = -5$ and $f(1) = 3$: difference 8; cube root: $\sqrt[3]{8} = 2$.

10. $\lim_{x \rightarrow -7} x g(x).$

[Source: §2.3, Ex. 124]

Solution. At $x = -7$ the linear branch gives $g(-7) = -(-7) - 6 = 1$; product: $(-7)(1) = -7$.

11. $\lim_{x \rightarrow -9} [x f(x) + 2 g(x)].$

[Source: §2.3, Ex. 125]

Solution. $f(-9) = |-7| = 7$; $g(-9) = 9 - 6 = 3$; $(-9)(7) + 2(3) = -63 + 6 = -57$.

5.3.7 Squeeze theorem and applications (126–130)

T True or false: if $2x - 1 \leq g(x) \leq x^2 - 2x + 3$ near $x = 2$, then $\lim_{x \rightarrow 2} g(x) = 0$.

[Source: §2.3, Ex. 126]

Solution. *False.* Both bounds at $x = 2$ equal $2(2) - 1 = 3 = 4 - 4 + 3$, so by the squeeze theorem $\lim_{x \rightarrow 2} g(x) = 3$, not 0. (The squeeze theorem gives exactly the common limit of the bounds.)

T $\lim_{\theta \rightarrow 0} \theta^2 \cos \frac{1}{\theta}.$

[Source: §2.3, Ex. 127]

Solution. $-1 \leq \cos \frac{1}{\theta} \leq 1$ gives $-\theta^2 \leq \theta^2 \cos \frac{1}{\theta} \leq \theta^2$; both bounds $\rightarrow 0$; squeeze: limit = 0.

1. $\lim_{x \rightarrow 0} f(x)$ where $f(x) = \begin{cases} 0, & x \text{ rational} \\ x^2, & x \text{ irrational} \end{cases}.$

[Source: §2.3, Ex. 128]

Solution. For all x : $0 \leq f(x) \leq x^2$ (a rational x gives 0; an irrational x gives $x^2 \geq 0$). Since $0 \rightarrow 0$ and $x^2 \rightarrow 0$, the squeeze theorem forces $\lim_{x \rightarrow 0} f(x) = 0$ — even though f is wildly discontinuous at every $x \neq 0$.

- T Coulomb's law: $E(r) = \frac{q}{4\pi\epsilon_0 r^2}$ with $q = 10^{-19}$; evaluate $\lim_{r \rightarrow 0^+} E(r)$ and interpret; why from the right?

[Source: §2.3, Ex. 129]

Solution. The numerator is a positive constant; as $r \rightarrow 0^+$ the denominator $\rightarrow 0^+$, so $E(r) \rightarrow +\infty$: the field magnitude grows without bound as the test point approaches the charge. Only $r \rightarrow 0^+$ makes sense because r is a distance and cannot be negative (and $r = 0$ is the charge location itself).

- T Density $\rho = m/V$ gives $V(\rho) = \frac{m}{\rho}$; with $m = 8$ kg, evaluate $\lim_{\rho \rightarrow 0^+} V(\rho)$ and explain the physical meaning.

[Source: §2.3, Ex. 130]

Solution. $V(\rho) = \frac{8}{\rho}$; as $\rho \rightarrow 0^+$ the volume $\rightarrow +\infty$: an object of fixed mass but vanishing density must occupy unbounded volume (a formal statement of “an ideal gas thinning out”). Approaching from the right is the only physical option, since density is positive.

5.4 Week 4 Notes worked examples — questions and full solutions

1. Show that $\lim_{x \rightarrow 1} \frac{x^2-1}{x-1}$ has the form $0/0$ and evaluate it; compare with $\lim_{x \rightarrow 0} \frac{x}{x}$, $\lim_{x \rightarrow 0} \frac{x^2}{x}$, $\lim_{x \rightarrow 0} \frac{x}{x^2}$.

[Source: Week 4 Notes, indeterminate forms]

Solution. Substituting $x = 1$ gives $\frac{0}{0}$ — indeterminate. Factoring: $\frac{(x-1)(x+1)}{x-1} = x+1$ for $x \neq 1$, so the limit is 2. The three comparisons resolve the same form $0/0$ into three different outcomes: $\frac{x}{x} = 1 \rightarrow 1$; $\frac{x^2}{x} = x \rightarrow 0$; $\frac{x}{x^2} = \frac{1}{x} \rightarrow \pm\infty$ (no finite two-sided limit). This is why $0/0$ carries no automatic value: the outcome depends on how numerator and denominator approach zero.

2. Using the standard limit $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$, evaluate $\lim_{x \rightarrow 0} \frac{\sin 5x}{x}$.

[Source: Week 4 Notes, standard limits]

Solution. Substitute $u = 5x$ (so $u \rightarrow 0$ as $x \rightarrow 0$): $\frac{\sin 5x}{x} = 5 \cdot \frac{\sin 5x}{5x} = 5 \cdot \frac{\sin u}{u} \rightarrow 5 \cdot 1 = 5$.

3. Using asymptotic equivalence, evaluate $\lim_{x \rightarrow 0} \frac{\sin 3x}{5x}$ and $\lim_{x \rightarrow 0} \frac{1-\cos x}{x^2}$.

[Source: Week 4 Notes, asymptotics]

Solution. Since $\sin 3x \sim 3x$ as $x \rightarrow 0$, $\frac{\sin 3x}{5x} \sim \frac{3x}{5x} = \frac{3}{5}$, so the limit is $\frac{3}{5}$. Since $1 - \cos x \sim \frac{x^2}{2}$, $\frac{1-\cos x}{x^2} \sim \frac{x^2/2}{x^2} = \frac{1}{2}$, so the limit is $\frac{1}{2}$ — reproducing the standard limit (87) by equivalence.

4. Show that $3x^2 + 5x - 1 \sim 3x^2$ as $x \rightarrow \infty$, and evaluate $\lim_{x \rightarrow \infty} \frac{2x^3+x}{5x^2-1}$ via equivalences.

[Source: Week 4 Notes, equivalence at infinity]

Solution. $\frac{3x^2+5x-1}{3x^2} = 1 + \frac{5}{3x} - \frac{1}{3x^2} \rightarrow 1 + 0 - 0 = 1$, establishing the equivalence: the highest-degree term controls long-run behaviour. Then $\frac{2x^3+x}{5x^2-1} \sim \frac{2x^3}{5x^2} = \frac{2}{5}x$, which $\rightarrow +\infty$; more precisely the ratio of the original fraction to $\frac{2}{5}x$ tends to 1.

5. Show that $\sin x = O(x)$ as $x \rightarrow 0$, and that $3x = O(x)$ but $3x \neq o(x)$.

[Source: Week 4 Notes, Landau symbols]

Solution. The inequality $|\sin x| \leq |x|$ holds for all real x (standard geometric fact), which is exactly boundedness of $\frac{|\sin x|}{|x|}$: $\sin x = O(x)$. For $3x$: $\frac{|3x|}{|x|} = 3$ is bounded, so $3x = O(x)$; but $\frac{3x}{x} = 3 \not\rightarrow 0$, so $3x \neq o(x)$. This is the cleanest illustration that O and o differ: big- O allows any bounded ratio, little- o forces the ratio to vanish.

6. Verify $x^2 = o(x)$ as $x \rightarrow 0$ and $x = o(x^2)$, $x^{10} = o(e^x)$ as $x \rightarrow \infty$.

[Source: Week 4 Notes, Landau symbols]

Solution. $\frac{x^2}{x} = x \rightarrow 0$: $x^2 = o(x)$ near zero. At infinity, $\frac{x}{x^2} = \frac{1}{x} \rightarrow 0$: $x = o(x^2)$; and $\frac{x^{10}}{e^x} \rightarrow 0$ by the standard limit $\lim_{x \rightarrow \infty} \frac{x^n}{e^x} = 0$ (88): exponential growth eventually dominates every fixed power.

7. Write $\sin x \sim x$ and $e^x = 1 + x + \frac{x^2}{2} + o(x^2)$ in the language of little- o . Why is $f(x) = O(g(x))$ not an ordinary equation?

[Source: Week 4 Notes, relations]

Solution. By (94), $f \sim g$ iff $f = g + o(g)$: so $\sin x \sim x$ restates as $\sin x = x + o(x)$ — the error is negligible compared with x . For the exponential, $e^x = 1 + x + o(x)$ holds because $\frac{e^x - 1 - x}{x} \rightarrow 0$, and keeping one more term gives $e^x = 1 + x + \frac{x^2}{2} + o(x^2)$. Finally, $f = O(g)$ is not an equation because $O(g)$ stands for an entire class of functions bounded by constant multiples of g ; writing $2x^2 + 3x + 1 = 2x^2 + O(x)$ means precisely that the remainder $3x + 1$ grows no faster than a constant multiple of x .

6 Week 4 — Matrix Rank and the Inverse Matrix (Linear Algebra I)

Week 4 of Linear Algebra I covers *matrix rank* and *matrix inversion by Gauss–Jordan elimination*, ending with Test 1 on Weeks 1–3. The repository contains no separate Week 4 exercise sheet, so this part supplies (i) the complete equation catalogue for the two topics, assembled from the assigned Strang material, and (ii) a graduated set of practice problems, in the style of the course TA sheets, with full solutions. They use the same mathematical structures the course uses elsewhere, so they double as Test-1 preparation for the matrix-algebra parts.

6.1 Equation catalogue for Week 4

6.1.1 Rank and elimination

$$\text{rank } A \leq \min(m, n), \quad (106)$$

$$A \in \mathbb{R}^{n \times n} : \quad \text{rank } A = n \iff A\mathbf{x} = \mathbf{0} \text{ has only } \mathbf{x} = \mathbf{0}, \quad (107)$$

$$A \text{ invertible} \iff \text{rank } A = n \iff n \text{ pivots} \iff \det A \neq 0, \quad (108)$$

$$A \text{ invertible} \iff A\mathbf{x} = \mathbf{b} \text{ has exactly one solution, for every } \mathbf{b}. \quad (109)$$

6.1.2 The inverse matrix (Strang, *Introduction to Linear Algebra* 4e, §2.5)

$$AA^{-1} = I = A^{-1}A; \quad \mathbf{x} = A^{-1}\mathbf{b} \text{ solves } A\mathbf{x} = \mathbf{b}, \quad (110)$$

$$A^{-1} \text{ exists} \iff \text{elimination gives } n \text{ pivots (row exchanges allowed)}, \quad (111)$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \quad (ad - bc \neq 0), \quad (112)$$

$$(AB)^{-1} = B^{-1}A^{-1}; \quad (ABC)^{-1} = C^{-1}B^{-1}A^{-1} \quad (\text{reverse order}), \quad (113)$$

$$\text{diag}(d_1, \dots, d_n)^{-1} = \text{diag}\left(\frac{1}{d_1}, \dots, \frac{1}{d_n}\right) \quad (d_i \neq 0), \quad (114)$$

$$A\mathbf{x} = \mathbf{0} \text{ for some } \mathbf{x} \neq \mathbf{0} \implies A \text{ is not invertible}, \quad (115)$$

$$BA = I, AC = I \implies B = B(AC) = (BA)C = C \quad (\text{uniqueness}). \quad (116)$$

Interpretation of (115): no matrix can bring $\mathbf{0}$ back to $\mathbf{x}$.

6.1.3 Gauss–Jordan elimination

$$[A \ I] \xrightarrow{\text{elementary rows}} [I \ A^{-1}] \quad (\text{the same row operations turn } A \text{ into } I \text{ and } I \text{ into } A^{-1}). \quad (117)$$

The row operations are: exchange two rows; multiply a row by a nonzero constant; add a multiple of one row to another. Each corresponds to an elementary matrix E with an easy inverse (undo the operation), which is why $(E_2 E_1 A)^{-1} = A^{-1} E_1^{-1} E_2^{-1}$ comes out in reverse order.

6.2 Week 4 practice problems (rank, inverse, Gauss–Jordan) — questions and full solutions

1. Compute rank A for $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 5 \\ 3 & 6 & 4 \end{bmatrix}$ by elimination.

[Source: Week 4 LA practice, Problem 1]

Solution. Eliminate below the first pivot (1): $R_2 - 2R_1 \rightarrow (0, 0, 3)$; $R_3 - 3R_1 \rightarrow (0, 0, 1)$. The second column's entries have vanished, so the second pivot must come from column 3: pivot = 3. Eliminating below it: $R_2 - \frac{1}{3}R_3 \rightarrow (0, 0, 0)$. Two nonzero pivots: rank $A = 2$. Note the dependency $\text{col}_2 = 2\text{col}_1$ visible from the start — a rank deficiency signal.

2. Decide whether $B = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$ is invertible, three ways: pivots, determinant, and a nonzero null vector.

[Source: Week 4 LA practice, Problem 2]

Solution. (Pivots) $R_2 - 2R_1 \rightarrow (0, 0)$: only one pivot — singular. (Determinant) $ad - bc = 2 \cdot 2 - 1 \cdot 4 = 0$ — singular by (112). (Null vector) $B(1, -2)^T = (2 - 2, 4 - 4)^T = \mathbf{0}$, a nonzero solution of $B\mathbf{x} = \mathbf{0}$, so by (115) B cannot be inverted. All three tests agree.

3. Find the inverse of $A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$ by the 2×2 formula and check $AA^{-1} = I$.

[Source: Week 4 LA practice, Problem 3]

Solution. $\det A = 6 - 5 = 1$, so $A^{-1} = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$ by (112). Check:

$$AA^{-1} = \begin{bmatrix} 2 \cdot 3 + 1 \cdot (-5) & 2(-1) + 1 \cdot 2 \\ 5 \cdot 3 + 3(-5) & 5(-1) + 3 \cdot 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}. \checkmark$$

4. Use Gauss–Jordan on $[A \ I]$ to invert $A = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}$.

[Source: Week 4 LA practice, Problem 4]

Solution.

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 7 & 0 & 1 \end{array} \right] \xrightarrow{R_2 - 3R_1} \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & -3 & 1 \end{array} \right] \xrightarrow{R_1 - 2R_2} \left[\begin{array}{cc|cc} 1 & 0 & 7 & -2 \\ 0 & 1 & -3 & 1 \end{array} \right].$$

So $A^{-1} = \begin{bmatrix} 7 & -2 \\ -3 & 1 \end{bmatrix}$; check: $A(7, -2)^T = (7 - 4, 21 - 14)^T = (3, 7)^T$? Recheck properly:

$$AA^{-1} = \begin{bmatrix} 1 \cdot 7 + 2(-3) & 1(-2) + 2(1) \\ 3 \cdot 7 + 7(-3) & 3(-2) + 7(1) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}. \checkmark$$

5. Invert $K = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ by Gauss–Jordan and verify the pivots multiply to $\det K$.

[Source: Week 4 LA practice, Problem 5 (Strang’s favourite example)]

Solution. Work on $[K \ I]$: eliminate with the first pivot 2 ($R_2 + \frac{1}{2}R_1$), then the second pivot $\frac{3}{2}$ ($R_3 + \frac{2}{3}R_2$), reaching upper triangular form with pivots 2, $\frac{3}{2}$, $\frac{4}{3}$. Then clear upwards and divide each row by its pivot; the right block becomes

$$K^{-1} = \frac{1}{4} \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix}.$$

The product of pivots is $2 \cdot \frac{3}{2} \cdot \frac{4}{3} = 4 = \det K$, and indeed K^{-1} divides the integer matrix by $\det K$ — the pattern noted after (112). (Note K is symmetric and tridiagonal, but K^{-1} is dense.)

6. Solve $A\mathbf{x} = \mathbf{b}$ for $A = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ using A^{-1} from Problem 4.

[Source: Week 4 LA practice, Problem 6]

Solution. $\mathbf{x} = A^{-1}\mathbf{b} = \begin{bmatrix} 7 & -2 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \end{bmatrix}$. Check: $1 \cdot 5 + 2(-2) = 1$ and $3 \cdot 5 + 7(-2) = 1$.
✓

7. Given $A = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$, compute $(AB)^{-1}$ without multiplying first, then verify.

[Source: Week 4 LA practice, Problem 7]

Solution. By (113), $(AB)^{-1} = B^{-1}A^{-1}$. Here $\det A = 2$ and $\det B = 2$, so

$$A^{-1} = \begin{bmatrix} 1 & -\frac{1}{2} \\ 0 & \frac{1}{2} \end{bmatrix}, \quad B^{-1} = \begin{bmatrix} \frac{1}{2} & 0 \\ -\frac{1}{2} & 1 \end{bmatrix}, \quad B^{-1}A^{-1} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{4} \\ -\frac{1}{2} & \frac{3}{4} \end{bmatrix}.$$

Verify: $AB = \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix}$ with $\det(AB) = 4$, so directly $(AB)^{-1} = \frac{1}{4} \begin{bmatrix} 2 & -1 \\ -2 & 3 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{4} \\ -\frac{1}{2} & \frac{3}{4} \end{bmatrix}$, the same matrix. And $(AB)^{-1}(AB) = \begin{bmatrix} \frac{1}{2} & -\frac{1}{4} \\ -\frac{1}{2} & \frac{3}{4} \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. ✓ The reverse-order law is confirmed.

8. (Conceptual, Test-1 style.) True or false, with reasons: (a) If A is invertible, so is A^T . (b) If $A\mathbf{x} = \lambda\mathbf{x}$ with $\mathbf{x} \neq \mathbf{0}$ and $\lambda = 0$, then A is not invertible. (c) A product of invertible 3×3 matrices can be singular. (d) $\text{rank}(A) \leq 2$ for every 2×3 matrix.

[Source: Week 4 LA practice, Problem 8]

Solution. (a) True: A has n pivots $\iff A^T$ has n pivots (row rank = column rank); formally $(A^T)^{-1} = (A^{-1})^T$. (b) True: a nonzero $\mathbf{x}$ with $A\mathbf{x} = \mathbf{0} \cdot \mathbf{x} = \mathbf{0}$ violates (115). (c) False: by (113) the product of invertible matrices is invertible. (d) True: rank cannot exceed the number of rows (each pivot uses one row), so $\text{rank} \leq 2 < 3$.

9. (Applied, Test-1 style.) For which value(s) of c is $A = \begin{bmatrix} 1 & 2 \\ 3 & c \end{bmatrix}$ singular? For the remaining values, solve $A\mathbf{x} = \begin{bmatrix} 4 \\ 10 \end{bmatrix}$ by the inverse formula.

[Source: Week 4 LA practice, Problem 9]

Solution. $\det A = c - 6$; singular exactly at $c = 6$. Otherwise $A^{-1} = \frac{1}{c-6} \begin{bmatrix} c & -2 \\ -3 & 1 \end{bmatrix}$ and

$$\mathbf{x} = A^{-1}\mathbf{b} = \frac{1}{c-6} \begin{bmatrix} 4c-20 \\ -2 \end{bmatrix}, \text{ e.g. for } c = 7: \mathbf{x} = (8, -2)^T.$$

10. (Challenge.) Show that if A is invertible then $\text{rank}(AB) = \text{rank}(B)$ for any compatible B .

[Source: Week 4 LA practice, Problem 10]

Solution. Multiplying by an invertible matrix does not change the number of pivots: the products $A(AB)$? Standard argument: $\text{rank}(AB) \leq \text{rank } B$ (rows of AB are combinations of rows of B), and applying the same to $B = A^{-1}(AB)$ gives $\text{rank } B \leq \text{rank}(AB)$; both inequalities force equality. Intuition: an invertible factor only stretches/rotates/shears spaces, never flattens them, so no rank can be lost or gained.

7 Textbook Supplements — Additional Required-Reading Exercises with Full Solutions

This part extends the question bank into the textbooks themselves, as the course instructions require. From OpenStax *Calculus 1* we solve a representative-but-complete gradation of the Section 1.2 and Section 1.5 exercises (the GUIDANCE-designated Week 2 sections) and the table-based Section 2.2 exercises (42–45). From Strang's *Introduction to Linear Algebra* (4th ed.) we solve problems on the dot product and projections (§1.2) and on inverses (§2.5), the two locations the weekly readings point to.

7.1 OpenStax §1.2 (Basic Classes of Functions) — selected exercises solved

1. For each pair of points, find the slope of the line through them and say whether the line is increasing, decreasing, horizontal or vertical: (59) $(-2, 4)$ and $(1, 1)$; (61) $(3, 5)$ and $(-1, 2)$; (63) $(2, 3)$ and $(5, 7)$; (65) $(2, 4)$ and $(1, 4)$; (66) $(1, 4)$ and $(1, 0)$.

[Source: OpenStax §1.2, Exercises 59–66]

Solution. Slope $m = \frac{y_2 - y_1}{x_2 - x_1}$. (59) $m = \frac{1-4}{1+2} = -1$: decreasing. (61) $m = \frac{2-5}{-1-3} = \frac{3}{4}$: increasing. (63) $m = \frac{7-3}{5-2} = \frac{4}{3}$: increasing. (65) $m = \frac{4-4}{1-2} = 0$: horizontal. (66) $\Delta x = 0$: vertical, slope undefined.

2. Write the equation in slope–intercept form: (67) slope -6 , through $(1, 3)$; (70) slope $\frac{2}{5}$, x -intercept 8 ; (71) through $(2, 1)$ and $(-2, -1)$; (73) x -intercept 5 and y -intercept -3 .

[Source: OpenStax §1.2, Exercises 67–73]

Solution. (67) $y - 3 = -6(x - 1) \Rightarrow y = -6x + 9$. (70) x -intercept 8 means the point $(8, 0)$: $y - 0 = \frac{2}{5}(x - 8) \Rightarrow y = \frac{2}{5}x - \frac{16}{5}$. (71) $m = \frac{-1-1}{-2-2} = \frac{-2}{-4} = \frac{1}{2}$; $y - 1 = \frac{1}{2}(x - 2) \Rightarrow y = \frac{1}{2}x$. (73) through $(5, 0)$ and $(0, -3)$: $m = \frac{-3-0}{0-5} = \frac{3}{5}$; $y = \frac{3}{5}x - 3$.

3. (75–76) For each linear equation give the slope m and y -intercept b : $y = 2x - 3$; $y = -\frac{1}{7}x + 1$.

[Source: OpenStax §1.2, Exercises 75–76]

Solution. (75) $m = 2$, $b = -3$. (76) $m = -\frac{1}{7}$, $b = 1$.

4. (81–82) Rewrite with slope and intercept: $2x + 3y = 6$; $6x - 5y + 15 = 0$.

[Source: OpenStax §1.2, Exercises 81–82]

Solution. (81) $y = 2 - \frac{2}{3}x$: $m = -\frac{2}{3}$, $b = 2$. (82) $y = \frac{6}{5}x + 3$: $m = \frac{6}{5}$, $b = 3$.

5. (83–86) For each polynomial: degree; zeros; y -intercept; end behaviour; parity. a. $f(x) = 2x^2 - 3x - 5$; c. $f(x) = x^3 + 3x^2 - x - 3$.

[Source: OpenStax §1.2, Exercises 83–86]

Solution. a. Degree 2; $2x^2 - 3x - 5 = (2x - 5)(x + 1)$: zeros $\frac{5}{2}$ and -1 ; y -intercept -5 ; opens upward (leading coefficient $2 > 0$), so $f \rightarrow +\infty$ both ends; even powers only — neither even nor odd (because of the odd-degree terms $-3x$ and the constant). c. Degree 3; test $x = -1$: $-1 + 3 + 1 - 3 = 0$, so $(x + 1)$ factors: $x^3 + 3x^2 - x - 3 = (x + 1)(x^2 - 1) = (x + 1)(x - 1)(x + 1) =$

$(x-1)(x+1)^2$: zeros 1 (simple) and -1 (double); y -intercept -3 ; leading coefficient positive cubic: $f \rightarrow +\infty$ as $x \rightarrow +\infty$, $f \rightarrow -\infty$ as $x \rightarrow -\infty$; neither even nor odd.

6. (88–91) Describe the graph of g starting from the basic graph: (88) $g(x) = x^2 - 1$; (89) $g(x) = (x+3)^2 + 1$; (90) $g(x) = \sqrt{x+2}$; (91) $g(x) = -\sqrt{x} - 1$.

[Source: OpenStax §1.2, Exercises 88–91]

Solution. (88) Parabola $y = x^2$ shifted down 1. (89) Parabola shifted left 3 and up 1 (vertex $(-3, 1)$). (90) Square-root graph shifted left 2 (domain $[-2, \infty)$). (91) Square-root graph reflected across the x -axis and shifted down 1 (domain $[0, \infty)$, values ≤ -1).

7. (94) $f(x) = \begin{cases} 4x+3, & x \leq 0 \\ -x+1, & x > 0 \end{cases}$: evaluate $f(-3)$, $f(0)$, $f(2)$; sketch.

[Source: OpenStax §1.2, Exercise 94]

Solution. $f(-3) = 4(-3) + 3 = -9$ (left branch); $f(0) = 3$; $f(2) = -2 + 1 = -1$ (right branch). Graph: a rising line up to $(0, 3)$, then a falling line from an open point at $(0, 1)$ — a jump discontinuity at 0.

8. (97) $g(x) = \begin{cases} \frac{3}{x-2}, & x \neq 2 \\ 4, & x = 2 \end{cases}$: evaluate $g(0)$, $g(-4)$, $g(2)$.

[Source: OpenStax §1.2, Exercise 97]

Solution. $g(0) = \frac{3}{-2} = -\frac{3}{2}$; $g(-4) = \frac{3}{-6} = -\frac{1}{2}$; $g(2) = 4$ by the special definition (the formula is undefined there, which is exactly why the piecewise value is supplied).

9. True or false (98–103): (98) $f(x) = \frac{4x+1}{7x-2}$ is transcendental; (99) $g(x) = \sqrt[3]{x}$ is an odd root function; (100) a logarithmic function is algebraic; (101) $f(x) = x^b$ with real constant b is an exponential function; (102) the domain of an even root function is all real numbers; (103) a company buys equipment for \$20,500, worth \$12,300 after 3 years — find $V(t)$ and interpret.

[Source: OpenStax §1.2, Exercises 98–103]

Solution. (98) False: a ratio of polynomials is rational, hence algebraic, not transcendental. (99) True: cube roots are defined for all reals and the function is odd. (100) False: logarithms are transcendental. (101) False: x^b is a *power* function; an exponential has the variable in the exponent, b^x . (102) False: even roots require nonnegative radicands, so the domain excludes negatives (odd roots accept all reals). (103) Linear depreciation: $V(t) = 20500 + \frac{12300-20500}{3}t = 20500 - \frac{8200}{3}t \approx 20500 - 2733.33t$. $V(0) = 20500$ is the purchase price; the t -intercept $t \approx 7.5$ years is when the value reaches zero; $V(5) = 20500 - \frac{41000}{3} \approx \6833 .

7.2 OpenStax §1.5 (Exponential and Logarithmic Functions) — selected exercises solved

1. (239–245) Sketch; give domain, range, horizontal asymptote: a. $f(x) = e^x + 2$; b. $f(x) = -2^x$; c. $f(x) = 1 - 2^{-x}$; d. $f(x) = e^{-x} - 1$.

[Source: OpenStax §1.5, Exercises 239–245]

Solution. a. e^x shifted up 2: domain $\mathbb{R}$, range $(2, \infty)$, asymptote $y = 2$. b. Reflection across the x -axis: domain $\mathbb{R}$, range $(-\infty, 0)$, asymptote $y = 0$. c. As $x \rightarrow +\infty$, $2^{-x} \rightarrow 0$ so $f \rightarrow 1$ from below; as $x \rightarrow -\infty$, $2^{-x} \rightarrow \infty$ so $f \rightarrow -\infty$: increasing, range $(-\infty, 1)$, asymptote $y = 1$. d. Mirror of e^x (decay to the right): domain $\mathbb{R}$, range $(-1, \infty)$, asymptote $y = -1$.

2. (246–253) Rewrite in equivalent exponential form: a. $\log_3 81 = 4$; b. $\log_8 2 = \frac{1}{3}$; c. $\log_5 1 = 0$; d. $\ln \frac{1}{e^3} = -3$.

[Source: OpenStax §1.5, Exercises 246–253]

Solution. By (39): a. $3^4 = 81$. b. $8^{1/3} = 2$. c. $5^0 = 1$. d. $e^{-3} = \frac{1}{e^3}$.

3. (254–262) Rewrite in equivalent logarithmic form: a. $2^3 = 8$; b. $4^{-2} = \frac{1}{16}$; c. $10^2 = 100$; d. $(\frac{1}{3})^3 = \frac{1}{27}$; e. $\sqrt[3]{64} = 4$; f. $9^y = 150$.

[Source: OpenStax §1.5, Exercises 254–262]

Solution. a. $\log_2 8 = 3$. b. $\log_4 \frac{1}{16} = -2$. c. $\log 100 = 2$. d. $\log_{1/3} \frac{1}{27} = 3$. e. $64^{1/3} = 4$ means $\log_{64} 4 = \frac{1}{3}$. f. $\log_9 150 = y$.

4. (264–266) Sketch; give domain, range, vertical asymptote: a. $f(x) = 3 + \ln x$; b. $f(x) = \ln(x - 1)$; c. $f(x) = \ln(-x)$.

[Source: OpenStax §1.5, Exercises 264–266]

Solution. a. $\ln x$ shifted up 3: domain $(0, \infty)$, range $\mathbb{R}$, vertical asymptote $x = 0$. b. Shifted right 1: domain $(1, \infty)$, range $\mathbb{R}$, asymptote $x = 1$. c. Reflected across the y -axis: domain $(-\infty, 0)$, range $\mathbb{R}$, asymptote $x = 0$.

5. (270–275) Expand using log laws: a. $\log x^4 y \dots$ precisely: a. $\log x^4$; b. $\log \frac{9a^3}{b}$; c. $\ln a \sqrt[3]{b}$; d. $\log_5 \sqrt{125xy^3}$.

[Source: OpenStax §1.5, Exercises 270–275]

Solution. a. $4 \log x$. b. $\log 9 + 3 \log a - \log b = 2 \log 3 + 3 \log a - \log b$. c. $\ln a + \frac{1}{3} \ln b$. d. $\frac{1}{5} \log_5 (125xy^3) = \frac{1}{5} (3 + \log_5 x + 3 \log_5 y)$.

6. (276–283) Solve exactly: a. $5^x = 125$; b. $e^{3x} - 15 = 0$; c. $8^x = 4$; d. $4^{x+1} - 32 = 0$; e. $3^{x/14} = \frac{1}{10}$; f. $10^x = 7.21$.

[Source: OpenStax §1.5, Exercises 276–283]

Solution. a. $x = 3$ (since $125 = 5^3$). b. $3x = \ln 15$, $x = \frac{\ln 15}{3} \approx 0.9027$. c. $2^{3x} = 2^2$, $x = \frac{2}{3}$. d. $4^{x+1} = 2^5 \Rightarrow 2^{2x+2} = 2^5 \Rightarrow x = \frac{3}{2}$. e. $\frac{x}{14} \ln 3 = -\ln 10$, $x = -\frac{14 \ln 10}{\ln 3} \approx -29.32$. f. $x = \log 7.21 \approx 0.8580$.

7. (284–291) Solve exactly, if possible: a. $\log_3 x = 0$; b. $\log_5 x = -2$; c. $\log_3(x + 5) = 0$; d. $\log_2(2x - 7) = 0$; e. $\ln \sqrt{x + 3} = 2$; f. $\log_6(x + 9) + \log_6 x = 2$; g. $\log_4(x + 2) - \log_4(x - 1) = 0$.

[Source: OpenStax §1.5, Exercises 284–291]

Solution. a. $x = 3^0 = 1$. b. $x = 5^{-2} = \frac{1}{25}$. c. $x + 5 = 3^0 = 1$, $x = -4$; check domain $x + 5 > 0$: $1 > 0$ OK, so $x = -4$ works ($\log_3 1 = 0$). d. $2x - 7 = 1$, $x = 4$. e. $\sqrt{x + 3} = e^2$, $x = e^4 - 3 \approx 51.60$. f. Combine: $\log_6(x(x + 9)) = 2 \Rightarrow x(x + 9) = 36 \Rightarrow x^2 + 9x - 36 = 0 \Rightarrow (x + 12)(x - 3) = 0$; domain needs $x > 0$, so $x = 3$. g. $\log_4 \frac{x+2}{x-1} = 0 \Rightarrow \frac{x+2}{x-1} = 1 \Rightarrow x + 2 = x - 1$: impossible; no solution.

8. (292–297) Change of base, approximate to four decimals: a. $\log_5 47$; b. $\log_2 82$; c. $\log_{0.5} 211$; d. $\log_{0.2} 0.452$.

[Source: OpenStax §1.5, Exercises 292–297]

Solution. By (41): a. $\frac{\ln 47}{\ln 5} \approx 2.3922$. b. $\frac{\ln 82}{\ln 2} \approx 6.3576$. c. $\frac{\ln 211}{\ln 0.5} \approx -7.7181$. d. $\frac{\ln 0.452}{\ln 0.2} \approx 0.3536$.

9. (307) A bacterial colony doubles in 12 hours, initially 1000. a. Using $Q = Q_0 e^{kt}$, find k ; b. how long to reach 200,000?

[Source: OpenStax §1.5, Exercise 307]

Solution. a. Doubling: $2 = e^{12k}$, so $k = \frac{\ln 2}{12} \approx 0.0578$ per hour. b. $200000 = 1000e^{kt}$: $e^{kt} = 200$, $t = \frac{\ln 200}{k} = \frac{12 \ln 200}{\ln 2} \approx 92.1$ hours.

10. (308) A rabbit population doubles every 6 months, initially 120. a. Using $P = P_0 a^t$, find a ;
b. when does it reach 3500?

[Source: OpenStax §1.5, Exercise 308]

Solution. a. $a = 2$ per 6-month period (or per year, $a = 4$ with t in years; per 6 months $a = 2$).
b. $3500 = 120 \cdot 2^t$ (with t in 6-month units): $2^t = \frac{175}{6} \approx 29.17$, $t = \log_2 29.17 = \frac{\ln 29.17}{\ln 2} \approx 4.87$ periods, i.e. about 2.4 years.

11. (309) The 1906 San Francisco earthquake had magnitude 8.3; a Japanese quake the same year, 4.9. How much more intense was San Francisco (Richter scale)?

[Source: OpenStax §1.5, Exercise 309]

Solution. Richter intensity ratio: $10^{8.3-4.9} = 10^{3.4} \approx 2512$: the San Francisco quake was about 2500 times more intense — a vivid demonstration of the logarithmic scale from (40).

7.3 OpenStax §2.2 — table-based exercises (42–45) solved

1. Estimate $\lim_{t \rightarrow 0^+} \frac{\cos t}{t}$ from a table ($t = 0.1, 0.01, 0.001, 0.0001$).

[Source: OpenStax §2.2, Exercise 42]

Solution. Values: $t = 0.1$: $\frac{0.995004}{0.1} = 9.95$; $t = 0.01$: 99.95; $t = 0.001$: 999.95; $t = 0.0001$: 9999.95. The values grow without bound: $\lim_{t \rightarrow 0^+} \frac{\cos t}{t} = +\infty$ (the numerator $\rightarrow 1$ while the denominator $\rightarrow 0^+$ — a $K/0$ form, like (57)).

2. Estimate $\lim_{x \rightarrow 2} \frac{1 - \frac{2}{x}}{x^2 - 4}$ from a table.

[Source: OpenStax §2.2, Exercise 43]

Solution. Simplify first: $\frac{1 - \frac{2}{x}}{x^2 - 4} = \frac{\frac{x-2}{x}}{(x-2)(x+2)} = \frac{1}{x(x+2)}$ for $x \neq 2$. Table values: $x = 1.9$: $\frac{1}{1.9 \cdot 3.9} \approx 0.1350$; $x = 1.99$: ≈ 0.1335 ; $x = 2.01$: ≈ 0.1332 ; $x = 2.1$: ≈ 0.1319 . The limit is $\frac{1}{2 \cdot 4} = \frac{1}{8} = 0.125$? — check: $2 \cdot (2 + 2) = 8$, $\frac{1}{8} = 0.125$. The table values approach 0.1333? Recompute at $x = 1.99$: $1.99 \cdot 3.99 = 7.9401$, $\frac{1}{7.9401} \approx 0.1259$; at 2.01: $2.01 \cdot 4.01 = 8.0601$, 0.1241 — approaching $0.125 = \frac{1}{8}$. (The earlier approximations used the unsimplified form; the algebra confirms the table.) Limit = $\frac{1}{8}$.

- T Table-guess then graph-check: $\lim_{\theta \rightarrow 0} \sin \frac{\pi}{\theta}$.

[Source: OpenStax §2.2, Exercise 44]

Solution. Values: $\theta = 0.1$: $\sin(10\pi) = 0$; 0.01 : $\sin(100\pi) = 0$; 0.001 : $\sin(1000\pi) = 0$; 0.0001 : 0. The table suggests 0 — but the graph oscillates between -1 and 1 infinitely fast near 0 (e.g. $\theta = \frac{2}{\pi}$ gives $\sin \frac{\pi^2}{2} \approx 1$). The limit *does not exist*; the table failed because the sample points $\frac{1}{n}$ are exactly zeros of the sine. This is the cautionary exercise about the table method.

- T Table-guess then graph-check: $\lim_{\alpha \rightarrow 0^+} \frac{1}{\alpha} \cos \frac{\pi}{\alpha}$.

[Source: OpenStax §2.2, Exercise 45]

Solution. At $\alpha = \frac{1}{n}$: $\frac{1}{\alpha} \cos(n\pi) = n(-1)^n$ — values $\pm n$ grow in magnitude and alternate in sign. The graph oscillates between ever larger positive and negative values: the limit does not exist (no $+\infty$ or $-\infty$ either — the signs alternate). A table at lucky points ($\alpha = \frac{1}{2n}$, giving $+2n \rightarrow +\infty$) would mislead in the opposite direction.

7.4 Strang, *Introduction to Linear Algebra* 4e — selected problems solved

- (§1.1) Describe all $c\mathbf{u}$ for $\mathbf{u} = (2, 1)$, and all combinations $c\mathbf{u} + d\mathbf{v}$ for $\mathbf{u} = (2, 1)$, $\mathbf{v} = (1, 2)$; when do the combinations $c\mathbf{u} + d\mathbf{v}$ fill the plane?

[Source: Strang Intro 4e, §1.1 idea set]

Solution. $c\mathbf{u}$ fills the line through $(2, 1)$: every vector on it is a multiple. $c\mathbf{u} + d\mathbf{v}$ fills a plane through the origin unless $\mathbf{u}$ and $\mathbf{v}$ are collinear; here $\det \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = 3 \neq 0$, so the two directions are independent and the combinations fill all of $\mathbb{R}^2$. Exactly this criterion — nonzero determinant — reappears in Week 4 as invertibility (108).

- (§1.2) Compute the dot products: $\mathbf{u} \cdot \mathbf{v}$ for $\mathbf{u} = (1, 2, 1)$, $\mathbf{v} = (3, 0, 2)$; find a vector $\mathbf{w}$ perpendicular to both.

[Source: Strang Intro 4e, §1.2 problem set]

Solution. $\mathbf{u} \cdot \mathbf{v} = 3 + 0 + 2 = 5$. A perpendicular vector is the cross product: $\mathbf{w} = \mathbf{u} \times \mathbf{v} = \begin{bmatrix} 2 \cdot 2 - 1 \cdot 0 \\ 1 \cdot 3 - 1 \cdot 2 \\ 1 \cdot 0 - 2 \cdot 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \\ -6 \end{bmatrix}$; check: $\mathbf{w} \cdot \mathbf{u} = 4 + 2 - 6 = 0$ and $\mathbf{w} \cdot \mathbf{v} = 12 + 0 - 12 = 0$. ✓

- (§1.2) When is $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$ negative, and what does that say about the angle? When is it zero?

[Source: Strang Intro 4e, §1.2 concept]

Solution. Negative exactly when $\cos \theta < 0$, i.e. the angle exceeds 90° (obtuse); zero when $\theta = 90^\circ$ — perpendicularity. This is the classification used in the Week 2 TA problems.

- (§2.5) Solve $A\mathbf{x} = \mathbf{b}$ with $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $\mathbf{b} = (3, 4)^T$ via A^{-1} .

[Source: Strang Intro 4e, §2.5 problem set]

Solution. $\det A = 3$: $A^{-1} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$; $\mathbf{x} = A^{-1}\mathbf{b} = \frac{1}{3} \begin{bmatrix} 2 \cdot 3 - 4 \\ -3 + 8 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 2/3 \\ 5/3 \end{bmatrix}$.
Check: $2 \cdot \frac{2}{3} + \frac{5}{3} = 3$, $\frac{2}{3} + 2 \cdot \frac{5}{3} = 4$. ✓

- (§2.5) Why does a nonzero solution of $A\mathbf{x} = \mathbf{0}$ rule out an inverse? Give the 2-sentence reason the Notes call “no matrix can bring $\mathbf{0}$ back to $\mathbf{x}$ ”.

[Source: Strang Intro 4e, §2.5 Note 4]

Solution. If $A\mathbf{x} = \mathbf{0}$ with $\mathbf{x} \neq \mathbf{0}$ and A had an inverse, then $\mathbf{x} = A^{-1}A\mathbf{x} = A^{-1}\mathbf{0} = \mathbf{0}$ — a contradiction. The zero vector has no “pre-image trouble” only when the map is one-to-one; a null vector proves it is not.

- (§2.5) If A and C are invertible 3×3 matrices, is $A + C$ invertible? Is AC ? Is $A^{-1}C$?

[Source: Strang Intro 4e, §2.5 problem set]

Solution. $A + C$: not necessarily (e.g. $C = -A$ gives 0); sums of invertibles can be singular. AC : yes, with $(AC)^{-1} = C^{-1}A^{-1}$ (113). $A^{-1}C$: yes, product of two invertibles.

7.5 Strang, *Linear Algebra and its Applications* 4e, §1.6 (Inverses and Transposes) — key results and a worked example

The assigned reading in §1.6 adds three items to the catalogue: the worked Gauss–Jordan inversion

of $A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix}$, the identity $\det A^{-1} = \frac{1}{\det A}$, and the transpose identities.

1. Carry out Gauss–Jordan for $A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix}$ and state A^{-1} .

[Source: Strang LAiA 4e, §1.6 Example 1]

Solution. Augment with I and eliminate: pivots are 2, -8 , 1 (after $R_2 - 2R_1$, $R_3 + R_1$, then $R_3 + \frac{3}{4}R_2$ of the reduced rows). Continuing upward to clear above the pivots and finally dividing rows by 2, -8 , 1 yields

$$A^{-1} = \begin{bmatrix} 12/8 & -5/8 & -6/8 \\ 8/8 & -3/8 & -2/8 \\ -1/8 & 1/8 & 1/8 \end{bmatrix} = \frac{1}{8} \begin{bmatrix} 12 & -5 & -6 \\ 8 & -3 & -2 \\ -1 & 1 & 1 \end{bmatrix}.$$

The determinant equals the pivot product $2(-8)(1) = -16$, and indeed A^{-1} displays the division by $\det A$ (up to sign placement of the cofactors — equivalently $A^{-1} = \frac{\text{adj } A}{\det A}$).

2. State the determinant-transpose identities used in §1.6: $\det A^T$, $\det(AB)$, $\det A^{-1}$.

[Source: Strang LAiA 4e, §1.6]

Solution. $\det A^T = \det A$; $\det(AB) = \det A \det B$ (Table 1); and, setting $B = A^{-1}$, $\det(A) \det(A^{-1}) = \det I = 1$, so

$$\det A^{-1} = \frac{1}{\det A},$$

which is exactly the statement “invertible $\iff \det A \neq 0$ ”, tying the Week 3 determinant catalogue to the Week 4 inverse catalogue.

3. Prove $(A^{-1})^T = (A^T)^{-1}$ and explain why $A^T A$ is symmetric with nonnegative diagonal.

[Source: Strang LAiA 4e, §1.6]

Solution. Transposing $AA^{-1} = I$ gives $(A^{-1})^T A^T = I$, which says exactly that $(A^{-1})^T$ is the inverse of A^T . For symmetry: $(A^T A)^T = A^T (A^T)^T = A^T A$; the diagonal entries are $\sum_k a_{ki}^2 \geq 0$, sums of squares — the fact behind CC^T in the Week 3 lecture problems.